From Robot and AI Threats to Restoration Politics: The Role of Perceived Marginalization and Nostalgia

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Why does technological disruption generate demands for restoration? I argue that automation threats become politically consequential when they disrupt expected life trajectories, erode valued roles in families and communities, and undermine recognition and social standing. These threats activate cultural grievances: perceived marginalization and nostalgia for a better past, which increase receptivity to restorationist appeals. I test this argument using a design-based experimental mediation analysis with double randomization of automation-risk exposure (robot- or AI-driven layoffs) and cultural-grievance encouragement. Both threats increase cultural grievances and support for exclusionary policies, including trade protectionism. Only the manufacturing-layoff threat increases anti-immigration attitudes. Encouraging cultural grievance shifts trade and immigration attitudes toward exclusion and points to a pathway that increases support for Trump. To assess external validity, I examine European observational evidence linking occupational automation risk to cultural grievances and radical-right voting. The findings identify a cultural mechanism linking automation threat to exclusionary and restoration politics.

Keywords: automation, layoffs, cultural grievances, nostalgia, perceived marginalization.

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AI and robotics are reshaping work and wages. Task-replacing technologies explain roughly one-half to two-thirds of recent U.S. wage changes (Acemoglu and Restrepo, 2022), while generative AI broadens exposure to clerical and professional work (Eloundou et al., 2024). Public concern over displacement is widespread,[1] and recently more than 200 experts have called for institutions that steer AI toward complementing workers and sharing its gains broadly (We-Must-Act-Now, 2026). Yet backlash rarely targets adopting firms or demands compensation and retraining (Gingrich, Wu, and Zhang, 2026). Instead, parties promising a return to a familiar order have advanced. Why does technological disruption travel with demands for restoration?

I argue that perceived automation risk can activate *cultural grievance*, expressed as marginalization and nostalgia for a better past. Respondents often described technological change in distinctly ambivalent terms. One observed, "Technology can be a blessing and a curse... I've liked what it has done for me at work, but I also know it's only a few steps away from being able to do my job"; another remarked, "I like the ease of work that is created through technology, but I see it quickly replacing so many jobs. It has been fun while it lasted though." Because occupations confer not only income but also identity, dignity, and social recognition, such threats can signal an irreversible loss of standing for "people like me." Marginalization captures perceived exclusion in the present, whereas nostalgia contrasts the present with a more valued past. These grievances can increase receptivity to appeals promising renewed recognition and the restoration of social order, thereby shifting voters toward exclusionary policies and the politicians—often right-wing populists—who champion them.

Economic and cultural accounts are not mutually exclusive: slow-moving cultural orientations cannot alone explain rapid political change, but economic disruptions may make identity and group standing politically salient (Rodrik, 2021). Studies associate economic shocks with authoritarian and nativist attitudes (Ballard-Rosa, Jensen, and Scheve, 2022; Carreras, Irepoglu Carreras, and Bowler, 2019), xenophobic and populist voting (Hays, Lim, and Spoon, 2019; Gonzalez-Rostani and Hays, 2026), and cultural discontent (Rhodes-Purdy, Navarre, and Utych, 2021*a*). Psychological research suggests

[1] For example, in the 2017 Eurobarometer, about three-quarters of respondents worried about job losses from robots and AI.

a similar mechanism, linking major disruptions to perceived social threat (e.g., Barauskaitė, Gineikienė, and Fennis, 2022; Routledge et al., 2008; Granulo, Fuchs, and Puntoni, 2019; Zhou et al., 2013). Yet the full economic–cultural pathway is difficult to identify. Controlling for post-treatment cultural attitudes can bias estimates (Agnolin, Colantone, and Stanig, 2024), while observational mediation requires strong assumptions (Imai, Keele, and Tingley, 2010). Most studies therefore identify only one link in the proposed chain.

I address these limitations by adapting experimental causal-chain designs from social psychology to political economy (Pirlott and MacKinnon, 2016; Bullock and Green, 2021; Spencer, Zanna, and Fong, 2005). The resulting encouragement–mediator design randomizes the treatment and the encouragement of the mediator. Respondents first review and rate two brief news stories for an online blog. They are randomly assigned to stories about job displacement by manufacturing automation or AI or to neutral technology stories unrelated to job loss. A random subset then writes a brief letter to the editor recalling either marginalization or a nostalgic memory, thereby encouraging cultural grievance. This double randomization tests the pathway from automation risk through grievance to policy preferences and voting intentions. Within the experimental survey, I supplement this design-based test with model-based mediation using observed variation in post-treatment grievance measures. Because a brief experimental intervention cannot reproduce workers' sustained experiences of labor-market change, I also examine whether the proposed relationships extend beyond the experimental setting using observational data from Western European countries.

Building on research linking technological change to populist backlash (Anelli, Colantone, and Stanig, 2021; Gonzalez-Rostani, 2026), this paper identifies a cultural mechanism underlying that relationship. I find that exposure to automation risk heightens nostalgia and perceived marginalization. Exploratory heterogeneity analyses further suggest that these cultural-grievance effects are stronger among White respondents, those without a college degree, and residents of communities experiencing greater employment losses. Automation can thus reshape individuals' perceptions of their social position, fostering feelings of cultural loss and exclusion that likely increase receptiveness to right-wing populist appeals. This mechanism identified here may help explain the broader puzzle

for embedded liberalism raised by Milner (2021): trade and technological change are associated with greater support for the populist right, yet existing social protections do not consistently temper the political backlash accompanying these disruptions. When technological disruption also blocks expected life trajectories and erodes valued roles within families and communities, social insurance and transfer programs aimed primarily at income loss may leave grievances about recognition and social standing unaddressed.

Second, the paper shows that cultural grievances link automation risk to political preferences. Respondents exposed to possible replacement by robots or AI become more supportive of exclusionary policies, expressing less favorable views of trade (especially among Whites) and, under some conditions, immigration. Heightened nostalgia and perceived marginalization, in turn, increase support for these policies and populist-right candidates. By independently randomizing technological threat and encouragements targeting the proposed mediators, the double-randomization design provides causal evidence on successive links between threat and political behavior and offers political scientists a template for testing other mechanisms. A further methodological innovation is the use of writing-based encouragement tasks from psychology to study constructs that cannot be manipulated directly, including identities, emotions, and cultural grievances. Variation in task engagement gauges treatment uptake and mediator activation. Moreover, embedding the treatments in active decision tasks and a realistic news-based cover story enhances ecological validity (Lelkes and Westwood, 2017).

Finally, the paper extends research on automation politics beyond its emphasis on blue-collar displacement from routine-task exposure, job loss, and local economic decline (Autor, 2013*b*; Colantone and Stanig, 2018; Broz, Frieden, and Weymouth, 2021; Baccini and Weymouth, 2021; Anelli, Colantone, and Stanig, 2021; Chaudoin and Mangini, 2025). Less is known about clerical, administrative, and professional workers increasingly exposed to AI displacement (Boix et al., 2026). Comparing threats from industrial robots and AI, I find that both heighten cultural grievances and some exclusionary attitudes, but AI threat does not consistently increase opposition to immigration. This divergence may reflect an occupational mismatch: AI primarily threatens highly educated, white-collar workers, whereas industrial robots threaten production and other less-skilled jobs (Webb,

2019; Eloundou et al., 2024). Because immigration concerns center on competition from less-skilled workers (Hainmueller and Hiscox, 2007, 2010), AI insecurity may not translate into broader anti-immigration sentiment.

These findings have implications for political demand and supply. On the demand side, perceived automation risk may activate cultural grievances among middle-class voters, a constituency central to party systems and democratic stability (Moore, 1966; Boix, 2003), potentially deepening polarization and threatening democratic institutions. On the supply side, politicians can mobilize these grievances. In recent U.S. elections, right-wing populists have combined pro-worker nationalism with exclusionary appeals (Gonzalez-Rostani, 2026), while protectionist policies framed as restoring jobs—especially tariffs—can attract voters when coupled with anti-elitism (Dai and Kustov, 2023; Neuner and Wratil, 2022; Balcazar, 2024). Meanwhile, automation has eroded employment in union-dense sectors (Agnolin et al., 2024), and left and social democratic parties have struggled to represent affected workers, widening representational gaps (Berman, 2021; Roberts, 2024). The compensation central to embedded liberalism may therefore be insufficient unless it also addresses the cultural grievances animating exclusionary demands (Milner, 2021; Mansfield, Milner, and Rudra, 2021).

Economic Change, Automation and Cultural Grievances

Economic Change and Cultural Activation: What We Know and What Is Missing

Existing research links major economic disruptions to political backlash through both material and symbolic channels. Classic accounts emphasize the political consequences of material interests (Scheve and Slaughter, 2001; O'Rourke et al., 2001; Mayda and Rodrik, 2005). More recent work shows that preferences also respond to perceptions and anxieties, even in the absence of direct losses. People may react politically to anticipated economic harm or perceived status decline (Guisinger, 2017; Mansfield, Mutz, and Brackbill, 2019; Owen and Johnston, 2017; Mansfield and Mutz, 2013).

In the United States, Mutz (2018) finds that perceived loss of group standing among traditionally high-status Americans predicted support for Donald Trump in 2016 more strongly than personal economic hardship. This finding need not place economic and cultural explanations in opposition. Identities and value orientations develop slowly,

whereas economic shocks occur at identifiable moments. Economic disruption may therefore activate existing concerns about recognition, group position, and social continuity, making them newly salient and consequential for political judgment (Rodrik, 2021).

Testing this pathway poses both identification and measurement challenges. It requires valid measures of cultural grievance and evidence for two successive causal links: economic threat must activate grievance, and grievance must subsequently shape political preferences. Existing approaches rarely establish both. Ethnographic and case-intensive studies reveal how communities interpret economic and social change, but their findings may not generalize beyond the communities studied (e.g., Gest, 2016; Cramer, 2016; Lamont, 2009; Newman and Skocpol, 2023). Aggregate analyses link local shocks to political change but cannot show changes in individual cultural orientations; surveys jointly measure insecurity, grievance, and behavior but cannot rule out confounding factors. Identifying a cultural mediator is especially demanding because it may share unobserved causes with the political outcome and, when measured after exposure, may itself be post-treatment (Agnolin, Colantone, and Stanig, 2024).

Existing research provides substantial evidence that economic disruption is accompanied by culturally and politically consequential responses. Long-term regional decline fostered anti-immigrant and Eurosceptic attitudes associated with support for Brexit (Carreras, Irepoglu Carreras, and Bowler, 2019). Economic decline can also heighten authoritarian values (Ballard-Rosa et al., 2021). Those living in relatively diverse regions facing more intense competition from Chinese imports similarly express more authoritarian values (Ballard-Rosa, Jensen, and Scheve, 2022). In addition, the loss of recognition and social integration derived from valued work can create reservoirs of support for radical parties (Gidron and Hall, 2020). Rising unemployment is associated with declining trust in national and European political institutions and greater support for anti-establishment parties (Algan et al., 2017), while institutional distrust figures prominently in explanations of the demand for populism (Guiso et al., 2017).

Other studies focus more directly on defensive and electoral responses. Exposure to international economic integration can increase protectionism and xenophobia (Bisbee et al., 2020; Gonzalez-Rostani and Hays, 2026). Among white voters in the United

States and the United Kingdom, experiences of economic decline are associated with greater support for radical-right or challenger candidates. This relationship appears both among people who subjectively perceive economic deprivation (Gest, Reny, and Mayer, 2018) and among those exposed to manufacturing layoffs (Baccini and Weymouth, 2021). Beyond individual economic experiences, broader pessimism about societal decline is also associated with stronger support for the radical right (Steenvoorden and Harteveld, 2018).

Taken together, these studies illuminate parts of the activation sequence, but evidence that economic shocks predict authoritarianism, distrust, or perceived status loss does not show that these attitudes shape political choice; conversely, evidence that grievance predicts populist support does not identify what activated it. A smaller literature traces both links. Regional import exposure is associated with xenophobic beliefs that mediate its relationship with right-wing populist support (Hays, Lim, and Spoon, 2019). Long-term local decline predicts Eurosceptic and anti-immigrant attitudes and, in turn, support for Brexit (Carreras, Irepoglu Carreras, and Bowler, 2019). Economic shocks may also generate negative emotions that foster cultural discontent and activate populist attitudes (Rhodes-Purdy, Navarre, and Utych, 2023). Whether automation follows this sequence remains unknown; this study addresses that gap.

Automation and the Mechanism Gap

Automation, digitalization, and AI are reshaping labor markets, but their political mechanisms remain understudied (Gallego and Kurer, 2022). Automation risk is associated with radical-right voting (Im et al., 2019; Anelli, Colantone, and Stanig, 2021; Milner, 2021), especially when workers lose ground in the occupational hierarchy (Kurer, 2020), and with weaker partisan attachment and political participation (Gonzalez-Rostani, 2024). In the United States, areas more exposed to computerization showed greater support for Donald Trump in 2016 (Frey, Berger, and Chen, 2017); in both the United States and Germany, workers in routine occupations were more likely to defect from establishment parties to populist challengers (Gonzalez-Rostani, 2026).

These responses do not fit a simple material-compensation account. In survey experiments, technologically induced layoffs increased demand for government action, but

support for transfers rose only when bad management was blamed, while internationally framed shocks favored trade protection (Di Tella and Rodrik, 2020). Across Europe, automation risk produces heterogeneous social-policy preferences (Busemeyer and Sahm, 2022; Kurer and Häusermann, 2022), and welfare compensation does not consistently reduce political backlash (Gingrich, 2019; Milner, 2021). The puzzle is why technological insecurity generates demands for protection or restoration even when material compensation is available.

Misdirected blame is one possibility. Workers at greater automation risk are more opposed to trade and immigration (Wu, 2022*a*; Mutz, 2021), and attributing manufacturing job loss to trade rather than automation shifts trade attitudes and heightens perceived threat to the national in-group (Mutz, 2021). Yet explicitly identifying automation as the threat still increased Republicans' support for immigration restrictions and Democrats' support for tariffs, while respondents remained reluctant to restrict technology itself (Wu, 2023). Source categorization also matters: foreign-origin technology can increase demand to "backpedal" against a shock (Chaudoin and Mangini, 2025). These studies establish that even correctly identified technological threats can elicit outsider-focused preferences, but they do not explain why this response occurs.

The remaining gap is whether automation activates cultural grievances that subsequently shape political choice. The closest study, the exploratory observational analysis by Anelli, Colantone, and Stanig (2021), associates vulnerability to robot adoption with radical-right support conditional on traditionalist and nativist predispositions. Whereas that study treats these orientations as moderators, I test whether automation threat heightens nostalgia and perceived marginalization and whether these grievances, in turn, shape political preferences.

AI provides a demanding scope test. Research on the politics of displacement has concentrated on manufacturing, routine work, and industrial decline (Autor, 2013*b*; Colantone and Stanig, 2018; Broz, Frieden, and Weymouth, 2021; Baccini and Weymouth, 2021), while AI's political effects remain poorly understood (Boix et al., 2026). Because generative AI exposure spans the wage distribution and may be greatest in higher-income

occupations (Eloundou et al., 2024), comparing threats from industrial robots and AI tests whether cultural activation extends beyond traditional industrial settings.

From Automation Threat to Restoration Politics

I argue that automation can produce political backlash by activating cultural grievance. Automation threat may do more than create economic insecurity. By jeopardizing employment, earnings, career prospects, or the value of one's skills, it can disrupt life plans and undermine perceived control and social status. Work provides both material security and socially valued roles through which people demonstrate competence and claim recognition as providers and community members (Hout, 2008). Threats to a valued occupation can therefore acquire cultural meaning before income falls, signaling that its contribution and way of life no longer command recognition. Automation may be especially likely to acquire this meaning because its broad reach can make change appear difficult to escape (Boix et al., 2026), while automatable tasks may seem gone for good, making prospective displacement evidence of lasting decline (Elchardus and Spruyt, 2016) and fostering alienation (Gonzalez-Rostani, 2024). By analogy with other economic threats, the resulting uncertainty may generate demand for protection before displacement occurs (Mansfield, Mutz, and Brackbill, 2019) and heighten the relevance of order, conformity, and decisive authority (Ballard-Rosa, Jensen, and Scheve, 2022; Ballard-Rosa et al., 2021). Applied to automation, cultural activation occurs when these signals make concerns about group standing, social order, and continuity politically salient; exclusionary responses therefore need not rest on mistaken causal beliefs.

I use the term *cultural grievance* to describe interpretations in which economic disruption signals that a valued in-group is losing recognition. Economic status concerns resources and labor-market opportunities; cultural status concerns whether "people like us" remain valued and central members of the national community. The relevant in-group may be occupational, local, or national, and material and cultural status can overlap. I distinguish a present-oriented grievance, perceived marginalization, from a backward-looking one, identity-relevant nostalgia.

Perceived marginalization is the belief that one's group is ignored, disrespected, or pushed toward society's fringes. Automation can activate this grievance when the

disappearance of valued work suggests that technological progress benefits others while rendering one's group dispensable, threatening voice and social worth (Gidron and Hall, 2020). This resembles Hirschman and Rothschild's "tunnel effect": frustration rises when expected mobility closes. When automation appears irreversible and unlikely to benefit one's group, technological change may thus be understood as exclusion rather than progress.

Identity-relevant nostalgia is the perceived gap between an in-group's current position and its position in an idealized collective past (Ferwerda, Gest, and Reny, 2025; Rogenhofer et al., 2023; Grosche and Rothmund, 2026). Rather than mere affection for the past, it is a collective judgment that "people like us" once held greater standing and that their values, contributions, or ways of life commanded broader recognition. The comparison may concern economic security, political influence, social recognition, or cultural authority and need not reflect objective decline. Under uncertainty, selective recollection can restore meaning and group continuity (e.g., Zhou et al., 2013, 2022). Technological disruption may be experienced as blocking a valued way of life in which work enabled people to pursue expected life plans, fulfill family and community responsibilities, and claim social standing. It may therefore evoke nostalgia for a past in which that way of life appeared more attainable and socially valued. Robotization may invoke the decline of industrial labor and manufacturing communities; AI may evoke comparable concerns among clerical, professional, and creative workers. Nostalgia thus identifies both what was lost and the social order that politics might restore. I therefore hypothesize:

Hypothesis 1. ***Cultural Grievance*** *Individuals exposed to automation threat are more likely to report heightened perceived marginalization and nostalgia.*

Why Cultural Grievance Can Favor the Populist Right

Anxiety alone does not determine a partisan destination, but cultural grievance can help us understand why right-populist appeals may be more resonant. I study this demand-side process in postindustrial democracies where right-populist options are available. Political supply is therefore a scope condition rather than an outcome; explaining how populist parties emerge or choose their strategies lies beyond this paper's scope. The *populist right* combines a moral opposition between a virtuous people and an unresponsive elite with

nativism, traditionalism, and promises to restore sovereignty or established boundaries (Mudde, 2007; Hawkins et al., 2018; Kaltwasser et al., 2017). Appeals to forgotten citizens and recovered control frame marginalization as elite neglect (Balcazar, 2024; Gonzalez-Rostani, 2026). By attributing decline to cosmopolitan elites, immigration, weakened boundaries, and the displacement of historically central groups (Rooduijn and Pauwels, 2011), right-populist actors also give nostalgia a restorationist destination (Müller and Proksch, 2024). These claims can recast correctly perceived technological change as evidence that established ways of life no longer command respect.

Cultural grievance may also make right-wing appeals more attractive because it aligns with conservative commitments and temporal outlooks. Conservatism is associated with resistance to change (Jost, 2017; Jost et al., 2003), greater emphasis on in-group loyalty, authority, and binding norms (Graham, Haidt, and Nosek, 2009; Pratto et al., 1994), and preferences for tradition, conformity, and the status quo (Schwartz et al., 2012; Jost, Banaji, and Nosek, 2004). These ideological dispositions make grievances about group displacement and devaluation especially compatible with right-wing identification. Furthermore, conservatives tend to be more past-oriented, whereas liberals tend to be more future-oriented (Robinson et al., 2015; Baldwin and Lammers, 2016); for example, Lammers and Uğurlar (2024) find that Western conservatives express greater nostalgia for the past. Accordingly, the two components of cultural grievance—marginalization and identity-relevant nostalgia—should resonate more strongly with right-wing projects of restoration than with progressive projects centered on future change.

Economic anxiety should generate broad demand for protection, including trade restrictions (Mansfield, Mutz, and Brackbill, 2019). Immigration restrictions additionally require a salient boundary between an in-group and a perceived out-group, while anti-elitism requires attributing neglect or decline to unresponsive authorities. When economic anxiety activates cultural grievance, right-populist options should become especially attractive because they combine these interpretations with a restorative political project. These arguments yield two further hypotheses:

Hypothesis 2. ***Populism and Exclusionary Policies*** *Individuals exposed to automation risk will be more likely to support right-wing populist candidates and their exclusionary policies.*

Hypothesis 3. ***Mediated Effect*** *The effect of automation exposure on support for right-wing populism and exclusionary policies is partly mediated by cultural grievances.*

I also examine whether these effects vary by technology. Threats from industrial robots and AI may both heighten cultural grievance and demand for economic protection, but their effects on immigration attitudes may differ. Robot-related insecurity fits narratives of offshoring, manufacturing decline, and competition with less-skilled labor. AI exposure, by contrast, is more concentrated among highly educated workers (Eloundou et al., 2024), who tend to hold more favorable immigration attitudes (Hainmueller and Hiscox, 2007). AI-related insecurity may therefore translate into exclusionary preferences less readily than robot-related insecurity.

Testing the Automation Pathway: Measures, Methods, and Results

To evaluate the hypotheses, I combine a survey experiment with cross-national observational analysis. The experiment provides internally valid estimates of immediate attitudinal responses to automation-related information and tests the roles of perceived marginalization and nostalgia. Its brief treatments, however, may not capture the cumulative effects of labor-market displacement. I therefore examine whether the same relationships hold across occupational risks and political contexts in European countries, providing a test of external validity. Together, the two designs test whether automation risk activates cultural grievances that increase support for exclusionary policies and for the politicians who promote them.

Study 1: Experimental Evidence

To date, only a limited number of studies have utilized survey experiments to explore the effects of job automation on individuals' attitudes. For instance, Wu (2022*b*) conducted a survey experiment comparing various sources of job threats. Her findings indicate a modest increase in support among Democrats for restricting technological integration. Additionally, Mutz (2021) observes a decline in support for international trade when

individuals were exposed to job loss resulting from automation. Furthermore, in the field of psychology, Yam et al. (2022) demonstrate that exposure to robots increased job insecurity, while Granulo, Fuchs, and Puntoni (2019) show that people preferred being replaced by robots over humans, reflecting a decline in pro-social sentiments when assessing their own job prospects.

I build on and advance previous experiments in several ways. First, unlike previous political science studies that focus on comparing various threat sources, my survey contrasts technology in a neutral context with technology leading to job losses, directly addressing the consequences of technological job replacement. Second, I introduce variations in the treatment to analyze the effects of incorporating robots, which predominantly affect blue-collar workers, and integrating AI, which primarily impacts white-collar workers. This approach provides new insights into the recent surge in AI's impact, an aspect previously overlooked due to an exclusive focus on manufacturing job losses. Third, I incorporate additional outcomes of interest, such as cultural grievances, to gain a comprehensive understanding of the consequences stemming from exposure to automation risk. Fourth, I implement active participation tasks and create an ecologically valid news consumption experience within the experiment rather than just prompting subjects. Lastly, my experiment is designed not only to examine the presence or absence of changes in political behavior due to automation risk but also to illuminate the mechanisms behind these changes.

Experimental Design and Procedures. I fielded the survey through CloudResearch, a widely used platform for online survey experiments in political behavior, between May 23 and 29, 2023 (Stagnaro et al., 2026). The sample comprises 3,133 U.S. citizens aged 18 or older who were either employed or seeking work.[2] I implement a design-based experimental mediation analysis with double randomization of the treatment (exposure to automation risk) and encouragement of the mediator cultural grievances.[3] This produces

[2]I requested CloudResearch's Census-Matched sampling option, which uses recruitment quotas based on age, gender, race, and Hispanic ethnicity to approximate U.S. Census benchmarks. Because respondents were recruited from an opt-in online panel, the resulting sample remains a nonprobability sample. I also implemented several measures to ensure data quality, including CAPTCHA to prevent spam and bots, location screening to limit participation to the US, attention checks, manipulation checks, survey timekeeping, a minimum time for some sections, and a minimum number of characters in writing exercises. To be considered for the analysis, respondents had to have a 90% survey approval HITS on CloudResearch and pass two attention checks. Appendix A.2 reports attrition and completion rates by assignment of treatment.

[3]For further reference of this method see Imai, Tingley, and Yamamoto (2013); Spencer, Zanna, and Fong (2005); Pirlott and MacKinnon (2016).

two tracks. In one track, respondents receive only the automation treatment or control; in the other, they are additionally randomized to mediator encouragement within treatment status.[4]

To provide an ecologically valid news consumption experience, in a realistic setting, I used a cover story and adopted the tasks for respondents based on a study conducted by Lelkes and Westwood (2017). Respondents were presented with two news articles[5] and given the responsibility of evaluating them to assist a startup online news content platform in deciding which article to publish. In the treatment condition, participants were exposed to two news articles discussing job displacement caused by automation. The two news focusing on either manufacturing jobs (robots treatment) or white-collar jobs affected by AI (AI treatment). One article highlighted an individual who recently lost their job as a result of technology incorporation, serving as a single identifiable victim, while the other article addressed the broader issue of upcoming layoffs and the overall impact on a community affected by technological change (every subject read both). Figure 1 illustrates the treatment task. The control condition featured two news articles related to technological advancements in a more neutral context. The following was the cover story used:

> The researchers hosting this survey are conducting it for the founders of an online news website about social change that launched about 3 months ago. In this short time period, their website has seen far more traffic than originally expected. Since their company is new to the online marketplace, they are conducting research on the topics and stories that consumers think are the most important. While most of the content appearing on their website homepage is selected by the editors, they have reserved certain slots for posts that the public can vote on. We would like your input regarding which of the following two articles should appear on next week's homepage.

[4]Both assignments were implemented in Qualtrics using simple randomization at the respondent level. Pretreatment characteristics were generally similar across experimental conditions; Appendix A.5 reports the full balance diagnostics.

[5]The news resembled the content that can be found in publications such as the *New York Times*. They were adapted from previous works such as Mutz (2021) and Chaudoin and Mangini (2025). Refer to Appendix A.6 for the news.

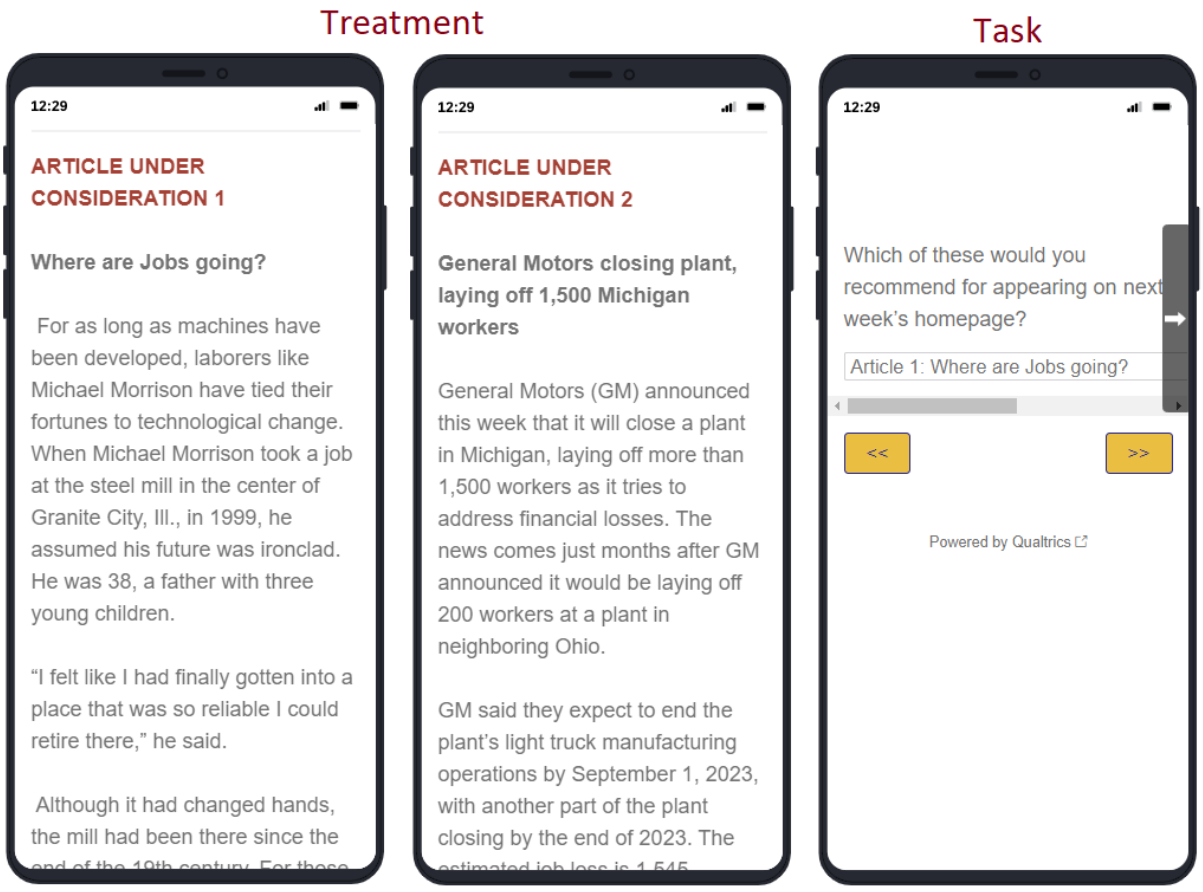


Figure 1: Illustration of the Treatment Task (Robot Exposure Condition)
Note: All respondents read two news articles and were asked to recommend one of them. The figure illustrates the robot exposure treatment. Other respondents were assigned to read articles about white-collar job displacement due to AI (AI treatment) or about technological development without labor-market implications (technology-neutral condition).

In the second experimental group, cultural grievances are randomly encouraged. Treatment and control conditions mirror those described above. To encourage cultural grievances, respondents are randomly assigned to complete a short autobiographical writing task designed to activate either nostalgic attitudes or feelings of perceived marginalization.[6] Respondents assigned to either task are classified as receiving a cultural-grievance encouragement. This pooling is intentional. Both nostalgia and perceived marginalization are theorized to activate the same underlying orientation: a sense of cultural loss and symbolic threat. Randomizing across these two prompts ensures meaningful variation in cultural grievance activation while preserving sufficient statistical power for downstream analyses. Importantly, pooling does not imply that these experiences are identical, but rather that they represent alternative pathways into the same latent grievance construct.[7]

[6]See examples in social psychology Xia, Wang, and Santana (2021), van Tilburg, Sedikides, and Wildschut (2015), and Newman et al. (2020), as well as in political science, for example, Rhodes-Purdy, Navarre, and Utych (2021*b*) and Busby, Gubler, and Hawkins (2019).

[7]Although the primary analysis pools grievance-inducing encouragement, post-treatment measures capture respondents' nostalgic sentiments and perceived marginalization directly. Hence, using model-

The cover story for this task was that the news organization was deciding whether to add a new section called "letter to the editors" which includes short passages from readers. Participants were prompted to think about a time in their life that made them feel a particular emotion. Those assigned to nostalgia saw a prompt that defined nostalgia, while those in the marginalization condition were prompted to recount an instance when they (or those similar to them) felt excluded or underappreciated by those who were different. The prompt explicitly asked subjects to:

> Take some time to think about the specifics of the situation - who was involved, what happened, and how you felt. We're interested in hearing about your personal experiences with [X] and what that experience was like for you.
>
> For the next 90 seconds, please write in detail about this event as if you were writing an anonymous letter to the editor. Your letter should be written with enough detail that someone reading this description could imagine the situation.

Following the reading and writing tasks, respondents indicated which article they would recommend for publication and whether the proposed letters-to-the-editor section should be added to the platform. These decisions reinforced the cover story and embedded the experimental tasks in a more natural news-consumption setting. Appendix A.4 provides additional details on the experimental procedures. The writing tasks also appear to have functioned as intended: validation analyses indicate high rates of prompt compliance, as well as the expected differences in emotional tone and lexical content across encouragement conditions. Moreover, comparisons by educational attainment provide little evidence that cognitive burden systematically shaped task performance (Appendix A.9).

Post-Treatment Questions. In the post-treatment section of the survey, I asked about respondents' political attitudes, public policy preferences, subjective exposure to risk, and cultural grievances. Specifically, I aim to assess whether the automation of jobs influences political behavior, such as support for right-wing populism and exclusionary policies. The political behavior questions encompassed various topics, such as support for

based mediation on observed post-treatment variables, I can assess whether these components operate through similar channels and produce comparable effects on populist attitudes.

a potential Trump candidacy in the 2024 presidential election, and evaluations of whether trade benefits American workers (core elements of neo-mercantilist foreign economic policies) or which policy should exist regarding legal immigrants.

Shifting attention to the observed mediators, I employed various questions previously utilized in psychology studies to measure nostalgic attitudes. To capture individual-level perceptions of nostalgia, respondents were asked to what extent they felt sentimental for the past (Newman et al., 2020). Additionally, questions related to collective nostalgia were included, such as whether many American traditions have been lost over time or if American identity is no longer what it used to be in the past (e.g, Smeekes, Sedikides, and Wildschut, 2023; Smeekes, Verkuyten, and Martinovic, 2015). Respondents indicated their level of agreement with these statements, allowing for the creation of an index representing nostalgic attitudes. To assess perceived marginalization, I examine individual-level perceptions of disrespect for the individual's own values, poor treatment in society, and collective perceptions of adversity faced by people similar to them in society (Rhodes-Purdy, Navarre, and Utych, 2021*b*).[8] Using these items on nostalgia and perceived marginalization, I constructed an index of cultural grievances.[9] These indexes serve two purposes: (1) to examine the effect of the treatment (automation risk exposure) on cultural grievances, and (2) to assess compliance with the encouragement of the mediators.

To serve as manipulation checks for the main treatment and to gain insight into the subjective perception of risk, I included several questions toward the end of the survey that aimed to assess respondents' concerns about the possibility of losing their current job due to automation, their outlook on future job prospects, and their general evaluations of the future of work in society. To assuage concerns that convenience samples may not yield results representative of the entire population, I have included summary statistics in the Appendix A.3 and their correlations with national survey samples. The sample aligns closely with the target population in terms of race and ethnicity, gender, and age, although it

[8]Specifically, respondents were asked to indicate their degree of agreement with statements such as "My values are not respected in this country," "People with values like mine are treated poorly in this society," and "Regardless of who is in political power, things are generally pretty bad for people like me."

[9]Cultural grievances are measured using three related indices: marginalization, nostalgia, and an overall cultural-grievances measure. The marginalization and nostalgia indices are constructed by summing their respective three items, while the overall index sums all six items. These measures exhibit good internal consistency. Appendix A.10 presents the complete item wording, additive construction of the indices, alternative PCA-based operationalizations, and the corresponding robustness analyses.

is more highly educated than the broader U.S. adult population. The primary experimental estimates are unweighted because causal identification comes from random assignment. Nevertheless, to assess whether differences in sample composition—particularly the overrepresentation of college-educated respondents—affect the findings, I reestimate the models using raking weights calibrated to the adult U.S. population.[10] The full survey instrument is reproduced in the Appendix.

Estimation Strategy. Survey experiments frequently identify treatment effects but often leave the mechanisms producing them unexplained.[11] Mediation analysis addresses this gap by examining how and why treatments affect outcomes.[12] It therefore constitutes my primary empirical strategy. Specifically, I examine cultural grievances as a mechanism linking automation-risk exposure to political attitudes, rather than as a moderator of that relationship. A moderation analysis would instead require interacting treatment assignment with a pretreatment measure of grievances. Because grievances are theorized as a response to automation risk and measured after treatment, I analyze them as a mediator.[13]

Following Baron and Kenny (1986), I model the effect of automation-risk exposure (X) on political attitudes and behavior (Y) through cultural grievances (M).Employing a structural equation model (equations 1-3), where i indexes subjects, α denotes intercepts, and ϵ represents zero-mean error terms from unobserved variables, this approach quantifies the total effect (ATE) of X on Y as c (equation 2), the direct effect (ADE) as d, and the mediated effect (ACME) through the product-of-coefficients method between the effects of X on M and M on Y (ab).

[10] I rake the sample to adult ACS benchmarks for sex, Hispanic origin, education, race/ethnicity, the interaction between education and broad race/ethnicity (White versus non-White), age group, and Census region. Following DeBell and Krosnick (2009), I cap the weights at 5 to limit the influence of extreme values. Similar weighting procedures are widely used in social-science research (Goldberg et al., 2019; Zhang et al., 2020; Penn, Petrolia, and Fannin, 2023; Wang et al., 2020).

[11] This challenge is known as the "black box" of causality. See Spencer, Zanna, and Fong (2005); Brady and Collier (2010); Imai et al. (2011).

[12] The method is widely used in social psychology: 59% of *JPSP* and 65% of *PSPB* articles published between 2005 and 2009 employed mediation analysis, as did 55 *JPSP* articles published in 2019 (Pirlott and MacKinnon, 2016).

[13] An interaction model may yield a zero coefficient even in the presence of strong mediation if the mediator's downstream effect is constant across treatment groups, or a non-zero coefficient if the treatment alters compliance behavior without altering the mediator itself (see Appendix F.6).

$$M_i = \alpha_1 + aX_i + \epsilon_{i1} \quad (1)$$
$$Y_i = \alpha_2 + cX_i + \epsilon_{i2} \quad (2)$$
$$Y_i = \alpha_3 + dX_i + bM_i + \epsilon_{i3} \quad (3)$$

Conventional model-based mediation measures rather than manipulates the mediator (Spencer, Zanna, and Fong, 2005). This approach requires assuming that no unobserved factors jointly affect M and Y. Otherwise, the error terms in equations 1 and 3 may be correlated, biasing the estimated mediator–outcome relationship (Spencer, Zanna, and Fong, 2005; Imai and Yamamoto, 2010; Bullock and Green, 2021; Bullock, Green, and Ha, 2010). A common misconception suggests that randomizing the independent variable can mitigate biases in mediation analysis (Bullock, Green, and Ha, 2010). However, while randomizing X ensures no systematic relationship with ϵ_1 or ϵ_3, it does not guarantee that M or Y are not systematically linked to these error terms, potentially introducing bias.[14]

In my *design-based experimental analysis*, I tackle this issue by experimentally manipulating both the treatment and the mediator. This approach allows for unbiased estimation of b and ensures temporal precedence from X to M and from M to Y (Pirlott and MacKinnon, 2016; Bullock, Green, and Ha, 2010). Initially, I estimate the total effect (equation 2), based on the expected outcome differences between treatment and control conditions. I conduct regressions for each outcome of interest—indicators of exclusionary policy preferences and right-wing populism—against the treatments. For the treatment-specific Robot and AI contrasts, I estimate separate two-arm models that retain the relevant treatment arm and the technology-neutral control group while excluding the alternative active treatment arm.[15] Subsequently, I regress the mediator—cultural grievances against the treatments, establishing the effects of exposure to automation risk on the mediators (equation 1). Third, I estimate the effect of experimentally induced grievances on the political outcomes (b) by comparing respondents assigned and not assigned to the encouragement. Because cultural grievances themselves cannot be assigned directly, the second randomization

[14]Measurement-of-mediation analysis design has been applied in top political science journals, including works based solely on observational data such as Karpowitz, Mendelberg, and Shaker (2012), and Hays, Lim, and Spoon (2019), as well as studies involving randomized treatment assignment in Tomz and Weeks (2020), Powers and Renshon (2023), and Young (2019).

[15]Under respondent-level simple randomization, these linear models identify arm-specific average treatment effects without cross-treatment contamination (Goldsmith-Pinkham, Hull, and Kolesár, 2024).

identifies the intent-to-treat effect of encouragement assignment. As a secondary analysis, I use encouragement assignment as an instrument for a constructed measure of uptake, defined by successful completion of the assigned task and a cultural-grievance level above the specified threshold. Finally, I calculate the mediated effect as the product of these coefficients (ab). Appendix F discusses the design's assumptions and limitations.

Evidence for the effect of Exposure to Automation Risk on Mediators

The paper's central claim is that the economic threat posed by technological displacement can alter perceptions of group standing and thereby activate cultural grievances. Its first empirical implication is that exposure to automation-related threat should heighten perceived marginalization and nostalgia for a valued past. I test this implication by comparing respondents who read about manufacturing workers displaced by robots or white-collar professionals displaced by AI with those who read neutral articles about technological development. The dependent variables are nostalgia, perceived marginalization, and a combined cultural-grievance index. The nostalgia index captures a backward-looking orientation toward a valued past, while the marginalization index captures perceived exclusion and declining social standing. All specifications adjust for pre-treatment gender, race, income, education, and occupation-level routine-task intensity (RTI).

The results presented in Figure 2 support **Hypothesis 1**, which predicts that the threat of automation heightens cultural grievances. The pooled treatment raises high cultural grievance from 19.7 percent under control to 26.0 percent—a 6.3-point, nearly one-third increase—and increases high perceived marginalization and identity-relevant nostalgia by 4.8 and 5.1 points, respectively. The treatment-specific estimates are substantively similar across the Robots and AI conditions.[16] The comparable effect in the AI condition shows that cultural grievance is not confined to blue-collar displacement.

Several complementary analyses reinforce the main interpretation. First, the findings are not an artifact of outcome dichotomization: models using the full continuous indices produce the same general pattern (Figure B.5). Nor are the results sensitive to index construction, as alternative measures derived using PCA yield substantively similar estimates (Figure A.4). Second, adjusting the main treatment-effect tests for multiple comparisons

[16]Formal Robots-versus-AI tests provide no evidence that the treatment effects differ for the mediators (Table C.12).

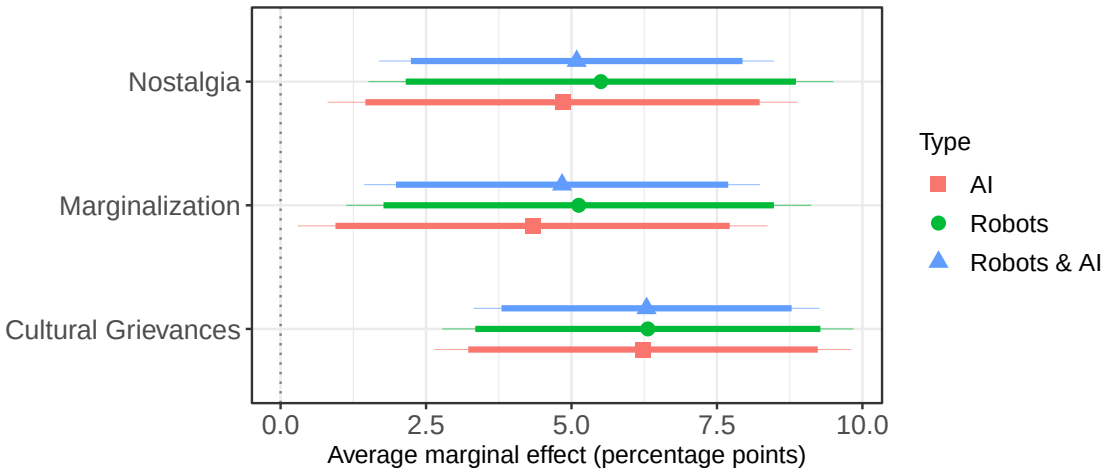


Figure 2: Effects of Exposure to Automation Risk on Mediators.
Note: This figure shows the effect of exposure to job-threatening news on the mediators against a control group of 1,182 subjects. 'AI' represents the treatment effect of AI-related job displacement (N=2,138), 'Robots' for machine-related displacement (N=2,177), and 'Robots & AI' pools the data (N=3,133). Results for each mediator are shown, with dummy variables representing scores above the median. 'Nostalgia' reflects sentimentality for the past, 'Marginalization' denotes a perceived exclusion - cultural discontent, and 'Cultural Grievances' merges both indices. Because these binary outcomes are estimated using logistic regression, the points report average marginal effects (AMEs), defined as the sample-average change in the predicted probability of scoring above the median when treatment changes from 0 to 1, with other covariates held at their observed values. Estimates are expressed in percentage points. The thick and thin bars report 90% and 95% confidence intervals, respectively. Detailed models (Table B.10), and without control variables (Figure D.7b) are in the Appendix. See supplementary material for the exact wording.

using the Benjamini–Hochberg procedure leaves the substantive conclusions unchanged (Appendix D.1). Third, I assess sample composition using population-calibrated raking weights, which yield estimates similar to the unweighted results (Appendix D.5). Fourth, reestimating the treatment effects with doubly robust augmented inverse-probability weighting (AIPW) reducing concerns about chance covariate imbalance and yields the same substantive conclusions (Appendix D.6). Finally, adding left–right self-identification to the models leaves the treatment estimates substantively unchanged, although left-identifying respondents exhibit lower baseline levels of nostalgia and marginalization (Table D.14).

A further analysis considers another theoretically relevant scope condition: respondents' local experience of economic decline. Plant closures can reduce employment and income among suppliers and other nearby businesses, extending economic distress beyond directly displaced workers (Baccini and Weymouth, 2021; Broz, Frieden, and Weymouth, 2021); local labor-market deterioration can also reshape political values and intensify authoritarian responses to economic disruption (Ballard-Rosa, Scheve, and Jensen, 2021; Ballard-Rosa, Jensen, and Scheve, 2022). These arguments imply that news of robot-driven manufacturing layoffs should be especially salient in communities that have previously experienced substantial employment loss. I evaluate this implication by linking respondents' ZIP codes to Census data on local employment change between 2002 and 2022. As Appendix D.4 shows, adjusting for local employment decline leaves

the estimated effect on cultural grievance substantively unchanged. Moreover, the Robots treatment produces a larger increase in cultural grievance in communities that experienced greater employment losses, a pattern driven primarily by heightened perceptions of marginalization rather than nostalgia. The corresponding interaction for the AI treatment is small and statistically indistinguishable from zero. This divergence is consistent with the possibility that the geography of past employment decline maps more closely onto vulnerability to robot-driven manufacturing displacement than to prospective displacement from AI.

Heterogeneous effects. Finally, I consider whether these effects on cultural grievances vary across respondent characteristics that previous literature identifies as moderators of economic shocks. Automation threats may be interpreted through pre-existing group positions, occupational vulnerabilities, and socio-demographic characteristics; these factors should shape both whether economic threat becomes cultural grievance and which component of grievance it activates. Because collective nostalgia compares the present with a valued group past, status-restoration accounts anticipate larger effects among White respondents than among respondents whose groups were historically excluded from that social order. This test more closely reflects my theoretical expectation that technological threat evokes feelings of marginalization and nostalgia among particular social groups rather than the population as a whole. The results by race therefore point to a potentially important boundary condition on both the activation and the political meaning of cultural grievances. Education and task content should also distinguish the two treatments. Robot-driven manufacturing displacement should be more salient to non-college respondents and workers in routine-intensive occupations, whose labor-market position and occupational identity are more closely tied to routine production work. The AI treatment, by contrast, describes threats to white-collar and professional work. Its education gradient may therefore be weaker or extend to college graduates, while conventional RTI may capture exposure to these emerging cognitive-task risks less completely. Pre-treatment ideology provides a further interpretive lens: respondents on the left and right may both react nostalgically to disruption, but differ in the collective past they value and in the political meaning they attach to it (Stefaniak et al., 2021; Jost, 2017).

I therefore examine heterogeneity by race, college education, occupational RTI, and ideology. Because the experiment was powered to identify average treatment effects in the full sample, subgroup estimates are necessarily less precise. I consequently emphasize directional patterns, treating these analyses as exploratory evidence about the argument's scope.

The subgroup analyses identify several differences, concentrated primarily in race and education (Appendix D.7). The clearest racial differences concern the backward-looking component of grievance: the estimated effects on nostalgia are larger among White than non-White respondents across the treatments. The combined cultural-grievance effect is also larger among White respondents in the pooled and AI comparisons. These patterns are consistent with research showing that the position of the racial group conditions political responses to economic decline (e.g., Baccini and Sattler, 2021; Gest, 2016; Jones, Owen, and Sung, 2026; Ballard-Rosa, Jensen, and Scheve, 2022). The education gradient is more circumscribed. The Robots treatment produces a larger increase in marginalization among respondents without a college degree than among college graduates, and the same ordering appears when the treatments are pooled, consistent with evidence that education conditions responses to economic disruption (Choi et al., 2024; Gidron and Hall, 2020). Under the AI treatment, however, the education difference is not statistically distinguishable from zero; if anything, the point estimate reverses, with a somewhat larger marginalization response among college graduates. Although this reversal is compatible with AI's distinctive reach into high-skill and professional work, the estimate is too imprecise to support a firm conclusion. By contrast, the RTI split yields no conventionally significant subgroup difference. Finally, stratifying by ideology also broadly recovers the main pattern (Figure D.8); the principal qualification is that the estimated effect of AI on nostalgia among conservatives is too imprecise to distinguish from zero.[17]

Evidence for the total effect of Exposure to Automation Risk on Politics

Having shown that exposure to automation risk activates cultural grievances, I turn to its political consequences. Exposure to news about automation risks increases support for protectionist and exclusionary policy positions: both treatments reduce support for trade

[17]Appendix D.7 reports the subgroup estimates, two-sided Wald tests of coefficient equality, and Benjamini–Hochberg-adjusted p-values.

openness, and the Robots treatment also increases anti-immigration attitudes (Figure 3). Neither treatment, however, produces a statistically detectable change in support for Trump. To put these results in context, the Robots treatment lowers trade favorability by 0.204 points, equivalent to 8.5% of the control-group mean and 0.16 control-group standard deviations (SD). The AI treatment produces a similar effect.

The effects on immigration, by contrast, depend on the type of technological threat. The Robots treatment increases anti-immigration attitudes by 3.8% relative to the control-group mean, or about one-tenth of a control-group SD, whereas the AI treatment produces no discernible change. A direct comparison confirms that the two treatment effects differ significantly (Table C.12). This asymmetry cannot be attributed to AI appearing less threatening. Both treatments increase automation concern (manipulation check): by 4.1% of the control mean (0.14 SD) under Robots and 7.7% (0.25 SD) under AI. One possibility is that robot-related displacement calls to mind industrial and manual employment, making perceived competition with lower-skilled immigrants a more salient frame. AI-related displacement may instead evoke risks to professional workers or threats originating from firms and technological systems. It may therefore heighten concern about automation without eliciting the same immigration response (Hainmueller and Hiscox, 2007).

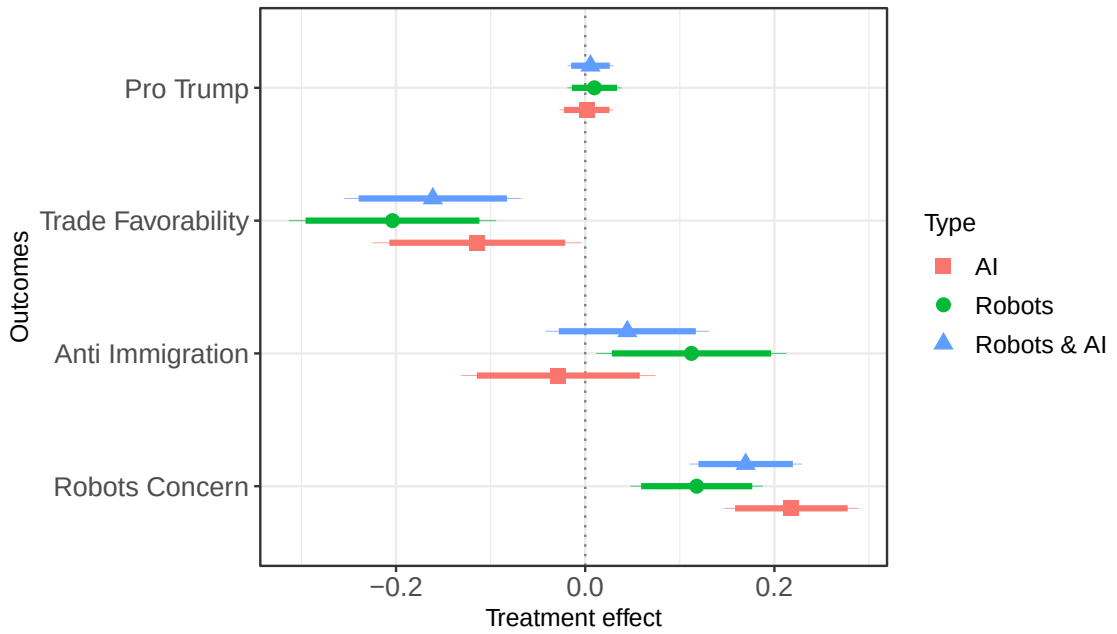


Figure 3: Total Effect of Exposure to Automation Risk (proxied as robots and AI).
Note: This figure shows the ATE of exposure to job-threatening news against a control group of 1,182 subjects. 'AI' represents the treatment effect of AI-related job displacement (N=2,138), 'Robots' for machine-related displacement (N=2,177), and 'Robots & AI' pools the data (N=3,133). Each row in the figure shows a set of results for a given outcome in a given treatment, distinguished by colors and shapes. Outcomes include 'Pro Trump' (willingness to vote for Trump), 'Trade favorability' (views on trade benefits for US workers), 'Anti-immigration' (preferences on legal immigration levels), and 'Robots concerns' (perceptions of automation risks), with 'Pro Trump' binary coded and others on a 1-5 scale. The thick and thin bars report 90% and 95% confidence intervals, respectively. Full models for the coefficients in this figure are available in Table C.11. Appendix presents estimation without control variables (Figure D.7a). See supplementary material for the exact wording.

Taken together, these results provide partial support for **Hypothesis 2**, which links automation risk to greater support for right-wing populist candidates and associated policies. Exposure shifts preferences toward trade protection and, in the Robots treatment, immigration restriction, but does not increase average support for Trump. Appendix Appendix C further shows that the Robots treatment heightens anti-elitist sentiment. This overall pattern suggests that narratives of economic displacement may increase support for exclusionary policies. Such policies, often coupled with anti-elitist appeals, are commonly espoused by the populist right. For example, Trump frames higher tariffs as a means of protecting vulnerable workers and "bringing jobs back."

Several additional analyses support the stability of these findings. The substantive conclusions are unchanged after correcting treatment-effect p-values using the Benjamini–Hochberg procedure (Appendix D.1), applying survey weights (Appendix D.5), or re-estimating the effects with doubly robust AIPW estimators (Appendix D.6). Moreover, treatment estimates remain largely unchanged after controlling for either pretreatment left–right identification (Table D.13) or local employment decline, with little evidence that employment decline moderates the effects (Table D.16). Beyond these robustness checks, we examine whether treatment effects vary across respondent characteristics. Because the study's power calculations were based on the full sample rather than individual subgroups, we treat these heterogeneity analyses as exploratory. The clearest heterogeneity appears by race, with larger estimated effects on trade protectionism and anti-elitism among White respondents (Figure D.13). For the political outcomes, differences by occupation and education are generally inconsistent and imprecisely estimated.[18]

The near-zero average effect on Trump support should not be interpreted as evidence that automation risk creates no pathway toward right-wing populism. The total effect could combine countervailing pathways, although this study does not identify an offsetting mechanism.[19] Exposure increases marginalization and nostalgia—grievances that right-populist appeals may channel through themes of restoration and exclusion—as well as

[18] Ideological heterogeneity is limited across policy outcomes. Trump support is the main exception: it increases among pre-treatment conservatives but not left-identifiers, a divergence that may help explain the null average effect (Figure D.8).

[19] One untested possibility is that some respondents attribute displacement to employer decisions or inadequate technology governance and may therefore favor regulation or worker protections rather than Trump. Future research should test such competing pathways directly.

support for immigration restriction and anti-elitism. The next section tests whether these cultural grievances mediate the effects of automation risk on political attitudes.

Evidence for Causal Mechanisms

So far, I have shown that exposure to news about job-threatening technological change activates cultural grievances and shifts policy preferences. I now examine whether these grievances mediate the effects of automation exposure on support for right-wing populism and exclusionary policies. I begin with a model-based mediation analysis of the survey data, using observed measures of nostalgia and perceived marginalization as proxies for cultural grievances. Figure 4 provides evidence of a mediated pathway to support for right-wing populism. Although the total effect on support for Trump is statistically indistinguishable from zero (as discussed in previous section), the estimated indirect effect through cultural grievances is positive. This pattern is consistent with cultural grievances increasing support for Trump. Cultural grievances also account for part of the decline in trade favorability. A similar mediated pattern emerges for preferences to restrict immigration. Substantively, the mediated effect accounts for approximately 39% of the total increase in anti-immigration attitudes and 13% of the total decline in trade favorability.

These results are consistent with **Hypothesis 3**, which posits that cultural grievances mediate the effects of automation threat on political attitudes. Their causal interpretation, however, depends on sequential ignorability. Although exposure is randomly assigned, in model-based mediation analysis cultural grievances are not; the analysis therefore assumes that, conditional on treatment and covariates, no unobserved factor jointly affects grievances and political outcomes. Because this assumption cannot be verified, Appendix E.1 assesses the sensitivity of the indirect effects to violations of this assumption. Additional checks yield substantively similar conclusions when applying survey weights, using doubly robust estimators, and adjusting p-values with the Benjamini–Hochberg procedure (Figure E.15). Although these analyses support the robustness of the findings, they cannot rule out unobserved mediator–outcome confounding. I therefore turn to an experimental test of the proposed pathway.

The experimental analysis addresses the central limitation of the model-based estimates by introducing randomized encouragement of the mediator. In addition to assigning

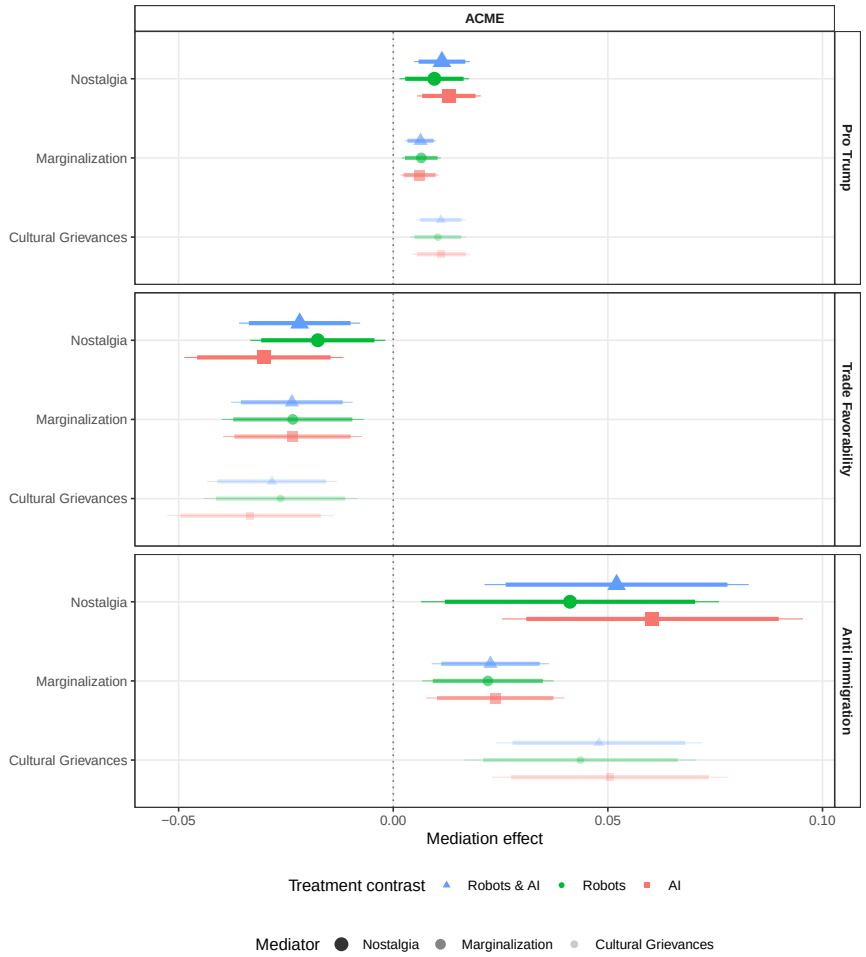


Figure 4: Average Causal Mediated Effects of Exposure to Automation Risk (proxied as robots and AI).

Note: This figure reports average mediated causal effects (ACMEs) of exposure to job-threatening news relative to a control group of 1,182 respondents. "AI" represents exposure to news about AI-related job displacement ($N = 2{,}138$), "Robots" represents exposure to news about machine-related job displacement ($N = 2{,}177$), and "Robots & AI" pools both treatment groups ($N = 3{,}133$). Colors and shapes distinguish these treatment contrasts. The mediators are nostalgia, marginalization, and cultural grievances, with cultural grievances combining the nostalgia and marginalization measures. The outcomes are "Pro Trump" (willingness to vote for Donald Trump), "Trade Favorability" (whether trade benefits U.S. workers), and "Anti Immigration" (preferences for reducing legal immigration). Pro Trump is binary, whereas the other outcomes are measured on 1–5 scales. All models adjust for control variables. Thick and thin bars report 90% and 95% confidence intervals, respectively. The corresponding average direct-effect (ADE) estimates appear in Figure E.14.

respondents to receive or not receive news about job-threatening automation, the design randomly assigns some respondents to complete a writing task intended to activate either nostalgia or perceived marginalization. Because both concepts are components of the hypothesized cultural-grievance mechanism, I pool respondents assigned to either writing task into a common encouragement condition.[20] I first estimate the intent-to-treat (ITT) effect of assignment to the cultural-grievance encouragement. These estimates, shown by the green circles in Figure 5, compare respondents randomly assigned to encouragement with those assigned to no encouragement, regardless of whether respondents completed

[20] I use the term "encouragement" because respondents' attitudes cannot be assigned directly. Some respondents assigned to a writing task may not complete it or may not exhibit the intended change in attitudes. This creates a problem of imperfect compliance rather than compromising the random assignment of the encouragement itself. See Imai, Tingley, and Yamamoto (2013).

the writing task or exhibited the intended mediator response. The point estimates indicate that encouragement reduced trade favorability and increased preferences for restricting legal immigration. The estimated effect on support for Trump is also positive, but less precisely estimated. Because encouragement assignment is randomized, these estimates identify the causal effect of being assigned to an intervention intended to activate cultural grievances.

The ITT estimates may understate the effect of actually taking up the encouragement because assignment does not guarantee either task completion or a corresponding change in the mediator. I therefore also report two-stage least-squares estimates, shown by the red squares in Figure 5. These models use random assignment to encouragement as an instrument for uptake. Uptake is defined by both completing the assigned writing task and exhibiting a cultural-grievance level above the specified compliance threshold.[21] Encouraged uptake of cultural grievances increased support for Trump by approximately 10.2 percentage points. This estimate is equivalent to 36% of the support rate in the no-encouragement group and to 0.23 SD. Encouraged uptake also reduced trade favorability by approximately 13 percentage points—27% of the no-encouragement mean and 0.26 SD—and increased support for reducing legal immigration by approximately 10.7 percentage points, equivalent to 39% of the no-encouragement mean and 0.24 SD. The 90% confidence intervals exclude zero for all three outcomes.

The experimental findings are robust to alternative definitions of uptake. Defining high mediator uptake using a simple above-median threshold produces estimates similar in direction and magnitude to the principal results (Appendix F.4.3). The conclusions also remain substantively unchanged when task completion is defined using only manually coded compliance (Appendix F.4.4), word-count thresholds at the 25th, 50th, or 75th percentiles, or measures combining manual coding and word count (Appendix F.4.2).

As a final test, I apply the product-of-coefficients approach described in *Estimation Strategy*. This approach combines the estimated effect of automation exposure on cultural

[21] Respondents are classified as exhibiting a high mediator level when there is at least a 70% probability—equivalent to $t = 1.036$—that their observed cultural-grievance level under encouragement exceeds the expected level among non-encouraged respondents with comparable education, gender, and race. Task completion is determined through manual coding of whether the response fulfills the assigned writing prompt. Appendix F.4 provides additional details and results using alternative thresholds.

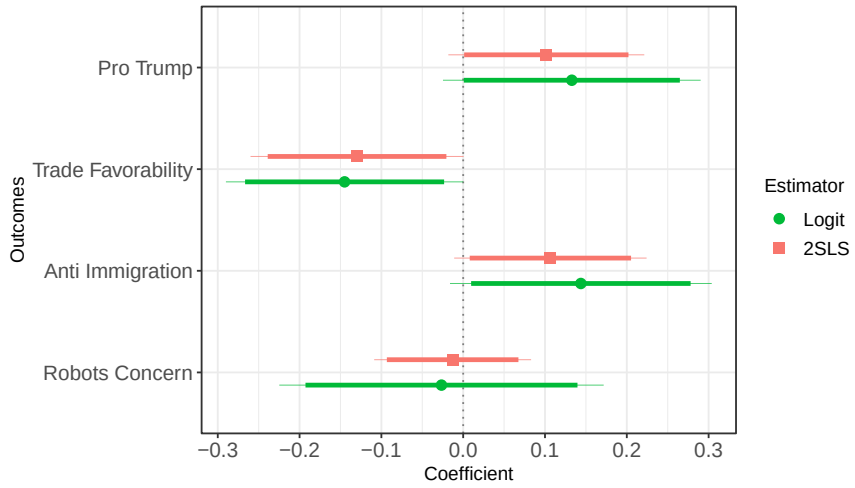


Figure 5: Mediator Effects on Outcomes: ITT and 2SLS Estimates

Note: Green circles report raw logit coefficients for assignment to the cultural-grievance encouragement. These coefficients are measured in log odds. Red squares report two-stage least squares estimates of the complier average causal effect, using random assignment to the encouragement as an instrument for compliance. All models use the same sample of 3,133 respondents, of whom 1,816 were assigned to encouragement and 1,317 were not, and adjust for gender, race, income, education, occupational RTI, and treatment assignment. Outcomes are binary indicators coded 0 or 1. "Pro Trump" measures willingness to vote for Donald Trump; "Trade Favorability" measures whether respondents view trade as beneficial to U.S. workers; "Anti Immigration" captures preferences for reducing legal immigration; and "Robots Concern" measures perceived risks from robots and artificial intelligence. Thick and thin bars report 90% and 95% confidence intervals, respectively. Full 2SLS models are reported in Table F.18. The supplementary material provides the exact question wording.

grievances (Figure 2) with the effect of the encouraged mediator on political attitudes (Figure 5). It thereby estimates the portion of the treatment effect transmitted through cultural grievances. I calculate uncertainty using the Delta Method (described in Appendix F.5). The resulting mediated-effect estimates imply that the cultural-grievance pathway increases support for Trump by approximately 3.7 percentage points, increases support for immigration restriction by 3.9 percentage points, and reduces trade favorability by 4.7 percentage points (Figure F.24). The corresponding 90% confidence intervals exclude zero. For the policy outcomes, the estimated mediated effect represents approximately 35% of the total change in anti-immigration attitudes and 26% of the total decline in trade favorability.[22] These results provide additional support for **Hypothesis 3**: perceived threats from automation affect political attitudes in part by activating cultural grievances.

Taken together, the model- and design-based analyses provide convergent evidence for the same mechanism. Exposure to job-threatening technological change activates nostalgia and perceived marginalization, and these grievances, in turn, increase support for Trump and exclusionary positions on trade and immigration. The two grievances may also be sequentially related: an exploratory analysis in Appendix E.3 is consistent with a pathway from marginalization to nostalgia, although the design does not permit a test of their temporal ordering.

[22]Appendix F.5 shows that these results are robust to alternative definitions of compliance.

STUDY 2: OBSERVATIONAL CROSS-SECTIONAL EVIDENCE

The experiment identifies the immediate attitudinal effects of brief exposure to information about automation, but it cannot establish whether these effects extend to sustained exposure to automation risk in real labor markets or translate into electoral behavior. To address these external-validity limitations, I complement the experimental findings with observational evidence from waves 1–7 (2002–2016) of the European Social Survey (ESS), covering thirteen West European countries.[23] I examine whether occupational exposure to automation risk is associated with attitudinal proxies conceptually related to the cultural grievances examined in the experiment and with reported vote choice for right-wing populist parties. Using waves 6 and 7, I then assess whether these proxy measures statistically mediate the association between automation risk and populist-right support. Finally, I replicate the analysis using the latest available wave (Wave 11), conducted after the release of GPT-3.5.

The ESS offers comprehensive occupational information through the International Standard Classification, which allows me to gauge individual-level risk exposure to automation. This approach assumes that workers' occupations and tasks shape their exposure to automation risk (e.g., Autor, 2013*a*). In particular, the independent variable is an occupation's probability of computerization, developed by Frey and Osborne (2017) using a Gaussian process classifier. The authors argue that "computerization is now spreading to domains commonly defined as non-routine" (p.258), and their measure has the uniqueness of providing an estimate of what recent technological change is likely to mean for the future of employment. This measure ranges from 0 to 1, with 0 representing no probability of computerization (e.g., recreational therapists), and 1 representing a fully computerizable task (e.g., telemarketers). As an alternative, I use the individual-level measure developed by Anelli, Colantone, and Stanig (2021), which combines individual characteristics with occupational risk predating the recent wave of production automation. I also use occupational suitability for machine learning (SML), a closer proxy for occupational exposure to AI (Brynjolfsson, Mitchell, and Rock, 2019).

[23]Austria, Belgium, Finland, France, Germany, Italy, the Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, and the United Kingdom.

Turning to the mediators, because the ESS items do not exactly match those in my experiment, I use conceptually related measures drawn from prior research. I proxy perceived status threat with three items assessing the cultural, economic, and societal consequences of immigration (e.g., Hays, Lim, and Spoon, 2019; Carreras, Irepoglu Carreras, and Bowler, 2019). Each ranges from 0 (negative) to 10 (positive). I proxy nostalgia with items measuring hope for the world's future and perceptions that life in the country is worsening, available in 2006 and 2012 and coded from 1 (strongly agree) to 5 (strongly disagree).[24]

The literature on political behavior discusses several other factors that may affect individuals' vote choices. Thus, I include individual-level controls for age, sex, years of education, location, being an ethnic minority, and employment characteristics (e.g. Frey, Berger, and Chen, 2017; Gingrich, 2019; Thewissen and Rueda, 2019). The model also includes changes in the stock of robots, unemployment rates, and immigrant exposure at the regional level.

I use model-based mediation analysis to examine the role of cultural grievances. Identification rests on the demanding sequential ignorability assumption (Imai et al., 2011; Keele, Tingley, and Yamamoto, 2015). This assumption requires the treatment (automation) to be independent of both the outcome and the mediator, given pretreatment covariates such as gender, and the mediator (cultural grievances) to be independent of outcomes, conditional on the treatment and covariates. I adjust for gender, age, education, NUTS-2 region, and country-year factors, and apply the sensitivity analysis of Imai et al. (2011) to assess robustness to unobserved mediator–outcome confounding. Because automation exposure and the proxy mediators are not randomly assigned in the ESS, I interpret these estimates as observational evidence consistent with the proposed pathway, not as causal mediation effects.

Observational Evidence Consistent with the Proposed Mechanism

I begin by examining the associations between automation risk, cultural grievances, and populist-right support (Table 1). Higher automation risk is associated with greater nostalgia, proxied by diminished optimism about the future (Columns 1–2); greater per-

[24]These items more directly capture collective pessimism than nostalgia, but prior research has used them for this purpose (e.g., Steenvoorden and Harteveld, 2018).

ceived marginalization, proxied by lower tolerance toward immigrants (Columns 3–5); and populist-right support (Column 6). These findings **support Hypotheses 1 and 2** regarding nostalgia, perceived marginalization, and populist-right support.[25]

	Nostalgia (Hyp. I)		Marginalization (Hyp. I)			Political Behavior (Hyp. II)
	(1)	(2)	(3)	(4)	(5)	(6)
	Life Better	Hopeful	Culture	Economy	Live	Radical Right
Frey & Osborne	-0.717***	-0.742***	-2.355***	-2.301***	-1.964***	3.560***
	(0.052)	(0.059)	(0.099)	(0.093)	(0.093)	(0.234)
Country-Year FE	✓	✓	✓	✓	✓	✓
NUTS FE	✓	✓	✓	✓	✓	✓
Observations	44326	44571	150245	149680	150516	63136
R^2 (p)	0.294	0.134	0.166	0.120	0.143	0.178
AIC	1.1e+05	1.2e+05	6.7e+05	6.6e+05	6.4e+05	2.6e+04

Standard errors clustered by region-year in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 1: Effects of Occupational Computerization Risk on Cultural Grievances and Radical-Right Support.

Note: This table reports the association between occupational exposure to computerization risk, measured using the probabilities developed by Frey and Osborne (2017), and the three sets of outcomes corresponding to the study's hypotheses. Columns 1–2 report the nostalgia outcomes: whether respondents believe that life is getting better and whether they feel hopeful about the future. These outcomes range from 1 to 5, with higher values indicating greater optimism. Columns 3–5 report the marginalization outcomes: evaluations of immigration's effects on the country's culture, economy, and overall quality of life. These outcomes range from 0 to 10, with higher values indicating more favorable attitudes toward immigration. Column 6 reports political behavior, measured as a binary indicator of voting for a radical-right party. Columns 1–5 report OLS coefficients, whereas Column 6 reports logit coefficients. All models use ESS survey weights and include demographic controls, country-year fixed effects, and NUTS-region fixed effects. Standard errors are clustered by region-year. Data are from ESS rounds 1–7. Full estimates are reported in Table G.19.

I next report the model-based mediation estimates. As shown in Figure 6, both indirect effects are in the predicted direction and statistically distinguishable from zero, supporting **Hypothesis 3**. Automation increases support for the populist right by about 3.5 percentage points through anti-immigration attitudes and by 1 percentage point through nostalgia. To contextualize these findings, the total impact of automation on support for the radical right is approximately 12.5 pp when comparing a probability shift from 0 to 1. Of this impact, 30 percent is mediated by anti-immigrant attitudes, influencing support for the populist right, while nostalgia accounts for 13 percent.

As robustness checks, I vary the pretreatment covariates and measure automation exposure using both the probability of computerization from Frey and Osborne (2017) and individual exposure from Anelli, Colantone, and Stanig (2021). The second-stage estimates in Appendix Tables G.30–G.32 remain substantively unchanged. I also assess sensitivity to violations of sequential ignorability. Appendix Table G.29 reports the residual correlation between the mediator and outcome equations required to reduce the ACME to zero: approximately 0.4 for marginalization and 0.1 for nostalgia. The marginalization-mediated estimate is therefore comparatively less sensitive; the nostalgia estimate is more fragile: modest unobserved mediator–outcome confounding could eliminate it. The

[25] Appendix G.2 reports additional robustness checks.

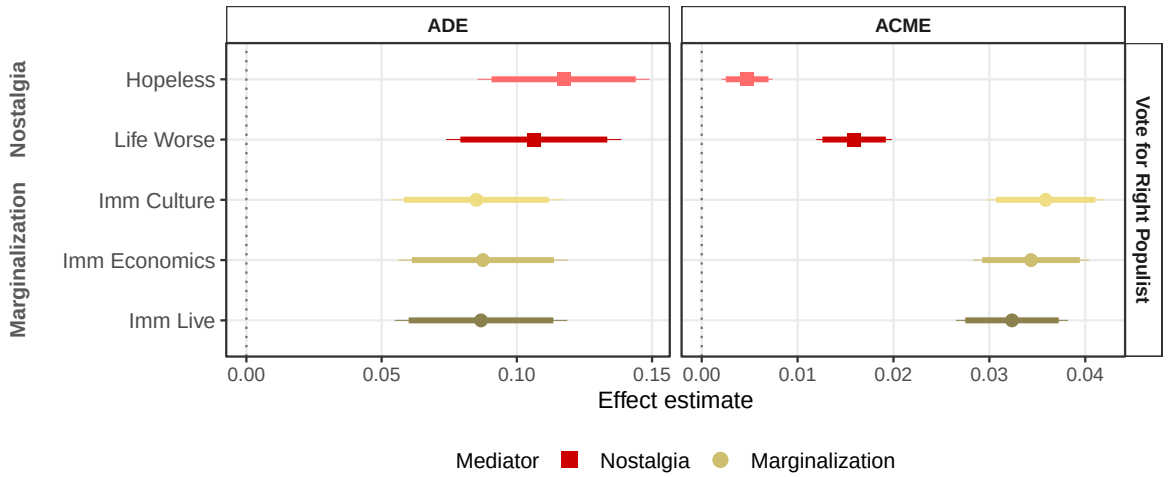


Figure 6: Mediated and direct effects of automation through cultural beliefs on support for the radical right.

Note: This figure reports the average direct effect (ADE, left) and average causal mediation effect (ACME, right) of automation exposure (following Frey and Osborne, 2017) on support for right-wing populism. Red squares identify nostalgia mediators; yellow circles identify perceived-marginalization mediators. Thick and thin lines show simulation-based 90% and 95% confidence intervals, respectively, calculated using a robust variance-covariance matrix. Immigration responses range from 0 (disagree) to 10 (agree), and future-sentiment responses from 1 (strongly agree) to 5 (strongly disagree). Sample sizes are: Culture ($N = 28{,}690$), Economy ($N = 28{,}576$), Live ($N = 28{,}638$), Life Worse ($N = 14{,}531$), and Hopeless ($N = 14{,}496$). Data come from ESS rounds 6–7. Full results appear in Table G.27 and Table G.28.

analysis does not establish whether such confounding is present, and sequential ignorability cannot be verified in observational data. For context, adding education—one of the strongest observed predictors—increases the mediator equation's R^2 by approximately 0.03. Although this increment is not directly comparable to the residual-correlation parameter, it may provide a useful descriptive benchmark for the additional explanatory power of a single observed covariate.

Finally, I assess whether the findings extend to a more recent, AI-salient period. I conduct a replication using ESS Round 11 (2023), the first round conducted after ChatGPT's public release. Because this round does not include the nostalgia items, the analysis is limited to the available immigration questions that proxy perceived marginalization. The results are substantively similar: automation exposure is associated with greater radical-right support, and perceived marginalization mediates a substantial share of this association. Full results appear in Table G.33 and Figure G.27.

Conclusion

How does automation risk reshape political demand? Combining experimental and observational evidence, this article shows how automation threat can increase feelings of marginalization and nostalgia, which in turn fuel support for exclusionary and restorative politics and the politicians who promote them—most commonly right-wing populists. The

experimental evidence traces both links in this chain: exposure to automation risk activates cultural grievances, while experimentally encouraged grievances shift political attitudes toward exclusionary policies and Trump support. Observational analyses across European countries reveal a consistent pattern, linking occupational automation risk to cultural grievances and support for the populist right. These findings complement ethnographic accounts of threatened identity and rightward realignment (Cramer, 2016; Newman and Skocpol, 2023) and are consistent with observational studies linking cultural grievances to populist support (Hays, Lim, and Spoon, 2019; Ballard-Rosa, Jensen, and Scheve, 2022). Methodologically, the study pairs mediator encouragement with a task-based measure of compliance, offering a strategy for studying causal mechanisms in survey experiments.

The results bridge economic and cultural accounts of populism by showing how an economic threat can acquire political force through concerns about status, recognition, and continuity. Consistent with research linking economic shocks to negative emotions and democratic discontent (Rhodes-Purdy, Navarre, and Utych, 2023), this demand-side pathway also has a supply-side implication. When mainstream parties leave affected groups underrepresented (Roberts, 2024) or social-democratic parties fail to offer credible responses to globalization (Berman, 2021), populist actors can mobilize these grievances by combining protectionism with restorative appeals. Social insurance alone may therefore leave losses of recognition and political voice unaddressed, preserving opportunities for exclusionary mobilization.

The evidence presented here, however, comes from postindustrial democracies in which established populist-right actors can channel automation-related nostalgia and feelings of marginalization. Whether this mechanism travels across political and economic contexts remains an open question. Where party supply differs, automation-related discontent may instead center on elite betrayal, foreign dependence, or blocked mobility and favor left-populist or anti-incumbent appeals. Responses may also vary with countries' positions in the global economy: AI could disrupt offshored service work in the Global South while altering demand for migrant labor and the salience of immigration in the Global North. A natural next step is to examine how party systems, labor-market institutions, and global economic integration shape these grievances and their political expression.

The design identifies a cultural pathway linking automation threat to political demand but does not test a parallel material pathway operating through perceived employment or income insecurity. Future research should examine whether these mechanisms operate independently or reinforce one another. These cultural responses are also unlikely to be uniform: pre-existing grievances and social position may condition both whether automation threat activates marginalization and nostalgia and how those grievances are expressed politically. The exploratory differences by race and education reported here point to potentially important boundary conditions that warrant further investigation.

Finally, AI may reshape politics through gains as well as losses, even when it does not eliminate jobs. AI-driven job creation and the automation of undesirable tasks may reduce marginalization and nostalgia, raising the question of whether perceived gains reduce these grievances as much as anticipated displacement heightens them. Yet these gains may coexist with new forms of control: augmentation may increase productivity while eroding autonomy, generating an "automation trap" (Jiang et al., 2025), whereas algorithmic surveillance and electronic monitoring may expand employer control (De Stefano and Taes, 2023; König, 2025). Examining how workers' everyday exposure to different applications of AI shapes political attitudes, alongside how workplace institutions govern these changes and distribute their benefits (Gingrich, Wu, and Zhang, 2026), would clarify how struggles over control shape the emerging politics of AI. The political consequences of AI will therefore depend not only on whether jobs disappear, but also on how technological change redistributes status, autonomy, and voice.

References

Acemoglu, Daron, and Pascual Restrepo. 2022. "Tasks, Automation, and the Rise in US Wage Inequality." *Econometrica* p. 75.

Agnolin, Paolo, Italo Colantone, and Piero Stanig. 2024. "In Search of the Causes of the Globalization Backlash, Between Methods and Evidence." *Comparative Political Studies* .

Agnolin, Paolo, Massimo Anelli, Italo Colantone, and Piero Stanig. 2024. "Robots Replacing Trade Unions: Novel Data and Evidence from Western Europe.".

Algan, Yann, Sergei Guriev, Elias Papaioannou, and Evgenia Passari. 2017. “The European Trust Crisis and the Rise of Populism.” *Brookings Papers on Economic Activity* 2017 (2): 309–400.

Anelli, Massimo, Italo Colantone, and Piero Stanig. 2021. “Individual vulnerability to industrial robot adoption increases support for the radical right.” *Proceedings of the National Academy of Sciences* 118 (47).

Angrist, Joshua D., Guido W. Imbens, and Donald B. Rubin. 1996. “Identification of Causal Effects Using Instrumental Variables.” *Journal of the American Statistical Association* 91 (434): 444–455.

Autor, David. 2013*a*. The” task approach” to labor markets: an overview. Technical report National Bureau of Economic Research.

Autor, David, David Dorn, Gordon Hanson, and Kaveh Majlesi. 2020. “Importing political polarization? The electoral consequences of rising trade exposure.”.

Autor, David H. 2013*b*. “The “task approach” to labor markets: an overview.” *Journal for Labour Market Research* 46 (3): 185–199.

Baccini, Leonardo, and Stephen Weymouth. 2021. “Gone For Good: Deindustrialization, White Voter Backlash, and US Presidential Voting.” *American Political Science Review* 115 (2): 550–567.

Baccini, Leonardo, and Thomas Sattler. 2021. Austerity, Economic Vulnerability, and Populism. SSRN Scholarly Paper ID 3766022 Social Science Research Network Rochester, NY: .

Balcazar, Carlos Felipe. 2024. “A theory of protectionist populism: The role of elite cues and identity on protectionism.”.

Baldwin, Matthew, and Joris Lammers. 2016. “Past-focused environmental comparisons promote proenvironmental outcomes for conservatives.” *Proceedings of the National Academy of Sciences* 113 (52): 14953–14957.

Balke, Alexander, and Judea Pearl. 1997. “Bounds on Treatment Effects from Studies with Imperfect Compliance.” *Journal of the American Statistical Association* 92 (439): 1171–1176.

Ballard-Rosa, Cameron, Amalie Jensen, and Kenneth Scheve. 2022. “Economic Decline, Social Identity, and Authoritarian Values in the United States.” *International Studies Quarterly* 66 (1): sqab027.

Ballard-Rosa, Cameron, Kenneth Scheve, and Amelie Jensen. 2021. “Economic Decline, Social Identity, and Authoritarian Values in the United States.”.

Ballard-Rosa, Cameron, Mashail A. Malik, Stephanie J. Rickard, and Kenneth Scheve. 2021. “The Economic Origins of Authoritarian Values: Evidence From Local Trade Shocks in the United Kingdom.” *Comparative Political Studies* 54 (13): 2321–2353.

Barauskaitė, Dovilė, Justina Gineikienė, and Bob M. Fennis. 2022. “Saved by the past? Disease threat triggers nostalgic consumption.” *Psychology & Marketing* 39 (8): 1433–1450.

Baron, Reuben M., and David A. Kenny. 1986. “The moderator–mediator variable distinction in social psychological research: Conceptual, strategic, and statistical considerations.” *Journal of Personality and Social Psychology* 51 (6): 1173–1182.

Benjamini, Yoav, and Yosef Hochberg. 1995. “Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing.” *Journal of the Royal Statistical Society Series B: Statistical Methodology* 57 (1): 289–300.

Berman, Sheri. 2021. “The Causes of Populism in the West.” *Annual Review of Political Science* 24 (Volume 24, 2021): 71–88. Publisher: Annual Reviews.

Bhattacharya, Oindrila. 2020. “The nostalgic process and experience : an examination of the cognitive and experiential nature of the nostalgic process at state and trait levels.”.

Bisbee, James, Layna Mosley, Thomas B. Pepinsky, and B. Peter Rosendorff. 2020. “Decompensating domestically: the political economy of anti-

globalism." *Journal of European Public Policy* 27 (7): 1090–1102. _eprint: https://doi.org/10.1080/13501763.2019.1678662.

Blackwell, Matthew, Jacob R. Brown, Sophie Hill, Kosuke Imai, and Teppei Yamamoto. 2023. "Priming bias versus post-treatment bias in experimental designs.".

Boix, Carles. 2003. *Democracy and Redistribution*. Cambridge University Press.

Boix, Carles, Michael Becher, Valentina Gonzalez-Rostani, and Daniel Stegmueller. 2026. "AI's Economy and Its Political and Institutional Consequences." *Cambridge University Press, APSA Taskforce - APSA Preprints* .

Brady, Henry E., and David Collier. 2010. *Rethinking Social Inquiry: Diverse Tools, Shared Standards*. Rowman & Littlefield Publishers.

Broz, J. Lawrence, Jeffry Frieden, and Stephen Weymouth. 2021. "Populism in Place: The Economic Geography of the Globalization Backlash." *International Organization* 75 (2): 464–494.

Brynjolfsson, Erik, Tom Mitchell, and Daniel Rock. 2019. "Machine learning and occupational change." *Unpublished manuscript, MIT* .

Bullock, John G., and Donald P. Green. 2021. "The Failings of Conventional Mediation Analysis and a Design-Based Alternative." *Advances in Methods and Practices in Psychological Science* 4 (4): 25152459211047227.

Bullock, John G., Donald P. Green, and Shang E. Ha. 2010. "Yes, but what's the mechanism?(don't expect an easy answer)." *Journal of personality and social psychology* 98 (4): 550.

Busby, Ethan C., Joshua R. Gubler, and Kirk A. Hawkins. 2019. "Framing and Blame Attribution in Populist Rhetoric." *The Journal of Politics* 81 (2): 616–630.

Busemeyer, Marius R., and Alexander H. J. Sahm. 2022. "Social Investment, Redistribution or Basic Income? Exploring the Association Between Automation Risk and Welfare State Attitudes in Europe." *Journal of Social Policy* 51 (4): 751–770.

Carreras, Miguel, Yasemin Irepoglu Carreras, and Shaun Bowler. 2019. "Long-Term Economic Distress, Cultural Backlash, and Support for Brexit." *Comparative Political Studies* 52 (9): 1396–1424.

Chaudoin, Stephen, and Michael-David Mangini. 2025. "Robots, Foreigners, and Foreign Robots: Policy Responses to Automation and Trade." *The Journal of Politics* p. 737849.

Choi, Jiwon, Ilyana Kuziemko, Ebonya Washington, and Gavin Wright. 2024. "Local economic and political effects of trade deals: Evidence from NAFTA." *American Economic Review* 114 (6): 1540–1575.

CloudResearch. 2024. "TurkPrime MTurk Toolkit by CloudResearch." *CloudResearch https://www.cloudresearch.com/products/turkprime-mturk-toolkit/* .

Colantone, Italo, and Piero Stanig. 2018. "The trade origins of economic nationalism: Import competition and voting behavior in Western Europe." *American Journal of Political Science* 62 (4): 936–953.

Cramer, Katherine J. 2016. *The politics of resentment: Rural consciousness in Wisconsin and the rise of Scott Walker*. University of Chicago Press.

Dai, Yaoyao, and Alexander Kustov. 2023. "The (in)effectiveness of populist rhetoric: a conjoint experiment of campaign messaging." *Political Science Research and Methods* pp. 1–8.

De Stefano, Valerio, and Simon Taes. 2023. "Algorithmic management and collective bargaining." *Transfer: European Review of Labour and Research* 29 (1): 21–36.

DeBell, Matthew, and Jon A. Krosnick. 2009. "Computing weights for American national election study survey data." *nes012427. Ann Arbor, MI, Palo Alto, CA: ANES Technical Report Series* .

Di Tella, Rafael, and Dani Rodrik. 2020. "Labour Market Shocks and the Demand for Trade Protection: Evidence from Online Surveys." *The Economic Journal* 130 (628): 1008–1030.

Egami, Naoki, and Erin Hartman. 2023. “Elements of external validity: Framework, design, and analysis.” *American Political Science Review* 117 (3): 1070–1088.

Elchardus, Mark, and Bram Spruyt. 2016. “Populism, Persistent Republicanism and Declinism: An Empirical Analysis of Populism as a Thin Ideology.” *Government and Opposition* 51 (1): 111–133.

Eloundou, Tyna, Sam Manning, Pamela Mishkin, and Daniel Rock. 2024. “GPTs are GPTs: Labor market impact potential of LLMs.” *Science* 384 (6702): 1306–1308.

Ferwerda, Jeremy, Justin Gest, and Tyler Reny. 2025. “Nostalgic deprivation and populism: Evidence from 19 European countries.” *European Journal of Political Research* 64 (3): 1506–1518.

Frey, Carl Benedikt, and Michael A. Osborne. 2017. “The future of employment: How susceptible are jobs to computerisation?” *Technological Forecasting and Social Change* 114: 254–280.

Frey, Carl Benedikt, Thor Berger, and Chinchih Chen. 2017. “Political machinery: Automation anxiety and the 2016 US presidential election.” *University of Oxford* .

Gallego, Aina, and Thomas Kurer. 2022. “Automation, Digitalization, and Artificial Intelligence in the Workplace: Implications for Political Behavior.” *Annual Review of Political Science* 25 (1): 463–484.

Gest, Justin. 2016. *The new minority: White working class politics in an age of immigration and inequality*. Oxford University Press.

Gest, Justin, Tyler Reny, and Jeremy Mayer. 2018. “Roots of the Radical Right: Nostalgic Deprivation in the United States and Britain.” *Comparative Political Studies* 51 (13): 1694–1719.

Gidron, Noam, and Peter A. Hall. 2017. “The politics of social status: economic and cultural roots of the populist right.” *The British Journal of Sociology* 68 (S1): S57–S84.

Gidron, Noam, and Peter A. Hall. 2020. “Populism as a problem of social integration.” *Comparative Political Studies* 53 (7): 1027–1059.

Gingrich, Jane. 2019. “Did state responses to automation matter for voters?” *Research & Politics* 6 (1): 2053168019832745.

Gingrich, Jane, Nicole Wu, and Baobao Zhang. 2026. “Workplace governance and labor perceptions of technological risks and benefits.” *Industrial and Corporate Change* 35 (1): 222–242.

Glynn, Adam N., and Kevin M. Quinn. 2010. “An introduction to the augmented inverse propensity weighted estimator.” *Political analysis* 18 (1): 36–56.

Goldberg, Matthew, Sander van der Linden, Matthew Ballew, Seth Rosenthal, and Anthony Leiserowitz. 2019. “Convenient but biased? The reliability of convenience samples in research about attitudes toward climate change.”.

Goldsmith-Pinkham, Paul, Peter Hull, and Michal Kolesár. 2024. “Contamination Bias in Linear Regressions.” *American Economic Review* 114 (12): 4015–4051.

Gonzalez-Rostani, Valentina. 2024. “Engaged robots, disengaged workers: Automation and political alienation.” *Economics & Politics* 36 (3): 1703–1730. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/ecpo.12307.

Gonzalez-Rostani, Valentina. 2026. “Elections, Right-Wing Populism, and Political-Economic Polarization: The Role of Institutions and Political Outsiders.” *The Journal of Politics* 88 (1): 125–144.

Gonzalez-Rostani, Valentina, and Jude C Hays. 2026. “Love of Variety? An Experimental Study of Heterogeneous Responses to Foreign Brands in the Marketplace.” *Political Science Research and Methods* .

Graham, Jesse, Jonathan Haidt, and Brian A. Nosek. 2009. “Liberals and conservatives rely on different sets of moral foundations.” *Journal of personality and social psychology* 96 (5): 1029.

Granulo, Armin, Christoph Fuchs, and Stefano Puntoni. 2019. “Psychological reactions to human versus robotic job replacement.” *Nature Human Behaviour* 3 (10): 1062–1069.

Grosche, Carla, and Tobias Rothmund. 2026. "Yesterday, all our troubles seemed so far away—(Re)conceptualizing nostalgic deprivation as a predictor for radical-right support." *Political Psychology* 47 (2): e70057. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/pops.70057.

Guisinger, Alexandra. 2017. *American opinion on trade : preferences without politics*. Oxford University Press.

Guiso, Luigi, Helios Herrera, Massimo Morelli, and Tommaso Sonno. 2017. "Demand and supply of populism.".

Hainmueller, Jens, and Michael J. Hiscox. 2007. "Educated Preferences: Explaining Attitudes Toward Immigration in Europe." *International Organization* 61 (2): 399–442.

Hainmueller, Jens, and Michael J. Hiscox. 2010. "Attitudes toward Highly Skilled and Low-skilled Immigration: Evidence from a Survey Experiment." *American Political Science Review* 104 (1): 61–84.

Hawkins, Kirk A., Ryan E. Carlin, Levente Littvay, and Cristóbal Rovira Kaltwasser. 2018. *The Ideational Approach to Populism: Concept, Theory, and Analysis*. Routledge. Google-Books-ID: hG5uDwAAQBAJ.

Hays, Jude, Junghyun Lim, and Jae-Jae Spoon. 2019. "The path from trade to right-wing populism in Europe." *Electoral Studies* 60: 102038.

Hicks, Raymond, and Dustin Tingley. 2011. "Causal Mediation Analysis." *The Stata Journal* 11 (4): 605–619.

Hirschman, Albert O., and Michael Rothschild. 1973. "The Changing Tolerance for Income Inequality in the Course of Economic Development." *The Quarterly Journal of Economics* 87 (4): 544–566.

Hopkins, Daniel J., Yotam Margalit, and Omer Solodoch. 2024. "Personal Economic Shocks and Public Opposition to Unauthorized Immigration." *British Journal of Political Science* 54 (3): 928–936.

Hout, Michael. 2008. “Objective and Subjective Aspects of Class Since the 1970s.” *Social class: How does it work?* p. 25.

Im, Zhen Jie, Nonna Mayer, Bruno Palier, and Jan Rovny. 2019. “The “losers of automation”: A reservoir of votes for the radical right?” *Research & Politics* 6 (1).

Imai, Kosuke, Dustin Tingley, and Teppei Yamamoto. 2013. “Experimental designs for identifying causal mechanisms.” *Journal of the Royal Statistical Society: Series A (Statistics in Society)* 176 (1): 5–51.

Imai, Kosuke, Luke Keele, and Dustin Tingley. 2010. “A general approach to causal mediation analysis.” *Psychological Methods* 15 (4): 309–334.

Imai, Kosuke, Luke Keele, Dustin Tingley, and Teppei Yamamoto. 2011. “Unpacking the Black Box of Causality: Learning about Causal Mechanisms from Experimental and Observational Studies.” *American Political Science Review* 105 (4): 765–789.

Imai, Kosuke, and Teppei Yamamoto. 2010. “Causal Inference with Differential Measurement Error: Nonparametric Identification and Sensitivity Analysis.” *American Journal of Political Science* 54 (2): 543–560.

Inglehart, Ronald. 2018. *Cultural Evolution: People’s Motivations are Changing, and Reshaping the World*. Cambridge University Press.

Jiang, Wei, Junyoung Park, Rachel Jiqiu Xiao, and Shen Zhang. 2025. AI and the extended workday: Productivity, contracting efficiency, and distribution of rents. Technical report National Bureau of Economic Research.

Jones, Daniel B., Erica Owen, and Rena Sung. 2026. “Relative Exposure to Negative Economic Shocks, Racial Animus, and Voting.” *International Organization* pp. 1–32.

Jost, John T. 2017. “Ideological Asymmetries and the Essence of Political Psychology.” *Political Psychology* 38 (2): 167–208. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/pops.12407.

Jost, John T., Jack Glaser, Arie W. Kruglanski, and Frank J. Sulloway. 2003. “Political conservatism as motivated social cognition.” *Psychological bulletin* 129 (3): 339.

Jost, John T., Mahzarin R. Banaji, and Brian A. Nosek. 2004. "A Decade of System Justification Theory: Accumulated Evidence of Conscious and Unconscious Bolstering of the Status Quo." *Political Psychology* 25 (6): 881–919. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1467-9221.2004.00402.x.

Kaltwasser, Cristóbal Rovira, Paul A. Taggart, Paulina Ochoa Espejo, and Pierre Ostiguy. 2017. *The Oxford Handbook of Populism*. Oxford University Press. Google-Books-ID: GcQ9DwAAQBAJ.

Karpowitz, Christopher F., Tali Mendelberg, and Lee Shaker. 2012. "Gender Inequality in Deliberative Participation." *The American Political Science Review* 106 (3): 533–547.

Keele, Luke, Dustin Tingley, and Teppei Yamamoto. 2015. "Identifying Mechanisms Behind Policy Interventions Via Causal Mediation Analysis." *Journal of Policy Analysis and Management* 34 (4): 937–963.

Knox, Dean, Will Lowe, and Jonathan Mummolo. 2020. "Administrative Records Mask Racially Biased Policing." *American Political Science Review* 114 (3): 619–637.

Kurer, Thomas. 2020. "The Declining Middle: Occupational Change, Social Status, and the Populist Right." *Comparative Political Studies* p. 0010414020912283.

Kurer, Thomas, and Silja Häusermann. 2022. "Automation risk, social policy preferences, and political participation." In *Digitalization and the welfare state*. Oxford University Press. Google-Books-ID: nt5eEAAAQBAJ.

König, Cornelius J. 2025. "Electronic Monitoring at Work.".

Lammers, Joris, and Pınar Uğurlar. 2024. "Political-Ideological Differences in Cultural Pessimism and Nostalgia Reflect People's Evaluation of Their Nation's Historical Developments." *Social Psychological and Personality Science* 15 (3): 370–380.

Lamont, Michèle. 2009. *The dignity of working men: Morality and the boundaries of race, class, and immigration*. Harvard University Press.

Lelkes, Yphtach, and Sean J. Westwood. 2017. "The Limits of Partisan Prejudice." *The Journal of Politics* 79 (2): 485–501.

Mansfield, Edward D., and Diana C. Mutz. 2013. “US versus Them: Mass Attitudes toward Offshore Outsourcing.” *World Politics* 65 (4): 571–608.

Mansfield, Edward D., Diana C. Mutz, and Devon Brackbill. 2019. “Effects of the Great Recession on American attitudes toward trade.” *British Journal of Political Science* 49 (1): 37–58.

Mansfield, Edward D., Helen V. Milner, and Nita Rudra. 2021. “The Globalization Backlash: Exploring New Perspectives.” *Comparative Political Studies* p. 00104140211024286.

Mayda, Anna Maria, and Dani Rodrik. 2005. “Why are some people (and countries) more protectionist than others?” *European Economic Review* 49 (6): 1393–1430.

Milner, Helen V. 2021. “Voting for Populism in Europe: Globalization, Technological Change, and the Extreme Right.” *Comparative Political Studies* 54 (13): 2286–2320.

Moore, Barrington. 1966. *Social Origins of Dictatorship and Democracy: Lord and Peasant in the Making of the Modern World*. Beacon Press.

Mudde, C. 2007. *Populist radical right parties in Europe*. Cambridge: Cambridge university press.

Mutz, Diana C. 2018. “Status threat, not economic hardship, explains the 2016 presidential vote.” *Proceedings of the National Academy of Sciences* 115 (19): E4330–E4339.

Mutz, Diana C. 2021. “(Mis)Attributing the Causes of American Job Loss: The Consequences of Getting It Wrong.” *Public Opinion Quarterly* 85 (1): 101–122.

Müller, Stefan, and Sven-Oliver Proksch. 2024. “Nostalgia in European party politics: a text-based measurement approach.” *British Journal of Political Science* 54 (3): 993–1005.

Neuner, Fabian G., and Christopher Wratil. 2022. “The Populist Marketplace: Unpacking the Role of “Thin” and “Thick” Ideology.” *Political Behavior* 44 (2): 551–574.

Newman, David B., Matthew E. Sachs, Arthur A. Stone, and Norbert Schwarz. 2020. “Nostalgia and well-being in daily life: An ecological validity perspective.” *Journal of Personality and Social Psychology* 118 (2): 325–347.

Newman, Lainey, and Theda Skocpol. 2023. *Rust Belt Union Blues: Why Working-Class Voters Are Turning Away from the Democratic Party*. Columbia University Press.

Norris, Pippa. 2004. *Electoral Engineering: Voting Rules and Political Behavior*. Cambridge University Press.

O'Rourke, Kevin H., Richard Sinnott, J. David Richardson, and Dani Rodrik. 2001. The determinants of individual trade policy preferences: International survey evidence [with comments and discussion]. In *Brookings trade forum*. pp. 157–206.

Owen, Erica, and Noel P. Johnston. 2017. "Occupation and the political economy of trade: job routineness, offshorability, and protectionist sentiment." *International Organization* 71 (4): 665–699.

Penn, Jerrod M., Daniel R. Petrolia, and J. Matthew Fannin. 2023. "Hypothetical bias mitigation in representative and convenience samples." *Applied Economic Perspectives and Policy* 45 (2): 721–743.

Pirlott, Angela G., and David P. MacKinnon. 2016. "Design approaches to experimental mediation." *Journal of Experimental Social Psychology* 66: 29–38.

Powers, Ryan, and Jonathan Renshon. 2023. "International Status Concerns and Domestic Support for Political Leaders." *American Journal of Political Science* 67 (3): 732–747.

Pratto, Felicia, Jim Sidanius, Lisa M. Stallworth, and Bertram F. Malle. 1994. "Social dominance orientation: A personality variable predicting social and political attitudes." *Journal of personality and social psychology* 67 (4): 741.

Rhodes-Purdy, Matthew, Rachel Navarre, and Stephen M. Utych. 2021*a*. "Measuring Simultaneous Emotions: Existing Problems and a New Way Forward." *Journal of Experimental Political Science* 8 (1): 1–14.

Rhodes-Purdy, Matthew, Rachel Navarre, and Stephen M. Utych. 2021*b*. "Populist Psychology: Economics, Culture, and Emotions." *The Journal of Politics* 83 (4): 1559–1572.

Rhodes-Purdy, Matthew, Rachel Navarre, and Stephen Utych. 2023. *The Age of Discontent: Populism, Extremism, and Conspiracy Theories in Contemporary Democracies*. Cambridge: Cambridge University Press.

Roberts, Kenneth M. 2024. "Structural and Institutional Sources of Populist Resiliency." In *Still the Age of Populism?* Routledge. Num Pages: 16.

Robins, James M., Andrea Rotnitzky, and Lue Ping Zhao. 1994. "Estimation of Regression Coefficients When Some Regressors are not Always Observed." *Journal of the American Statistical Association* 89 (427): 846–866.

Robinson, Michael D., Deirdre M. Cassidy, Ryan L. Boyd, and Adam K. Fetterman. 2015. "The politics of time: Conservatives differentially reference the past and liberals differentially reference the future." *Journal of Applied Social Psychology* 45 (7): 391–399.

Rodrik, Dani. 2021. "Why Does Globalization Fuel Populism? Economics, Culture, and the Rise of Right-Wing Populism." *Annual Review of Economics* 13 (1): 133–170.

Rogenhofer, Julius Maximilian, Koen Abts, Oliver Klein, and Paul Bertin. 2023. "The resentful undergrowth of nostalgia: Ontological insecurity, relative deprivation and powerlessness." *The British Journal of Sociology* 74 (2): 173–188. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/1468-4446.12991.

Rooduijn, Matthijs, and Teun Pauwels. 2011. "Measuring Populism: Comparing Two Methods of Content Analysis." *West European Politics* 34 (6): 1272–1283.

Routledge, Clay, Jamie Arndt, Constantine Sedikides, and Tim Wildschut. 2008. "A blast from the past: The terror management function of nostalgia." *Journal of Experimental Social Psychology* 44 (1): 132–140.

Scheve, Kenneth F., and Matthew J. Slaughter. 2001. "Labor Market Competition and Individual Preferences Over Immigration Policy." *The Review of Economics and Statistics* 83 (1): 133–145.

Schwartz, Shalom H., Jan Cieciuch, Michele Vecchione, Eldad Davidov, Ronald Fischer, Constanze Beierlein, Alice Ramos, Markku Verkasalo, Jan-Erik Lönnqvist, and Kursad Demirutku. 2012. "Refining the theory of basic individual values." *Journal of personality and social psychology* 103 (4): 663.

Smeekes, Anouk, Constantine Sedikides, and Tim Wildschut. 2023. "Collective nostalgia: Triggers and consequences for collective action intentions." *British Journal of Social Psychology* .

Smeekes, Anouk, Maykel Verkuyten, and Borja Martinovic. 2015. "Longing for the country's good old days: National nostalgia, autochthony beliefs, and opposition to Muslim expressive rights." *British Journal of Social Psychology* 54 (3): 561–580.

Sobel, Michael E. 1982. "Asymptotic confidence intervals for indirect effects in structural equation models." *Sociological methodology* 13: 290–312.

Spencer, Steven J., Mark P. Zanna, and Geoffrey T. Fong. 2005. "Establishing a causal chain: Why experiments are often more effective than mediational analyses in examining psychological processes." *Journal of Personality and Social Psychology* 89 (6): 845–851.

Stagnaro, Michael N., James N. Druckman, Adam J. Berinsky, Antonio A. Arechar, Robb Willer, and David G. Rand. 2026. "Representativeness and response validity across nine opt-in online samples." *Nature Human Behaviour* pp. 1–14.

Steenvoorden, Eefje, and Eelco Harteveld. 2018. "The appeal of nostalgia: the influence of societal pessimism on support for populist radical right parties." *West European Politics* 41 (1): 28–52.

Stefaniak, Anna, Michael Jeremy Adam Wohl, Constantine Sedikides, Anouk Smeekes, and Tim Wildschut. 2021. "Different Pasts for Different Political Folk: Political Orientation Predicts Collective Nostalgia Content." *Frontiers in Political Science* 3: 15.

Thewissen, Stefan, and David Rueda. 2019. "Automation and the Welfare State: Technological Change as a Determinant of Redistribution Preferences." *Comparative Political Studies* 52 (2): 171–208.

Tomz, Michael R., and Jessica L. P. Weeks. 2020. “Human Rights and Public Support for War.” *The Journal of Politics* 82 (1): 182–194.

van Tilburg, Wijnand A. P., Constantine Sedikides, and Tim Wildschut. 2015. “The mnemonic muse: Nostalgia fosters creativity through openness to experience.” *Journal of Experimental Social Psychology* 59: 1–7.

Wang, Yan, Yilan Huang, Kathy Phipps Di Xiong, Steve Y. Lee, Jie Shen, Abrey Daniel, Maritza Cabezas, James J. Crall, and Honghu Liu. 2020. “Enhancing sampling weights using raking method.” *JSM Proceedings, Survey Research Methods Section. American Statistical Association* 202: 136–143.

We-Must-Act-Now. 2026. “We Must Act Now. A Statement on AI’s Transformation of the Economy (https://www.wemustactnow.ai).”.

Webb, Michael. 2019. “The Impact of Artificial Intelligence on the Labor Market.”.

Wu, Nicole. 2022*a*. “Misattributed blame? Attitudes toward globalization in the age of automation.” *Political Science Research and Methods* 10 (3): 470–487.

Wu, Nicole. 2022*b*. ““Restrict foreigners, not robots”: Partisan responses to automation threat.” *Economics & Politics* .

Wu, Nicole. 2023. ““Restrict foreigners, not robots”: Partisan responses to automation threat.” *Economics & Politics* 35 (2): 505–528. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/ecpo.12225.

Xia, Lan, Joyce (Feng) Wang, and Shelle Santana. 2021. “Nostalgia: Triggers and its role on new product purchase intentions.” *Journal of Business Research* 135: 183–194.

Yam, Kai Chi, Pok Man Tang, Joshua Conrad Jackson, Runkun Su, and Kurt Gray. 2022. “The rise of robots increases job insecurity and maladaptive workplace behaviors: Multimethod evidence.” *Journal of Applied Psychology* .

Young, Lauren E. 2019. “The psychology of state repression: Fear and dissent decisions in Zimbabwe.” *American Political Science Review* 113 (1): 140–155.

Zhang, Baobao, Matto Mildenberger, Peter D. Howe, Jennifer Marlon, Seth A. Rosenthal, and Anthony Leiserowitz. 2020. "Quota sampling using Facebook advertisements." *Political Science Research and Methods* 8 (3): 558–564.

Zhou, Ling, Tao Wang, Qin Zhang, and Yupeng Mou. 2013. "Consumer insecurity and preference for nostalgic products: Evidence from China." *Journal of Business Research* 66 (12): 2406–2411.

Zhou, Xinyue, Constantine Sedikides, Tiantian Mo, Wanyue Li, Emily K. Hong, and Tim Wildschut. 2022. "The Restorative Power of Nostalgia: Thwarting Loneliness by Raising Happiness During the COVID-19 Pandemic." *Social Psychological and Personality Science* 13 (4): 803–815. Publisher: SAGE Publications Inc.

ONLINE APPENDIX

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A SURVEY

A.1 Pre-Registration

The paper was pre-registered in OSF. Reference eliminated for anonymity PAP attached as supplementary material in the submission.

A.2 Attrition

Qualtrics recorded 3,882 clicks, and after eligibility (age, citizenship, and employment), 3,362 satisfied the screening requirements and continued to the demographic portion of the survey. Among this post-screening sample, 3,354 answered the pretreatment political questions, and 3,299 received treatment assignment. Thus, 63 post-screening respondents left before their treatment assignment could be identified.

Following treatment assignment, 3,288 respondents answered the recommendation question immediately after the news treatment. Eleven respondents stopped before this question. Encouragement was randomized at the next stage, but another 25 respondents stopped before their encouragement condition became observable. Among the remaining 3,263 respondents, 971 entered the nostalgia condition, 931 entered the marginalization condition, and 1,361 received no encouragement.

Table A.1 reports attrition after being assigned to treatments or encouragement. The resulting analytic sample contains 3,133 respondents.

Group	Assigned	Completed	Did not complete	Attrition rate	Comparison	Fisher p-value
Panel A: Treatment assignment						
Control	1,248	1,182	66	5.3%	Reference	–
Robots	1,060	995	65	6.1%	Robots vs. Control	0.417
AI	991	956	35	3.5%	AI vs. Control	0.051
Total	3,299	3,133	166	5.0%	–	–
Panel B: Encouragement condition						
Nostalgia	971	927	44	4.5%	Nostalgia vs. No encouragement	0.122
Marginalization	931	889	42	4.5%	Marginalization vs. No encouragement	0.118
No encouragement	1,361	1,317	44	3.2%	Reference	–
Total	3,263	3,133	130	4.0%	–	–

Table A.1: Analytic Retention by Treatment and Encouragement Condition

Note: Completed denotes inclusion in the final analytic sample of 3,133 respondents. Panel A includes the 3,299 respondents whose treatment assignment is observable. Panel B includes the 3,263 respondents whose encouragement condition is observable; encouragement assignment cannot be recovered for the 36 respondents who stopped before reaching this stage. Group-specific p-values are from two-sided Fisher exact tests.

A.3 Descriptive statistics

In terms of representativeness, approximately 71% of participants in the sample are White, closely aligning with the 2016 voter demographics: according to the CCS, White voters constituted approximately 74% of the 2016 electorate. The sample's mean age is approximately 44, also mirroring the average age of 2016 voters. Moreover, the age and racial composition of respondents across political affiliations generally follows the patterns observed in the broader electorate. The sample is more highly educated than the general population: approximately 60.5% of respondents hold a bachelor's degree or higher, compared with 36.2% of U.S. adults age 25 or older in the 2023 American Community Survey. The share of respondents who are unemployed and seeking work is also higher—approximately 8.6%, compared with 3.4% of the U.S. civilian labor force age 20 or older in May 2023.[26] Overall, although the education differences should be considered when generalizing the results, the sample compares favorably with the broader voting population on several important demographic characteristics, particularly race and age.

	Mean	Median	S.D.	Min.	Max	Obs.
% Female	55.44	100.00	49.71	0	100	3133
% White	71.47	100.00	45.17	0	100	3133
Income	7.79	8.00	3.45	1	14	3133
% Unemployed	8.55	0.00	27.97	0	100	3133
% Bachelors	60.5	100	49.00	0	100	3133
Age	44.25	42.00	12.20	20	88	3133
Ideological spectrum	4.47	5.00	2.97	0	10	3133

Table A.2: Descriptive statistics pre-treatment variables.

Income is an ordinal scale from 1 (less than $10,000) to 14 (greater than $200,000), where 5 is $40,000–49,999. Ideological Spectrum is an ordinal scale from 1 (Very Liberal) to 10 (Very Conservative), where 5 is Moderate

[26] During the initial screening, I excluded working-age respondents who were neither employed nor seeking work (e.g., students and retirees), so this difference is likely reasonable.

	Mean	Median	S.D.	Min.	Max	Obs.
% Definitely support Trump 2024	13.25	0.00	33.90	0	100	3133
Anti-Elitism	4.07	4.00	1.00	1	5	3129
Nostalgia Index	10.08	10.00	3.11	3	15	3133
% High Nostalgia	34.92	0.00	47.68	0	100	3133
Marginalization Index	9.09	9.00	3.23	3	15	3133
% High Marginalization	36.00	0.00	48.01	0	100	3133
Anti Immigration	3.01	3.00	1.21	1	5	3133
Trade Favorability: American Workers	46.16	48.00	26.09	0	100	3133
% High Trade Favorability	46.89	0.00	49.91	0	100	3133
Concerns about Robots Index	14.77	15.00	4.21	5	25	3103

Table A.3: Descriptive statistics outcome variables.

Anti-Elitism is measured on an ordinal scale from 1 (strongly disagree with anti-elite statement) to 5 (strongly agree). The nostalgia index incorporates three statements assessing nostalgia, scored from 1 (low) to 5 (high), with higher numbers indicating greater levels of nostalgia. Similarly, the marginalization index consists of three statements evaluating marginalization, also scored from 1 (low) to 5 (high). Anti-immigration attitudes is assessed on an ordinal scale from 1 (strongly disagree with the anti-immigration statement) to 5 (strongly agree). Trade favorability is rated on a scale from 1 (trade is very bad) to 100 (trade is very good). The concerns about robots index encompasses five questions related to apprehensions about robots and artificial intelligence, rated from 1 (low) to 5 (very concerned).

A.4 Experimental Design and Procedure

This appendix section provides additional detail on the experimental design and implementation. The study was presented as an evaluation task for a news website. Respondents completed a sequence of steps involving (i) exposure to randomly assigned news content about technology, (ii) a recommendation task, (iii) a second randomization that encouraged (or not) cultural grievances through a short writing exercise, and (iv) post-treatment measures. Together, these features implement a double-randomized design intended to identify both the total effect of exposure to automation risk and the role of cultural grievances as a mediating mechanism.

A.4.1 Overview of the Double Randomization. Figure A.1 summarizes the design. The first randomization assigns respondents to a technology framing condition. The second randomization assigns respondents either to a cultural-grievances encouragement task (nostalgia or marginalization) or to no encouragement.

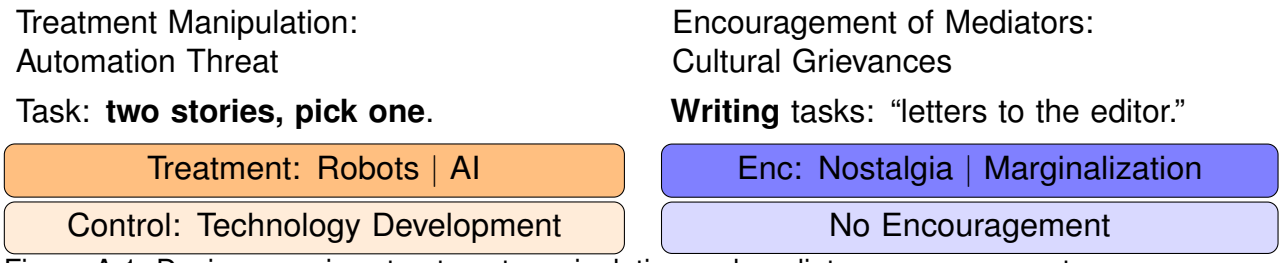


Figure A.1: Design overview: treatment manipulation and mediator encouragement.

A.4.2 Conceptual Structure: Treatment, Mediator, and Outcomes. Figure A.2 illustrates the causal structure the design is intended to probe. The first randomization identifies the total effect of exposure to automation risks on political behavior ($T \rightarrow Y$) as well as the effect on cultural grievances ($T \rightarrow M$). The second randomization encourages cultural grievances, allowing additional leverage on the mediator-to-outcome relationship ($M \rightarrow Y$).

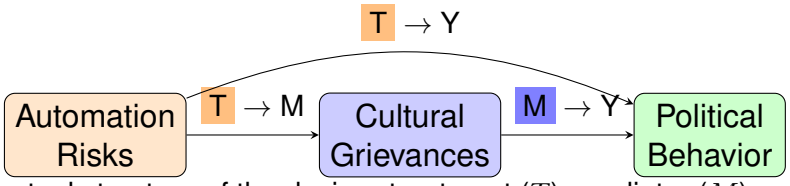


Figure A.2: Conceptual structure of the design: treatment (T), mediator (M), and outcomes (Y).

A.4.3 Step-by-Step Procedure.

1. **Pre-treatment measures.** Baseline questions were administered prior to any experimental stimuli.
2. **Treatment randomization (automation framing).** Respondents were assigned to one of the technology conditions described in the main text (e.g., automation risk vs. technology-neutral development).

3. **News task.** Respondents read two short articles and were asked to recommend one of them.
4. **Encouragement randomization (cultural grievances).** Respondents were assigned either to a writing exercise designed to activate nostalgia or marginalization, or to no writing encouragement.
5. **Post-treatment measures.** Respondents completed outcome measures and post-treatment items capturing the mediator components.

A.5 Covariate Balance

Table A.4 reports the distribution of pretreatment covariates across the three treatment arms and the three encouragement conditions. The group means and proportions are generally similar across assignments. Table A.5 reports Pearson chi-squared tests for five pairwise contrasts: the pooled treatment, robots, and AI conditions relative to the control group, and the marginalization and nostalgia encouragements relative to no encouragement.

Across the 50 comparisons reported in Table A.5, four are statistically significant at the $p < .05$ level. Taken together, these results are consistent with generally successful randomization, although they also reveal a few imbalances. The adjusted specifications therefore include pretreatment covariates—including race, income, education, sex, and occupational routine-task intensity—to improve precision and account for these observed differences.

Table A.4: Distribution of Pretreatment Covariates across Experimental Conditions

	Treatment assignment			Encouragement assignment		
Covariate	Control	Robots	AI	Not encouraged	Marginalization	Nostalgia
N	1182	995	956	1317	889	927
Female (%)	53.81	55.68	57.22	55.43	55.79	55.12
White (%)	73.18	68.94	71.97	70.84	72.44	71.41
Age 18–29 (%)	8.97	8.34	8.79	8.88	7.99	9.17
Age 30–44 (%)	50.51	48.54	47.59	50.11	49.04	47.36
Age 45–64 (%)	34.43	35.28	34.52	33.41	34.65	36.68
Age 65+ (%)	6.09	7.84	9.10	7.59	8.32	6.80
Income	7.85	7.71	7.80	7.75	7.87	7.77
Education	4.39	4.42	4.53	4.40	4.47	4.47
Bachelor's degree (%)	58.88	59.70	64.33	59.68	60.85	62.35
Unemployed (%)	8.38	6.63	10.77	9.26	8.10	7.98
Ideology	4.34	4.55	4.54	4.31	4.55	4.62
Democrat (%)	55.25	54.37	53.97	54.90	53.88	54.80
Higher RTI (%)	26.14	24.92	25.63	26.73	25.53	24.06

Note: Entries report group means or percentages, as indicated. The treatment-assignment columns compare respondents assigned to the control, robots, and AI conditions. The encouragement-assignment columns compare respondents assigned to no encouragement, the marginalization encouragement, and the nostalgia encouragement. Democrat is an indicator for respondents who identify with the Democratic Party. Higher RTI identifies respondents whose occupational routine-task intensity is at or above the sample's 75th percentile. The analysis includes respondents retained in the final analytic sample.

Table A.5: Pairwise Tests of Pretreatment Covariate Balance

	Treatment assignment			Encouragement assignment	
Covariate	Pooled vs. control	Robots vs. control	AI vs. control	Marg. vs. none	Nostalgia vs. none
Female	0.152	0.382	0.115	0.866	0.886
White	0.098	0.030*	0.531	0.415	0.769
Age group	0.089	0.366	0.061	0.748	0.385
Income	0.146	0.199	0.302	0.614	0.002*
Education	0.120	0.800	0.016*	0.545	0.601
Bachelor's degree	0.086	0.700	0.010*	0.581	0.202
Unemployed	0.781	0.126	0.059	0.344	0.290
Ideology	0.328	0.775	0.140	0.420	0.151
Democrat	0.561	0.683	0.557	0.638	0.964
Higher RTI	0.587	0.517	0.787	0.532	0.154
Significant at p $< .05$	0 / 10	1 / 10	2 / 10	0 / 10	1 / 10

Note: Entries are unadjusted p-values from Pearson chi-squared tests. The first three columns compare the pooled treatment, robots, and AI conditions with the control group. The final two columns compare the marginalization and nostalgia encouragements with the no-encouragement condition. $^*p < .05$.

A.6 Experimental News Articles

This section reproduces the full text of the news articles shown to respondents in the experimental task. Respondents read two articles and were asked to recommend one of them. Articles were randomly assigned according to treatment condition.

A.6.1 Robot Treatment.

Where Are Jobs Going? When Michael Morrison took a job at the steel mill in the center of Granite City, Illinois, in 1999, he assumed his future was secure. He was 38, a father with three young children.

"I felt like I had finally gotten into a place that was so reliable I could retire there," he said.

Although the mill had changed hands, it had been there since the end of the nineteenth century. For those willing to sweat, the mill was a reliable means of supporting a family. Mr. Morrison began by shoveling slag out of the furnaces, working his way up to crane driver. From inside a cockpit tucked in the rafters of the building, he manned the controls, guiding a 350-ton ladle that spilled molten iron. It was a difficult job requiring perpetual focus, and he was paid accordingly.

Now his job has been eliminated due to automation. Robots now man the same machine that Mr. Morrison once operated. As the company website describes, "Of the 74 machines that were operating in the factory, 63 now run on their own with no human intervention."

Mr. Morrison has not been able to find other work, and he has no idea how he will pay for his children's college educations. "When they don't need me anymore," he said, "I'm nothing."

Adapted from Mutz (2021).

General Motors Closing Plant, Laying Off 1,500 Michigan Workers General Motors (GM) announced this week that it will close a plant in Michigan, laying off more than 1,500 workers as it tries to address financial losses. The news comes just months after GM announced it would be laying off 200 workers at a plant in neighboring Ohio.

GM said it expects to end the plant's light truck manufacturing operations by December 1, 2023, with another part of the plant closing by the end of 2023. The estimated job loss is 1,545 workers.

"We are conscious of the impact this decision will have on our employees, their families, and the local community, and we are announcing it to provide them with as much time as possible to prepare for this transition," the CEO said in a press release. "These decisions are never easy, nor are they taken lightly."

Economic analysts say that the auto manufacturing industry in the United States faces various challenges, including automation. Based on industry analysis, automation is the leading cause of job losses, affecting not only blue-collar workers (47% of jobs in the U.S. are at risk of becoming automated in the next 10 years). High-tech companies have developed computer software and advanced robotics that allow machines to perform many of the same tasks that auto plant workers previously did. Factories like this one get shut down as employees are replaced with advanced robotics developed for the auto industry.

When asked about the layoffs, Carter Douglas, 46, said, "I was heartbroken when I was told I wasn't coming back." He started as an entry-level worker but had worked his way up over the past 20 years with the company. "I know the times change, but losing this job will really make things rough for my family and me."

Adapted from Chaudoin and Mangini (2025).

A.6.2 AI Treatment.

Artificial Intelligence Reshaping White-Collar Jobs Artificial intelligence (AI) is swiftly infiltrating higher-level positions traditionally held by white-collar workers, such as accountants, software programmers, human resources specialists, freelance writers, and lawyers. According to Meta Platforms CEO Mark Zuckerberg, the integration of AI and other technological advancements is enhancing operational efficiency for companies, resulting in job losses that will be permanent.

Michael Morrison, a software engineer, recently shared his personal account of being impacted by the rise of AI-driven automation. Initially expecting long-term job security, Morrison believed his position would support his family and fund his children's college education. However, his role was replaced by AI systems in March when his company implemented ChatGPT. He now struggles to find alternative employment. "I'm facing difficulties finding a job because many companies are reducing their hiring," Morrison explained.

A concerned co-worker, preferring anonymity, stated, "I work as a front-end developer and designer. Technology is constantly evolving in this field, but I hadn't been concerned until recently. I kept learning new languages and frameworks. However, with the emergence of ChatGPT, which can write all sorts of code, my worries started to grow."

In response to the growing influence of AI, several major companies have announced layoffs, specifically targeting tasks that can be automated through AI systems like ChatGPT and Google Bard. Furthermore, a decrease in hiring is anticipated, indicating a long-lasting shift in employment practices.

Meta Implements Layoffs, Embraces AI, and Sparks Industry Reflection Meta (formerly Facebook) announced new layoffs this week as part of a strategic reorganization that embraces the integration of AI technologies. This move reflects a broader trend in the industry, with companies like Amazon and Lyft considering similar approaches.

Mark Zuckerberg, Meta's CEO, highlighted the shift, stating, "We need fewer people to do the same thing." This acknowledges that AI can efficiently handle tasks previously done by humans, such as content writing and code development.

Over 10,000 employees will be affected by these layoffs, following a prior round impacting 3,000 workers. The specific departments impacted have not been disclosed, but the stated goal is to streamline operations, enhance productivity, and maintain Meta's position as a technological frontrunner.

As Meta leads the way, other prominent companies are also contemplating similar measures. Goldman Sachs estimates that smaller companies will soon introduce generative AI into their procedures, resulting in further layoffs. They estimate that AI could impact around 300 million jobs worldwide. The process is expected to be disruptive. "I honestly don't think that it's going to be years. It feels more like months," Mr. Argenti said.

AI is anticipated to replace workers in a wide array of sectors. In past technological shifts, blue-collar and frontline workers typically bore the brunt of job losses. This time, the impact is concentrated among office and professional workers.

A.6.3 Technology-Neutral Control.

Boston Dynamics' First Consumer Product: A Battle Bot? Boston Dynamics' quadruped robot dog—SpotMini—was first announced in 2016. The robot, equipped with sensors, cameras, mechanical legs, and a grabber arm, is a notable achievement in robotics engineering and comes with a five-digit price tag.

Its exact purpose as a consumer product has remained ambiguous. However, at Amazon's inaugural conference in Las Vegas, Boston Dynamics CEO Marc Raibert suggested that the robot could be used for entertainment. He described a future in which Spot robots could engage in networked battles, creating a novel gaming experience.

Beta testing has also hinted at the robot's use in playground environments. One video showed four robots playing with a ball before clumsily tumbling. In a live demonstration, a Spot robot unexpectedly collapsed. Nonetheless, controlling the robot is intuitive, relying on a D-pad similar to standard remote-controlled toys.

The anticipated commercial release year for these entertainment-oriented robots is 2024.

CNH Industrial's Electric Tractor with Autonomous Features CNH Industrial recently unveiled the New Holland T4 Electric Power, the industry's first all-electric light utility tractor prototype with autonomous features, at its Tech Day in Phoenix, Arizona.

Michael Morrison, a member of the development team, described the prototype as "disruptive technology" and a "world-first result." He explained that the T4 Electric Power is designed for lower-horsepower operations, including mixed farming, livestock, municipal, orchard, and specialty applications.

The tractor combines power, precision, and performance. Morrison emphasized his pride in contributing to "years of hard work developing revolutionary technology," and highlighted that he felt part of a broader movement empowering customers to grow and build a better world.

The tractor includes electrical outlets for daily farm tasks such as welding and drilling and can serve as a backup power generator. Beyond environmental benefits, it allows farmers to reduce operating costs by up to 90 percent by eliminating diesel fuel costs and related maintenance. Its autonomous features enable remote operation and real-time monitoring through integrated sensors and camera systems.

Commercial production of this first-generation utility tractor is expected to begin at the end of 2023, with a broader product offering to follow.

A.7 Emotions evoked by the treatments

When examining the negative emotions evoked by the treatment (uneasiness, anger, and fear), I observed a statistically significant difference between the treatment and control groups. For instance, the probability of individuals in the treatment group indicating that they felt enthusiastic about technological change was only 6.4 percent, whereas in the control condition, it was approximately 77.5 percent.

A.8 Mediator Encouragement Tasks

This section reproduces the full text of the writing tasks used to encourage cultural grievances. Respondents were randomly assigned to one of the following autobiographical writing exercises and were instructed to write for 90 seconds.

Nostalgia Now we would like your input to evaluate the potential for a new section in our news website called "**Letters to the Editor**." In this section, readers would be able to submit letters about their own experiences, thoughts, and opinions.

For the next week, we are focusing on the theme of **nostalgia**, which is a sentimental longing for a person, place, or time from the past. Think about all the details of the experience—who was with you, what was happening, and how you felt. Try to immerse yourself in the memory to the point where you can almost feel as if you're experiencing it again in the present. We want to know if readers have experienced nostalgia, and what those experiences were like.

For the next 90 seconds, please write in detail about this event as if you were writing an anonymous letter to the editor. Your letter should be written with enough detail that someone reading this description could imagine the situation.

Marginalization Now we would like your input to evaluate the potential for a new section in our news website called "**Letters to the Editor**." In this section, readers will be able to submit letters about their own experiences, thoughts, and opinions.

Over the next week, we will be exploring the concept of "**marginalization**" and the experiences related to it. Specifically, we would like you to reflect on a time in your life when you felt that you, or people like you, were either excluded or underappreciated socially, politically, or economically by those who were different from you. Take some time to think about the specifics of the situation—who was involved, what happened, and how you felt. We're interested in hearing about your personal experiences with marginalization and what that experience was like for you.

For the next 90 seconds, please write in detail about this event as if you were writing an anonymous letter to the editor. Your letter should be written with enough detail that someone reading this description could imagine the situation.

A.9 Validation of the Encouragement Task

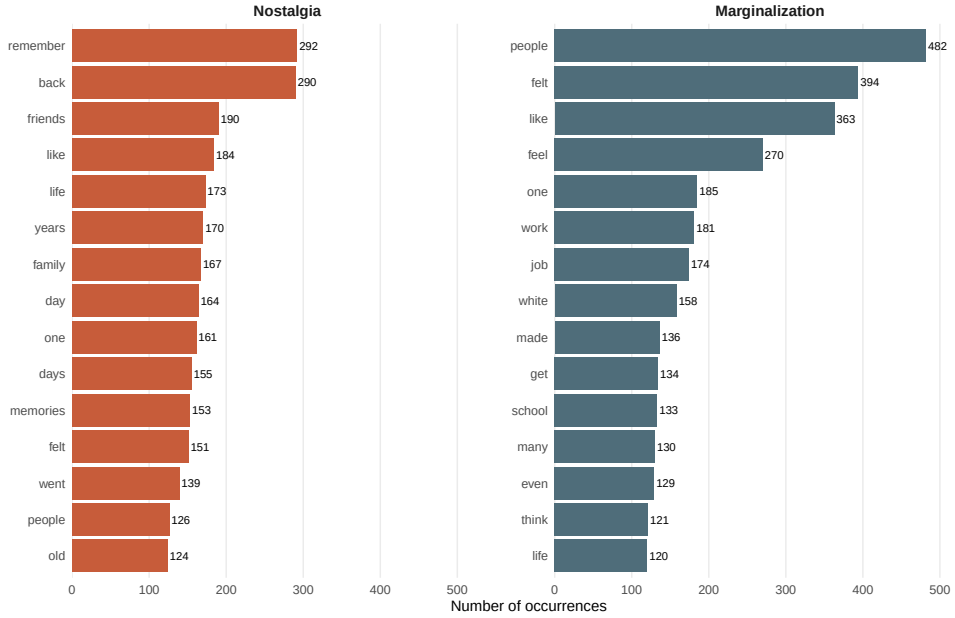
To assess the validity of the encouragement task, I examine task completion, response length, prompt compliance, economic content, and emotional tone. Two research assistants hand-coded prompt compliance, economic content, and emotional tone, and the author audited their coding. As Table A.6 shows, all 1,816 respondents assigned to an encouragement condition submitted a response, and 85.7% were coded as complying with the assigned prompt. The responses also differ in the expected direction: nostalgia responses are predominantly positive or combine positive views of the past with negative views of the present—a pattern consistent with nostalgia serving as a coping mechanism—whereas marginalization responses are predominantly negative. The lexical patterns shown in Figure A.3 reinforce this distinction. Nostalgia responses emphasize memories, friends, and family, while marginalization responses emphasize people, feelings, work, and jobs.

Table A.6: Validation of the Cultural-Grievance Encouragement Tasks

Measure	Nostalgia	Marginalization	Cultural Grievances
Completion and length			
Assigned	927	889	1,816
Text submitted	927	889	1,816
Mean words	67.4	74.8	71.0
Median words	56.0	61.0	58.0
75th percentile of words	83.5	91.0	88.0
Maximum words	577	470	577
Content validation			
Hand-coded complier share	88.6%	82.7%	85.7%
Economic-reference share	3.0%	19.3%	10.7%
Emotions			
Negative	3.1%	78.2%	39.9%
Neutral	8.2%	7.1%	7.7%
Positive	67.3%	2.1%	35.4%
Positive past / negative now	16.0%	2.4%	9.3%
Unclear or ambiguous	4.4%	6.2%	5.3%

Notes: Cultural grievances pools the nostalgia and marginalization conditions. Hand-coded compliance and emotion entries are percentages of assigned respondents. The economic-reference share is the percentage of topic-matched texts containing an economic reference. Unclear and ambiguous emotion codes are combined.

Figure A.3: Most Frequent Words in the Encouragement-Task Responses



Notes: The figure reports the 15 most frequent substantive words within each assigned writing condition after removing standard English stop words. Numbers are raw word-occurrence counts.

To assess whether the task posed cognitive barriers, Table A.7 compares respondents with and without a college degree separately within each prompt condition. The analysis identifies some differences: college-educated respondents wrote somewhat shorter nostalgia responses, were less likely to comply with the nostalgia prompt, more often used a neutral tone in those responses, and made fewer economic references under the marginalization condition. Crucially, however, these differences are not consistent across the two prompts or across indicators of completion, effort, compliance, and content. Taken together, the results provide limited evidence that cognitive burden systematically shaped respondents' performance.

Table A.7: Encouragement-Task Validation by College Education

	Nostalgia		Marginalization	
Measure	No college	College	No college	College
Completion and length				
Assigned	349	578	348	541
Text submitted	100.0%	100.0%	100.0%	100.0%
Mean words	72.0	64.6**	76.2	74.0
Median words	59.0	54.0	58.0	62.0
75th percentile of words	88.0	81.8	92.0	91.0
Maximum words	577	503	470	423
Content validation				
Hand-coded complier share	93.1%	85.8%***	83.3%	82.3%
Economic-reference share	4.0%	2.4%	23.4%	16.7%**
Emotions				
Negative	3.4%	3.0%	81.7%	81.3%
Neutral	4.0%	10.9%***	6.0%	8.3%
Positive	70.2%	66.6%	2.1%	2.3%
Positive past / negative now	18.1%	14.9%	3.0%	2.1%
Unclear or ambiguous	4.3%	4.6%	7.2%	6.0%

Notes: College denotes a bachelor's degree or higher. Shares are calculated among respondents with a nonmissing code; the economic-reference share is restricted to responses coded as matching the assigned topic. Stars appended to the College estimate denote the unadjusted comparison with No college: $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$. Binary measures use Pearson chi-squared tests without a continuity correction; mean word count uses Welch's t-test. The remaining response-length statistics and assigned counts are descriptive.

A.10 Construction and Measurement Robustness of Cultural Grievances

A.10.1Survey Items and Additive-Index Construction. Respondents indicated their agreement with six post-treatment statements on a five-point scale. The original response codes were recoded to range from 1 to 5, with higher values consistently indicating greater perceived marginalization or nostalgia. The questions were:

Marginalization

M1. "My values are not respected in this country."

M2. "People with values like mine are treated poorly in this society."

M3. "Regardless of who is in political power, things are generally pretty bad for people like me."

Nostalgia

N1. "Many American traditions have been lost over time."

N2. "American identity is no longer what it used to be."

N3. "People like me will not represent American identity in the future."

The main specification uses transparent additive indices because all items share the same response scale. The marginalization and nostalgia indices sum their respective three items and therefore range from 3 to 15. The overall cultural-grievance index sums all six items and ranges from 6 to 30:

$$M_i = M_{1i} + M_{2i} + M_{3i},$$
$$N_i = N_{1i} + N_{2i} + N_{3i},$$
$$G_i = M_i + N_i.$$

For the binary outcomes used in the main experimental models, respondents are classified as having high marginalization when $M_i > 10$, high nostalgia when $N_i > 11$ (percentile 66th), and high overall cultural grievances when $G_i > 23$. Respondents at or below each threshold form the corresponding low-or-moderate category.

A.10.2 Internal Consistency. I first assess whether the six questions behave as a coherent scale using Cronbach's alpha. Table A.8 reports the reliability of the full battery and of each theoretically defined subscale. The six-item alpha is 0.850, indicating high internal consistency. The marginalization and nostalgia subscales also exhibit good consistency, with alpha coefficients of 0.833 and 0.793, respectively. Removing individual items does not materially improve the reliability of the overall battery.

Table A.8: Internal Consistency of the Cultural-Grievance Measures

Scale	N	Items	Average inter-item covariance	Cronbach's alpha
Cultural grievances (six items)	3133	6	0.745	0.850
Marginalization (items M1-3)	3133	3	0.964	0.833
Nostalgia (items N1-3)	3133	3	0.850	0.793

Notes: All six items range from 1 to 5, with higher values indicating stronger marginalization or nostalgia. Alpha is calculated from the raw item covariance matrix. The standardized coefficients are nearly identical and are reported in the reproducibility appendix.

A.10.3 Alternative Construction Using Principal Component Analysis. As an alternative to equal-weight addition, I estimate three principal component analyses (PCAs): one using the three marginalization questions, one using the three nostalgia questions, and one using all six questions. Each item is centered and standardized before estimation, so the PCA is based on the item correlation matrix. I retain the first principal component (PC1) and orient its sign so that higher scores indicate stronger cultural grievances.

PC1 explains 75.1% of the variation in the marginalization items, 71.0% in the nostalgia items, and 57.4% in the overall six-item battery. As shown in Table A.9, every item loads positively and strongly on its subscale PC1 and on the overall PC1. In the overall PCA, the second component explains an additional 16.9% of the variation. Its positive marginalization loadings and negative nostalgia loadings recover the theoretically expected distinction between the two subdomains. The results therefore support both a common cultural-grievance dimension and the paper's practice of reporting marginalization and nostalgia separately.

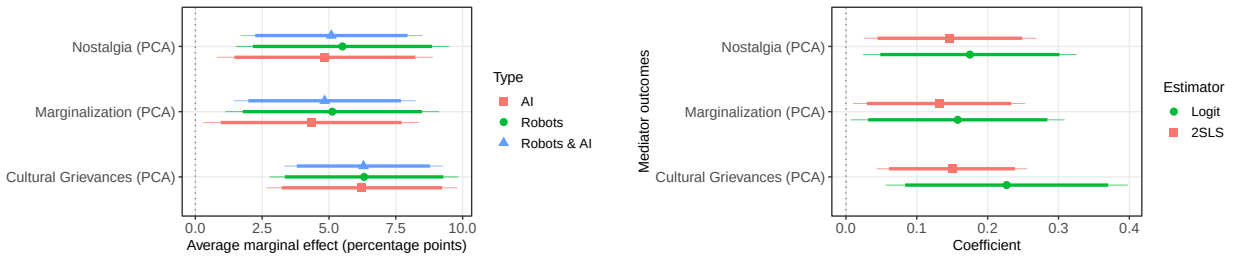
Table A.9: Principal-Component Loadings for the Cultural-Grievance Battery

Survey item	Domain	Subscale PC1 loading	Overall PC1 loading	Overall PC2 loading
My values are not respected in this country	Marginalization	0.888	0.819	0.318
People with values like mine are treated poorly in this society	Marginalization	0.903	0.817	0.360
Things are generally pretty bad for people like me	Marginalization	0.805	0.689	0.445
Many American traditions have been lost over time	Nostalgia	0.860	0.691	-0.565
American identity is no longer what it used to be	Nostalgia	0.857	0.721	-0.493
People like me will not represent American identity in the future	Nostalgia	0.809	0.796	-0.145

Notes: Entries are correlations between each standardized item and the indicated component. "Subscale PC1" refers to the PC1 from the three-item marginalization or nostalgia PCA. "Overall PC1" and "Overall PC2" are estimated from all six items. Component signs are oriented so that overall PC1 increases with cultural grievances.

To preserve the interpretation of the original binary models, I also define PCA-based high-score groups using cutoffs that reproduce the number of respondents classified as high by each additive measure. This PCA extension shows that allowing the data to determine the item weights does not change the substantive classification produced by the original equal-weight indices.

A.10.4 Robustness of the Experimental Results. Figure A.4 replaces each additive outcome with its corresponding PCA-based classification. Panel (a) re-estimates the treatment effects as average changes in predicted probability. Panel (b) re-estimates the randomized-encouragement Logit models and the 2SLS models in which randomized encouragement instruments for the published compliance measure. The direction, magnitude, and statistical significance of every estimate are unchanged.



(a) Treatment effects using PCA-based classifications

(b) Encouragement Logit and 2SLS estimates using PCA-based classifications

Figure A.4: Robustness of the Experimental Results to PCA-Based Measures

Notes: Panel (a) reports average marginal effects in percentage points for AI, robots, and the pooled robots-and-AI treatment. Models adjust for gender, race, income, education, routine-task intensity, and mediator encouragement. Panel (b) reports raw Logit coefficients on randomized mediator encouragement and 2SLS coefficients on the compliance measure, instrumented by randomized encouragement, with the published controls. Thick and thin intervals denote 90% and 95% confidence intervals, respectively. Logit and 2SLS coefficients are displayed together to match the main figure but are expressed on different scales. The full sample is $N = 3{,}133$; treatment-specific comparisons use the corresponding treatment and control observations.

B Effects of T on M, refers to model in Figure 2

The following are the tables related to the main text figures.

	Dependent variable:								
	Robots			AI			Robots & AI		
	Nostalgia	Marginalization	Cultural Grievances	Nostalgia	Marginalization	Cultural Grievances	Nostalgia	Marginalization	Cultural Grievances
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Treat	0.248***	0.231**	0.367***	0.220**	0.197**	0.363***	0.230***	0.219***	0.366***
	(0.092)	(0.092)	(0.104)	(0.094)	(0.093)	(0.106)	(0.079)	(0.079)	(0.091)
Female	−0.044	0.030	−0.031	0.036	0.048	−0.042	0.009	0.028	−0.019
	(0.093)	(0.093)	(0.106)	(0.094)	(0.094)	(0.107)	(0.077)	(0.077)	(0.087)
White	0.092	−0.291***	−0.064	0.041	−0.287***	0.032	0.091	−0.301***	−0.009
	(0.102)	(0.100)	(0.115)	(0.105)	(0.103)	(0.119)	(0.085)	(0.083)	(0.095)
Education: highschool	−0.506	−1.082	−0.934	−0.217	−1.046	−1.393*	−0.787	−1.205*	−1.250*
	(0.783)	(0.859)	(0.789)	(0.728)	(0.757)	(0.760)	(0.641)	(0.696)	(0.646)
Education: incomplete college	−0.891	−1.394	−1.239	−0.414	−0.997	−1.621**	−1.022	−1.331*	−1.475**
	(0.778)	(0.855)	(0.784)	(0.722)	(0.750)	(0.753)	(0.636)	(0.692)	(0.642)
Education: technical	−0.703	−1.004	−1.089	−0.347	−0.658	−1.388*	−0.917	−0.984	−1.329**
	(0.782)	(0.858)	(0.788)	(0.726)	(0.753)	(0.756)	(0.639)	(0.694)	(0.644)
Education: college	−1.107	−1.340	−1.505*	−0.776	−0.953	−1.743**	−1.335**	−1.349*	−1.729***
	(0.777)	(0.854)	(0.783)	(0.720)	(0.747)	(0.750)	(0.635)	(0.690)	(0.640)
Education: gradschool	−0.991	−1.221	−1.329*	−0.807	−0.867	−1.862**	−1.257**	−1.242*	−1.671***
	(0.783)	(0.859)	(0.790)	(0.726)	(0.753)	(0.758)	(0.639)	(0.694)	(0.646)
Education: other	−1.186	−2.160	−1.375	−1.587	−0.555	−2.108	−1.593	−0.927	−1.592
	(1.397)	(1.445)	(1.405)	(1.316)	(1.114)	(1.330)	(1.036)	(0.992)	(1.040)
RTI	0.013*	0.013*	0.021**	0.005	0.002	0.002	0.010*	0.009	0.010
	(0.007)	(0.007)	(0.008)	(0.007)	(0.007)	(0.008)	(0.006)	(0.006)	(0.007)
Income dummy	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Encouragement dummy	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,177	2,177	2,177	2,138	2,138	2,138	3,133	3,133	3,133
Log Likelihood	−1,377.356	−1,377.210	−1,136.916	−1,345.605	−1,345.092	−1,111.423	−1,997.818	−1,993.138	−1,673.287
Note:									*p<0.1; **p<0.05; ***p<0.01

Table B.10: Effects of Automation-Threat Treatments on Proposed Mediators

Note: This table reports the effects of exposure to job-threatening news on the proposed mediators relative to the neutral technology control group ($N = 1{,}182$). Columns 1–3 compare the Robots condition with the control condition (analysis $N = 2{,}177$); Columns 4–6 compare the AI condition with the control condition (analysis $N = 2{,}138$); and Columns 7–9 pool the Robots and AI conditions and compare them with the control condition (analysis $N = 3{,}133$). Each outcome is an indicator equal to one for scores above the sample median. Nostalgia captures sentimentality for the past, Marginalization captures perceived social exclusion, and Cultural Grievances combines the two dimensions. These are the models underlying Figure 2.

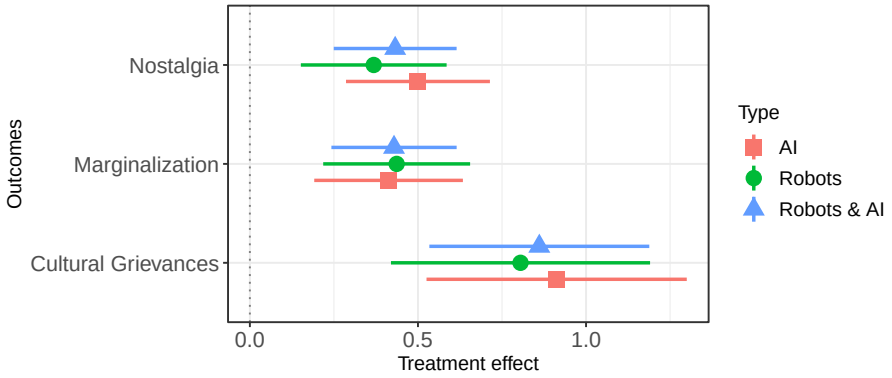


Figure B.5: Treatment Effects of Exposure to Automation Risk on mediators (full continuous indices). *Note*: This figure shows the effect of exposure to job-threatening news on the mediators against a control group of 1,182 subjects. 'AI' represents the treatment effect of AI-related job displacement (N=2,138), 'Robots' for machine-related displacement (N=2,177), and 'Robots & AI' pools the data (N=3,133). Results for each mediator are shown, with the full indices, derived from the sum of various questions, resulting in a range of 3 to 15 for marginalization and nostalgia, and cultural grievances is the addition of these two indices.

C Average Total Effect

C.1 ATE, refers to model in Figure 3

Table C.11: Effects of Automation-Threat Treatments on Political Attitudes

	Dependent variable:																	
	Robots						AI						Robots & AI					
	Pro Taxing Rich	Anti Elitism	Pro Trump	Trade Favorability	Anti Immigration	Robots Concern	Pro Taxing Rich	Anti Elitism	Pro Trump	Trade Favorability	Anti Immigration	Robots Concern	Pro Taxing Rich	Anti Elitism	Pro Trump	Trade Favorability	Anti Immigration	Robots Concern
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
Treat	0.048	0.110**	0.010	−0.204***	0.112**	0.118***	0.002	0.056	0.002	−0.114**	−0.028	0.218***	0.027	0.087**	0.006	−0.161***	0.045	0.170***
	(0.045)	(0.043)	(0.015)	(0.056)	(0.052)	(0.036)	(0.047)	(0.044)	(0.015)	(0.057)	(0.053)	(0.036)	(0.039)	(0.037)	(0.012)	(0.048)	(0.044)	(0.031)
Female	0.190***	0.132***	−0.007	−0.230***	0.089*	−0.140***	0.178***	0.053	−0.002	−0.168***	0.063	−0.110***	0.174***	0.093**	−0.001	−0.182***	0.070	−0.137***
	(0.045)	(0.044)	(0.015)	(0.057)	(0.052)	(0.036)	(0.048)	(0.045)	(0.015)	(0.057)	(0.053)	(0.037)	(0.038)	(0.036)	(0.012)	(0.047)	(0.044)	(0.030)
White	−0.034	−0.077	0.038**	−0.125**	0.086	−0.126***	−0.033	−0.026	0.041**	−0.178***	0.218***	−0.081**	−0.033	−0.027	0.039***	−0.166***	0.147***	−0.098***
	(0.049)	(0.048)	(0.016)	(0.062)	(0.057)	(0.040)	(0.053)	(0.049)	(0.016)	(0.063)	(0.059)	(0.041)	(0.042)	(0.040)	(0.013)	(0.051)	(0.048)	(0.033)
Education: highschool	−0.503	0.103	0.201	0.097	0.308	−0.316	−0.493	−0.312	0.066	−0.524	−0.349	0.125	−0.465	−0.193	0.103	−0.159	−0.321	−0.207
	(0.397)	(0.385)	(0.130)	(0.501)	(0.459)	(0.319)	(0.390)	(0.366)	(0.121)	(0.469)	(0.435)	(0.299)	(0.324)	(0.307)	(0.104)	(0.399)	(0.368)	(0.252)
Education: incomplete college	−0.394	0.119	0.124	0.055	0.061	−0.199	−0.456	−0.126	−0.001	−0.577	−0.479	0.206	−0.380	−0.096	0.035	−0.158	−0.536	−0.086
	(0.394)	(0.383)	(0.130)	(0.498)	(0.456)	(0.317)	(0.387)	(0.363)	(0.120)	(0.465)	(0.432)	(0.297)	(0.321)	(0.305)	(0.103)	(0.396)	(0.365)	(0.250)
Education: technical	−0.436	0.192	0.135	0.172	0.130	−0.284	−0.443	−0.228	0.030	−0.507	−0.314	0.144	−0.416	−0.134	0.071	−0.137	−0.379	−0.145
	(0.396)	(0.385)	(0.130)	(0.500)	(0.459)	(0.318)	(0.389)	(0.365)	(0.121)	(0.468)	(0.434)	(0.298)	(0.322)	(0.306)	(0.103)	(0.397)	(0.367)	(0.251)
Education: college	−0.462	−0.042	0.105	0.205	−0.077	−0.296	−0.420	−0.299	−0.028	−0.359	−0.661	0.210	−0.405	−0.286	0.011	−0.002	−0.683*	−0.138
	(0.394)	(0.382)	(0.129)	(0.497)	(0.456)	(0.316)	(0.385)	(0.362)	(0.120)	(0.463)	(0.430)	(0.295)	(0.320)	(0.304)	(0.103)	(0.395)	(0.364)	(0.250)
Education: gradschool	−0.464	−0.115	0.075	0.420	−0.278	−0.230	−0.363	−0.379	−0.065	−0.261	−0.881**	0.160	−0.388	−0.366	−0.024	0.141	−0.888**	−0.134
	(0.396)	(0.385)	(0.130)	(0.500)	(0.459)	(0.318)	(0.388)	(0.364)	(0.121)	(0.467)	(0.433)	(0.297)	(0.322)	(0.306)	(0.103)	(0.397)	(0.367)	(0.251)
Education: other	−2.347***	−0.941	0.289	0.111	−0.451	−0.480	−1.035*	−0.524	0.205	0.088	−1.098*	0.342	−1.326***	−0.714	0.166	0.062	−1.193**	−0.066
	(0.649)	(0.630)	(0.213)	(0.820)	(0.752)	(0.522)	(0.585)	(0.549)	(0.182)	(0.704)	(0.652)	(0.448)	(0.490)	(0.465)	(0.157)	(0.604)	(0.558)	(0.382)
RTI	0.0002	0.008**	0.002	−0.003	0.007*	0.012***	−0.001	0.009***	0.001	−0.001	0.004	0.014***	−0.0002	0.007**	0.001	−0.001	0.004	0.013***
	(0.004)	(0.003)	(0.001)	(0.004)	(0.004)	(0.003)	(0.004)	(0.003)	(0.001)	(0.004)	(0.004)	(0.003)	(0.003)	(0.003)	(0.001)	(0.004)	(0.003)	(0.002)
Income dummy	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Encouragement dummy	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,177	2,174	2,177	2,177	2,177	2,158	2,138	2,135	2,138	2,138	2,138	2,113	3,133	3,129	3,133	3,133	3,133	3,103
R^2	0.038	0.033	0.023	0.029	0.031	0.064	0.025	0.020	0.023	0.021	0.034	0.070	0.025	0.021	0.019	0.022	0.030	0.062
Note:																		*p<0.1; **p<0.05; ***p<0.01

Note: This table reports the effects of exposure to job-threatening news on political attitudes relative to the neutral technology control condition. Columns 1-6 compare the Robots condition with the control condition; Columns 7–12 compare the AI condition with the control condition; and Columns 13–18 pool the Robots and AI conditions and compare them with the control condition. Outcomes include 'Pro Trump' (willingness to vote for Trump), 'Trade favorability' (views on trade benefits for US workers), 'Anti-immigration' (preferences on legal immigration levels), with 'Pro Trump' binary coded and others on a 1-5 scale. Standard errors are reported in parentheses.

C.2 Robots vs AI Comparison

Table C.12 directly compares the Robots and AI conditions across the three grievance measures and four main outcomes. After correcting for multiple comparisons, the Robots condition produces lower robot concern but greater anti-immigration sentiment than the AI condition, while the remaining outcomes do not differ significantly.

Table C.12: Comparisons of the Robots and AI treatment conditions

Outcome	Robots - AI	SE	95% CI	p-value	BH-adjusted p-value	N
Nostalgia	0.007	0.022	[-0.036, 0.050]	0.743	0.988	1951
Marginalization	0.004	0.022	[-0.039, 0.046]	0.863	0.988	1951
Cultural Grievances	-0.000	0.020	[-0.039, 0.038]	0.988	0.988	1951
Robots Concern	-0.103	0.037	[-0.175, -0.031]	0.005	0.022	1935
Anti Immigration	0.148	0.054	[0.042, 0.255]	0.006	0.022	1951
Trade Favorability	-0.098	0.059	[-0.213, 0.017]	0.096	0.225	1951
Pro Trump	0.007	0.015	[-0.023, 0.037]	0.653	0.988	1951

Notes: Entries report covariate-adjusted OLS differences between respondents assigned to the Robots and AI conditions (Robots minus AI). Positive estimates indicate higher outcome values under Robots; negative estimates indicate higher values under AI. Models adjust for gender, race, income, education, routine-task intensity, and encouragement assignment. The table reports conventional standard errors, 95% confidence intervals, two-sided p-values, and Benjamini–Hochberg-adjusted p-values across the seven outcomes.

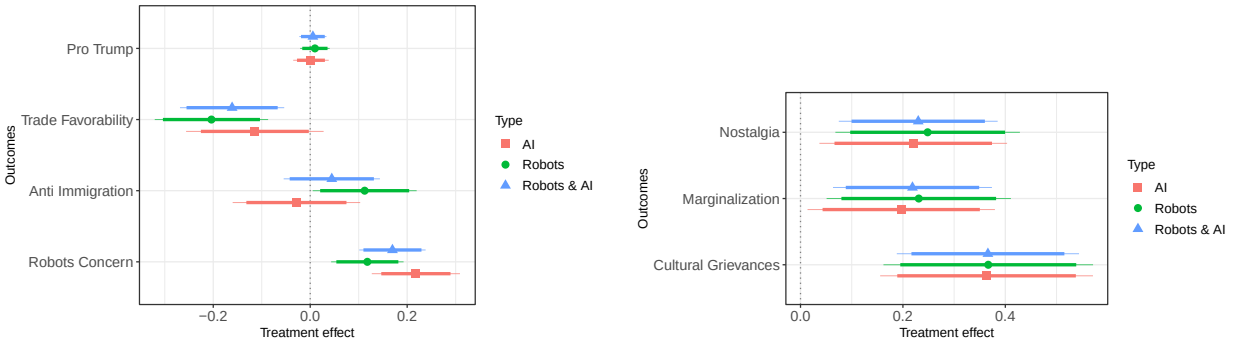
D Robustness Checks and Additional Analysis - Automation Risk Treatment

D.1 Benjamini–Hochberg correction.

As an additional check on multiple hypothesis testing, I apply the Benjamini–Hochberg (BH) procedure (Benjamini and Hochberg, 1995), which controls the false discovery rate (FDR), defined as the expected proportion of false rejections among all rejected hypotheses. For each family of tests, I compute the corresponding p-values, rank them from smallest to largest, and compare each ordered p-value $p_{(i)}$ to $\alpha i/m$, where m is the number of hypotheses in the family. Let

$$k = \max\left\{i : p_{(i)} \leq \alpha i/m\right\}.$$

All hypotheses with ranks $1, \ldots, k$ are then rejected; if no such k exists, no null hypotheses are rejected. Figure D.6 presents the results; the substantive conclusions remain unchanged.



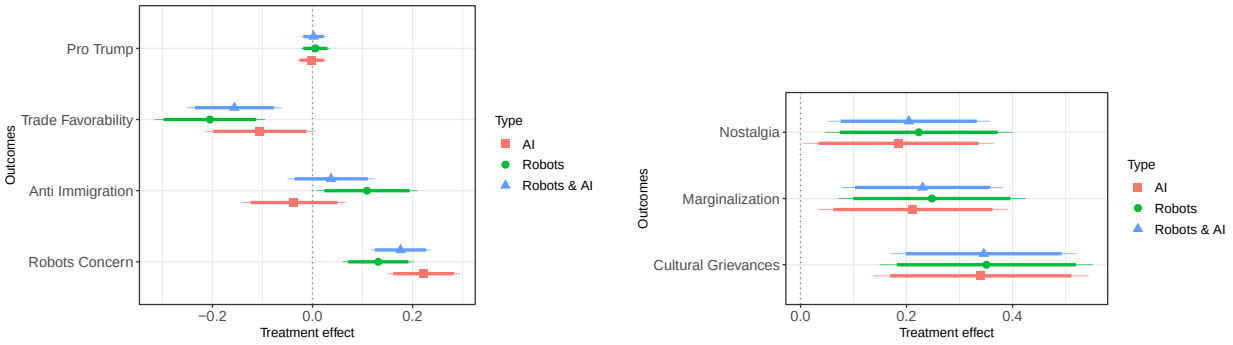
(a) Treatment effects on main outcomes. (b) Treatment effects on mediators.

Figure D.6: Benjamini–Hochberg robustness check for multiple hypothesis testing.

The left panel reports the main treatment effects and the right panel reports the treatment effects on the mediators. The overall substantive conclusions remain unchanged after applying the BH procedure.

D.2 Specifications without Additional Covariates

Figure D.7 re-estimates the treatment effects without demographic or occupational covariates. The results closely mirror the adjusted specifications.



(a) Treatment effects on the main outcomes. (b) Treatment effects on the mediators.

Figure D.7: Treatment effects from specifications without additional covariates.

The left panel reports OLS treatment effects on the main outcomes, while the right panel reports logit coefficients for the binary mediator outcomes. All specifications exclude demographic and occupational covariates. The Robots and AI specifications compare each treatment condition separately with the control condition; the pooled specification compares assignment to either automation treatment with control. Thick and thin bars report 90% and 95% confidence intervals, respectively.

D.3 Accounting for ideology controls

Tables D.13 and D.14 show that the main findings are robust to including ideology (measured pre-treatment) as an additional control. In particular, I construct a dummy variable for left ideology, set to 1 for respondents with an ideology of less than 5 and 0 otherwise. Adding this control does not materially alter the results: the treatment coefficients remain very similar in sign, magnitude, and statistical significance across both the outcome and mediator models. The relationship between exposure to automation risk, cultural grievances, and right-wing populist attitudes is therefore not driven by this simple measure of pre-existing ideology.

	Dependent variable:																	
	Robots						AI						Robots & AI					
	Pro Taxing Rich	Anti Elitism	Pro Trump	Trade Favorability	Anti Immigration	Robots Concern	Pro Taxing Rich	Anti Elitism	Pro Trump	Trade Favorability	Anti Immigration	Robots Concern	Pro Taxing Rich	Anti Elitism	Pro Trump	Trade Favorability	Anti Immigration	Robots Concern
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
Treat Robots	0.082**	0.112***	0.002	−0.197***	0.078*	0.114***												
	(0.039)	(0.043)	(0.014)	(0.056)	(0.047)	(0.036)												
Treat AI							0.013	0.056	−0.001	−0.112**	−0.040	0.217***						
							(0.042)	(0.044)	(0.014)	(0.057)	(0.048)	(0.036)						
Treat Both													0.048	0.088**	0.0004	−0.157***	0.023	0.167***
													(0.034)	(0.037)	(0.012)	(0.048)	(0.041)	(0.030)
Left	0.976***	0.041	−0.230***	0.184***	−0.969***	−0.111***	0.982***	0.049	−0.227***	0.182***	−0.999***	−0.093**	0.987***	0.046	−0.231***	0.177***	−0.977***	−0.112***
	(0.039)	(0.043)	(0.014)	(0.056)	(0.047)	(0.036)	(0.042)	(0.044)	(0.014)	(0.057)	(0.048)	(0.036)	(0.034)	(0.036)	(0.011)	(0.047)	(0.039)	(0.030)
Demographic Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Income dummy	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Encouragement dummy	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,177	2,174	2,177	2,177	2,177	2,158	2,138	2,135	2,138	2,138	2,138	2,113	3,133	3,129	3,133	3,133	3,133	3,103
R^2	0.251	0.034	0.134	0.034	0.189	0.068	0.225	0.020	0.133	0.026	0.199	0.073	0.238	0.022	0.132	0.026	0.190	0.066
Note:																		*p<0.1; **p<0.05; ***p<0.01

Table D.13: Robustness to ideology controls: treatment effects on political outcomes.

Note: This table re-estimates the main outcome models after adding a left-ideology dummy as a control. Columns are reported separately for the Robots, AI, and pooled Robots & AI treatments. The dependent variables are pro-taxing-the-rich attitudes, anti-elitism, Trump support, trade favorability, and anti-immigration attitudes.

	Dependent variable:								
	Robots			AI			Robots & AI		
	Nostalgia	Marginalization	Cultural Grievances	Nostalgia	Marginalization	Cultural Grievances	Nostalgia	Marginalization	Cultural Grievances
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Treat Robots	0.223**	0.215**	0.348***						
	(0.097)	(0.093)	(0.108)						
Treat AI				0.227**	0.192**	0.371***			
				(0.099)	(0.095)	(0.110)			
Treat Both							0.222***	0.209***	0.362***
							(0.083)	(0.080)	(0.094)
Left	−1.493***	−0.624***	−1.262***	−1.459***	−0.710***	−1.266***	−1.434***	−0.630***	−1.251***
	(0.101)	(0.094)	(0.116)	(0.102)	(0.096)	(0.117)	(0.083)	(0.078)	(0.095)
Demographic Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Income dummy	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Encouragement dummy	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,177	2,177	2,177	2,138	2,138	2,138	3,133	3,133	3,133
Log Likelihood	−1,258.047	−1,354.790	−1,071.679	−1,233.720	−1,316.972	−1,047.010	−1,835.674	−1,959.825	−1,577.376
Note:									*p<0.1; **p<0.05; ***p<0.01

Table D.14: Robustness to ideology controls: treatment effects on mediators.
Note: This table re-estimates the mediator models after adding the same left-ideology dummy. The dependent variables are nostalgia, marginalization, and the combined cultural-grievance measure.

Figure D.8 reports heterogeneous effects by ideology. The results do not show a uniform ideological split across all outcomes, but they do suggest some differences in where the treatment resonates most strongly. On the mediator side, the treatment appears to generate larger increases in nostalgia among left respondents, while marginalization is positive in both groups and somewhat less clearly differentiated. On the political outcome side, by contrast, the treatment does not appear to move left respondents in a more pro-Trump direction, whereas the corresponding effects are somewhat more positive among non-left respondents.

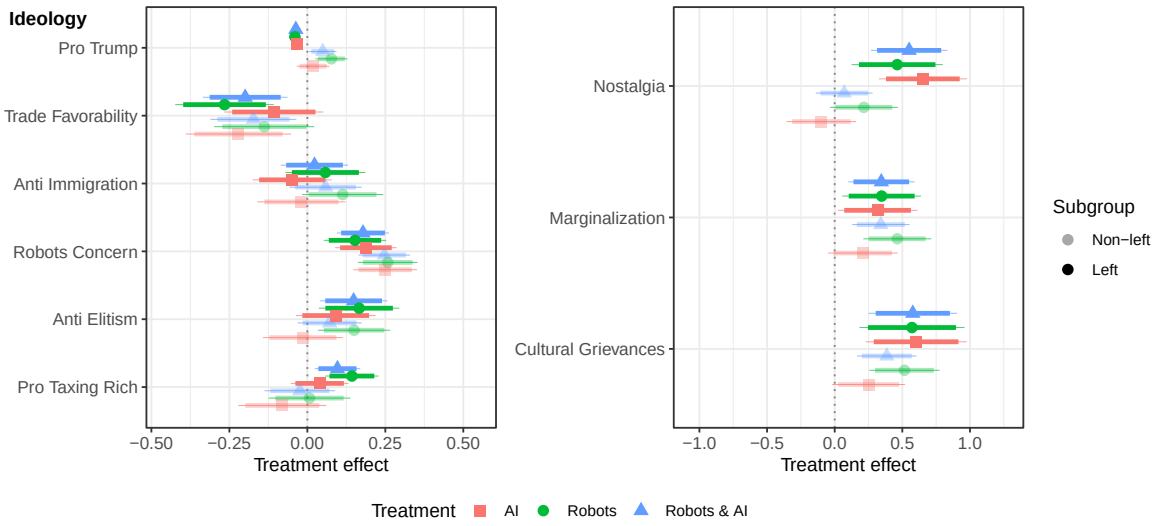


Figure D.8: Heterogeneous treatment effects by ideology.
Note: The figure re-estimates the baseline treatment-effect specifications separately for left and non-left respondents. The left panel reports political and policy outcomes, and the right panel reports mediator outcomes.

D.4 Local Economic Decline.

To assess whether the treatment effects on the mediators are stronger in economically declining places, I merge respondents' ZIP codes to a ZIP/ZCTA-level measure of local employment change built from the U.S. Census Bureau's LEHD Origin-Destination Employment Statistics (LODES) Workplace Area Characteristics files. The measure uses JT03 private primary jobs and defines the local employment decline rate as $-\frac{JT03_{2022,z}-JT03_{2002,z}}{JT03_{2002,z}}$, so that higher values indicate larger local employment losses.

Appendix Table D.15 and Figure D.9 report the interaction models with the full set of individual-level controls regarding the effect of automation threats on cultural grievances (associated with results in Figure 2). The interaction between the Robots treatment and the local employment decline rate is positive for Cultural Grievances ($b = 0.366$, $p < 0.05$) and Marginalization ($b = 0.283$, $p < 0.10$), while it is close to zero for Nostalgia ($b = 0.004$, $SE = 0.149$). Substantively, these positive interaction coefficients imply that the effect of the Robots treatment is stronger in places that experienced larger local employment decline over 2002–2022.

Table D.15: Robots and AI treatments by local employment decline, 2002–2022

	Robots treatment			AI treatment		
	Cultural Grievances	Marginalization	Nostalgia	Cultural Grievances	Marginalization	Nostalgia
Treatment	0.395***	0.281***	0.197*	0.502***	0.281**	0.218**
	(0.124)	(0.108)	(0.107)	(0.124)	(0.109)	(0.108)
Local decline rate (standardized)	-0.125	-0.067	-0.011	-0.126	-0.077	-0.009
	(0.085)	(0.079)	(0.080)	(0.085)	(0.079)	(0.079)
Treatment × Local decline rate (standardized)	0.366**	0.283*	0.004	0.078	-0.018	-0.039
	(0.184)	(0.161)	(0.149)	(0.103)	(0.098)	(0.097)
N	1630	1630	1630	1589	1589	1589
Female	Yes	Yes	Yes	Yes	Yes	Yes
White	Yes	Yes	Yes	Yes	Yes	Yes
RTI control	Yes	Yes	Yes	Yes	Yes	Yes
Income FE	Yes	Yes	Yes	Yes	Yes	Yes
Education controls	Yes	Yes	Yes	Yes	Yes	Yes
Encouragement dummy	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Entries are logit coefficients with standard errors in parentheses. The dependent variables are indicators for high levels of Cultural Grievances, Marginalization, and Nostalgia. The local employment decline rate is standardized, with higher values indicating larger local employment losses between 2002 and 2022. The Robots models estimate the effect of the Robots treatment relative to its control condition, while the AI models estimate the effect of the AI treatment relative to its control condition. All models include controls for gender, race, routine-task intensity, income, education, and encouragement assignment. *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

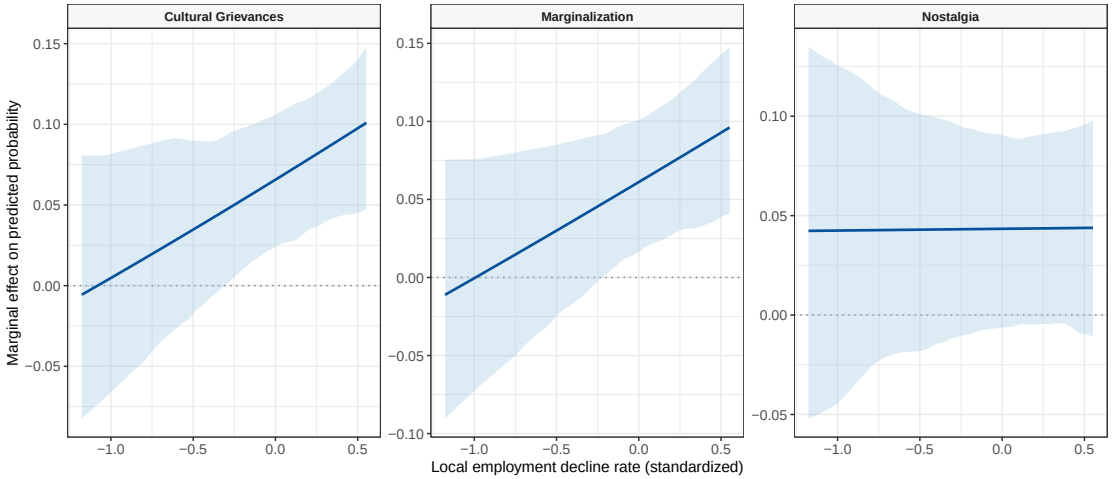


Figure D.9: Marginal effect of the Robots treatment by local employment decline rate, 2002–2022.
Note: Higher values of the moderator indicate larger local employment losses. The figure reports the estimated marginal effect of the Robots treatment on the probability of Cultural Grievances, Marginalization, and Nostalgia across the observed range of local employment decline. Shaded bands show 95% confidence intervals. The corresponding regression coefficients are reported in Table D.15.

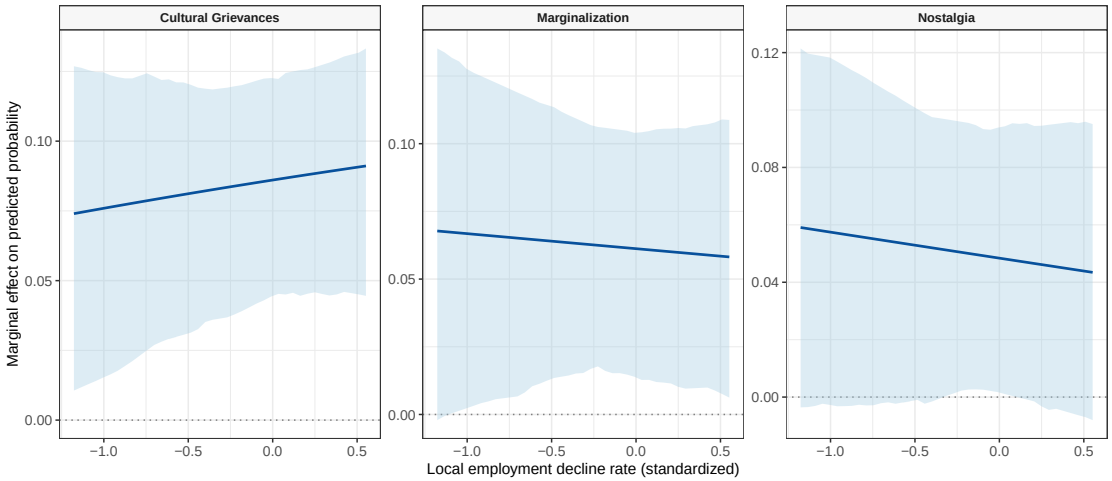


Figure D.10: Marginal effect of the AI treatment by local employment decline rate, 2002–2022.
Note: Higher values of the moderator indicate larger local employment losses. The figure reports the estimated marginal effect of the AI treatment on the probability of Cultural Grievances, Marginalization, and Nostalgia across the observed range of local employment decline. Shaded bands show 95% confidence intervals. The corresponding regression coefficients are reported in Table D.15.

Local Decline and Politics Results in Table D.16 show that the findings regarding preferences toward exclusionary politics presented in Figure 3 are robust to including local employment decline as a control. They nevertheless provide limited evidence that local decline systematically conditions the political effects of automation treatments. For the Robots treatment, greater local decline is associated with a larger increase in anti-elitism but a smaller increase in concern about robots, although both interaction estimates are only marginally significant. One possible explanation for the latter pattern is that respondents in areas with greater long-term employment losses already had relatively high baseline concern about technological change, leaving less scope for the treatment to increase those concerns further. None of the interactions involving the AI treatment is statistically significant. Overall, the treatment effects appear broadly similar across places with different levels of long-run employment decline, with only suggestive evidence of heterogeneous responses to the Robots treatment.

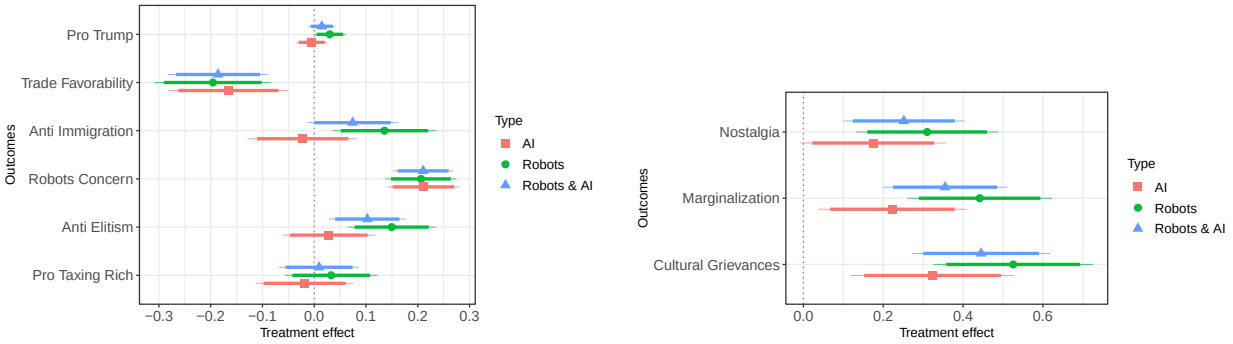
Table D.16: Robots and AI treatments by local employment decline: Political and policy outcomes, 2002–2022

	Robots treatment						AI treatment					
	Pro Trump	Trade Favorability	Anti Immigration	Robots Concern	Anti Elitism	Pro Taxing Rich	Pro Trump	Trade Favorability	Anti Immigration	Robots Concern	Anti Elitism	Pro Taxing Rich
Treatment	0.011	-0.195***	0.134**	0.108**	0.109**	0.032	0.010	-0.138**	-0.008	0.222***	0.079	0.027
	(0.017)	(0.066)	(0.060)	(0.042)	(0.051)	(0.052)	(0.017)	(0.066)	(0.062)	(0.043)	(0.052)	(0.056)
Local decline rate (standardized)	-0.016	-0.024	-0.029	0.033	-0.006	0.084**	-0.017	-0.020	-0.031	0.039	-0.006	0.087**
	(0.013)	(0.049)	(0.044)	(0.031)	(0.038)	(0.039)	(0.013)	(0.048)	(0.045)	(0.031)	(0.038)	(0.041)
Treatment × Local decline rate (standardized)	-0.016	0.104	-0.015	-0.106*	0.119*	-0.039	-0.015	0.046	-0.000	-0.037	-0.026	-0.043
	(0.024)	(0.092)	(0.084)	(0.059)	(0.072)	(0.074)	(0.016)	(0.060)	(0.057)	(0.039)	(0.047)	(0.051)
N	1630	1630	1630	1622	1629	1630	1589	1589	1589	1573	1588	1589
Female	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
White	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
RTI control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Income FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Education controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Encouragement dummy	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Entries are OLS coefficients with standard errors in parentheses. The dependent variables are Pro Trump, Trade Favorability, Anti Immigration, Robots Concern, Anti Elitism, and Pro Taxing Rich. The local employment decline rate is standardized, with higher values indicating larger local employment losses between 2002 and 2022. The Robots models estimate the effect of the Robots treatment relative to its control condition, while the AI models estimate the effect of the AI treatment relative to its control condition. All models include controls for gender, race, routine-task intensity, income, education, and encouragement assignment. *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

D.5 Survey weights

Because the experimental data come from an online convenience sample,[27] the weighted specifications re-benchmark the sample to better approximate the adult U.S. population. I construct raking weights using adult ACS benchmarks. Raking iteratively adjusts respondent weights so that the weighted sample matches known population distributions. I implement the procedure with the `anesrake` package and cap the resulting weights at 5. The weighting targets are sex, Hispanic origin, four-category education, five-category race/ethnicity, an eight-cell interaction between education and broad race/ethnicity (White versus non-White), age group, and Census region. Region is assigned from respondents' ZIP codes. The original experimental sample contains $N = 3{,}133$ respondents, of whom $N = 3{,}117$ receive weights. Eight respondents are excluded because their education responses cannot be mapped to the four-category education target, which also prevents construction of the education-by-race target. An additional eight respondents are excluded because Census region is unavailable.



Panel A. Weighted treatment effects on the main political outcomes and the manipulation check.

Panel B. Weighted treatment effects on cultural grievances.

Figure D.11: Weighted estimates using ACS-raked survey weights.

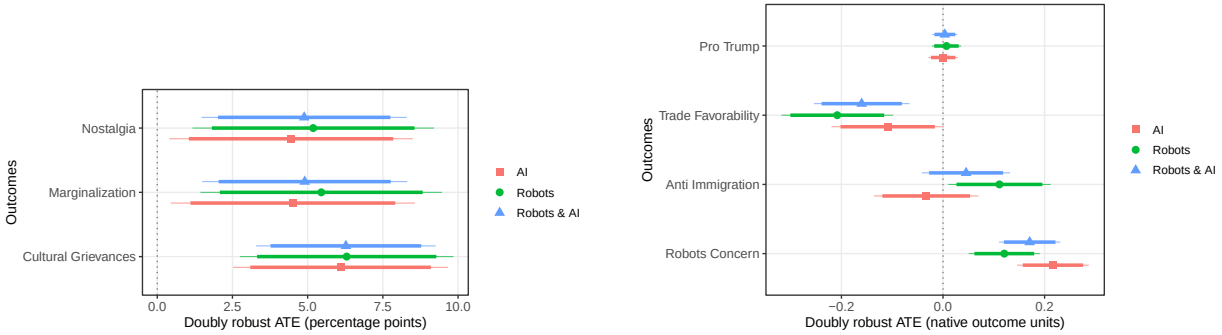
Note: Panel A reports weighted treatment effects for the main political outcomes and the automation-risk manipulation check. Panel B reports weighted treatment effects for the mediator outcomes. Both panels use the weighted analysis sample with $N = 3{,}117$. Relative to the initial experimental sample ($N = 3{,}133$), 16 observations are excluded from the weighting stage: 8 because education is missing, which also prevents construction of the education-by-race weighting target, and 8 because Census region cannot be assigned even after geographic imputation. All weights are constructed by raking to ACS benchmarks and are capped at 5.

[27] While some sample characteristics resemble those of the population, others, such as education, do not (see Appendix A.3).

D.6 Doubly Robust Estimation

As an additional robustness check, experimental effects are re-estimated using augmented inverse-probability weighting (AIPW), which synthesizes two distinct estimation strategies: a treatment-assignment (propensity) model and an outcome regression model. Rather than relying on either approach alone, AIPW combines them so that the resulting estimate remains consistent even if one of the two component models is misspecified, provided the other is correctly specified — a property commonly referred to as double robustness (Egami and Hartman, 2023; Glynn and Quinn, 2010; Robins, Rotnitzky, and Zhao, 1994). This offers a form of built-in redundancy: because the analysis relies on the assignment model and the outcome model jointly rather than on either in isolation, the design is less vulnerable to bias arising from misspecification of any single stage.

In this application, random assignment of treatment provides a natural anchor for one of the two components. Because the probability of treatment does not depend on covariates by construction, the propensity model is correctly specified without any need for covariate-based modeling; the outcome model is included primarily to increase precision. Nuisance models are estimated using ten-fold cross-fitting to avoid overfitting bias, and standard errors are computed from the AIPW influence function using standard-normal critical values. Figure D.12 presents the results. Panel a shows effects on the binary cultural-grievance measures, and Panel b presents the main experimental outcomes. Across specifications, the doubly robust estimates closely track the point estimates and confidence intervals from the original regression specifications, indicating that the substantive conclusions are not an artifact of a particular modeling choice for covariate adjustment.



(a) Cultural-grievance outcomes (b) Main experimental outcomes

Figure D.12: Doubly Robust Estimates of the Experimental Effects

Note: The figure reports cross-fitted augmented inverse-probability-weighted estimates. Panel (a) reports risk differences for binary cultural-grievance outcomes, expressed in percentage points. Panel (b) reports effects in each outcome's native units: Robots Concern and Anti Immigration are measured on 1–5 scales, Trade Favorability is measured on a 0–5 scale, and Pro Trump is binary, so an estimate of 0.01 for that outcome represents a one-percentage-point change. The models adjust for gender, race, income, education, occupational RTI, and encouragement assignment. The pooled Robots & AI contrast uses the full experimental sample; the Robots-only and AI-only contrasts compare the relevant treatment condition with the common control condition and exclude the other active treatment arm. Thick and thin bars represent 90% and 95% confidence intervals, respectively.

D.7 Heterogeneous Effects

Figure D.13 presents heterogeneous treatment effects by race, education, and exposure to automation risk, estimated separately within each subgroup. These results should be interpreted cautiously because the experiment was not powered to detect subgroup differences. Once the sample is divided across race, education, and exposure-to-automation categories, statistical power declines rapidly. I therefore treat these estimates as suggestive rather than conclusive. Even so, they provide preliminary evidence that the effects of automation-related treatment exposure may not be uniform across respondents, helping understand the scope conditions of the results. I therefore supplement visual comparisons of the subgroup estimates with two-sided Wald tests of coefficient equality. Overall, this analysis presents suggestive evidence of heterogeneous effects by race and education.

The greatest differences appear in the race split. Before adjustment, 11 of the 27 race contrasts have $p < 0.05$. These include differences in support for taxing the rich under the pooled and AI treatments; anti-elitism under the AI treatment; trade favorability under the pooled, Robots, and AI treatments; cultural grievances under the pooled and AI treatments; and nostalgia under the pooled, Robots, and AI treatments. Five of these differences remain robust after BH adjustment. Relative to non-White respondents, White respondents have a lower pooled-treatment coefficient for trade favorability (difference $= -0.365$, BH-adjusted $p = 0.016$), a larger AI-treatment coefficient for anti-elitism (difference $= 0.360$, BH-adjusted $p = 0.011$), a lower AI-treatment coefficient for trade favorability (difference $= -0.527$, BH-adjusted $p = 0.003$), a larger AI-treatment coefficient for cultural grievances (difference $= 0.949$, BH-adjusted $p = 0.011$), and a larger AI-treatment coefficient for nostalgia (difference $= 0.641$, BH-adjusted $p = 0.047$). These patterns remain consistent with scholarship emphasizing group-specific translations of economic threat into political

attitudes (Baccini and Weymouth, 2021; Hopkins, Margalit, and Solodoch, 2024; Gest, 2016; Autor et al., 2020; Choi et al., 2024; Jones, Owen, and Sung, 2026).

The education split produces some differences before adjustment. These involve concern about robots under the pooled and AI treatments, Trump support under the pooled and Robots treatments, support for taxing the rich under the Robots treatment, and marginalization under the pooled and Robots treatments. Three education differences remain robust after BH adjustment. Regarding cultural grievances, the Robots-treatment coefficient for marginalization is substantially smaller among college respondents (college minus non-college difference $= -0.950$, BH-adjusted $p < 0.001$). The marginalization result is consistent with the argument that technological disruption may be more likely to activate feelings of marginalization among respondents with fewer educational resources (Gidron and Hall, 2020, 2017). Regarding the political outcomes, the pooled and Robots-treatment coefficients for Trump support are larger among college respondents than among non-college respondents (differences $= 0.081$, BH-adjusted $p = 0.016$, and $= 0.103$, BH-adjusted $p = 0.012$, respectively).

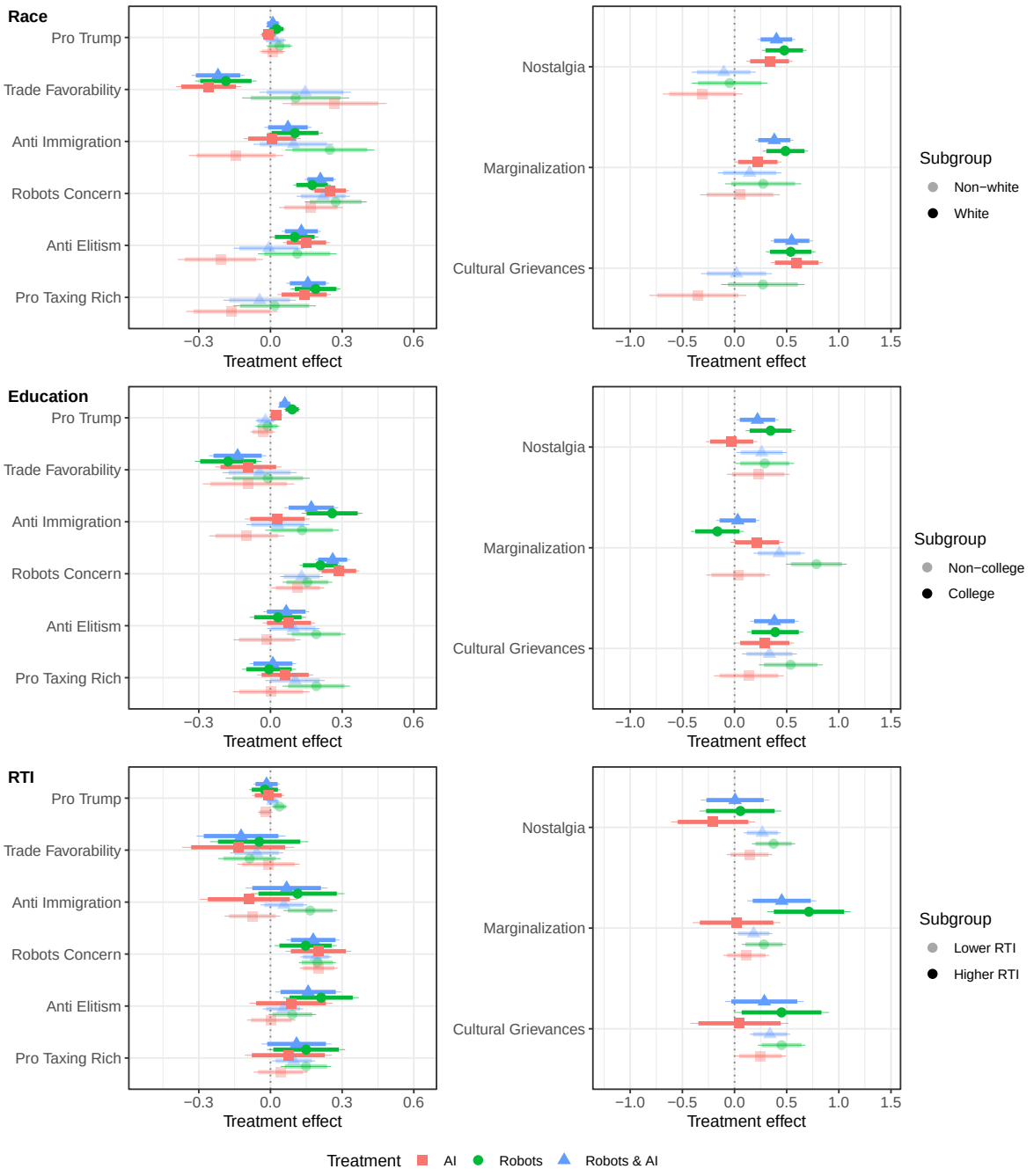


Figure D.13: Heterogeneous treatment effects by race, education, and automation exposure.
Note: Each row re-estimates the baseline treatment-effect specifications within subgroups defined by race, education, and occupational routine task intensity (RTI). The left column reports political and policy outcomes, and the right column reports mediator outcomes. Points denote estimated effects and bars show 90% confidence intervals.

By contrast, the occupational RTI split provides little evidence of differential treatment effects. None of the tests reaches $p < 0.05$, even before multiple-testing adjustment. Although some subgroup point estimates differ visually, their confidence intervals and formal contrasts are too imprecise to conclude that respondents in higher-RTI occupations respond differently. Occupational risk remains theoretically relevant (e.g., Kurer, 2020; Gonzalez-Rostani, 2026; Thewissen and Rueda, 2019), but this analysis does not provide statistically distinguishable treatment effects by RTI.

E Model-Based Inference with Survey Data

Figure E.14 presents the estimated direct and indirect effects for several outcomes using model-based mediation analysis of the survey data.

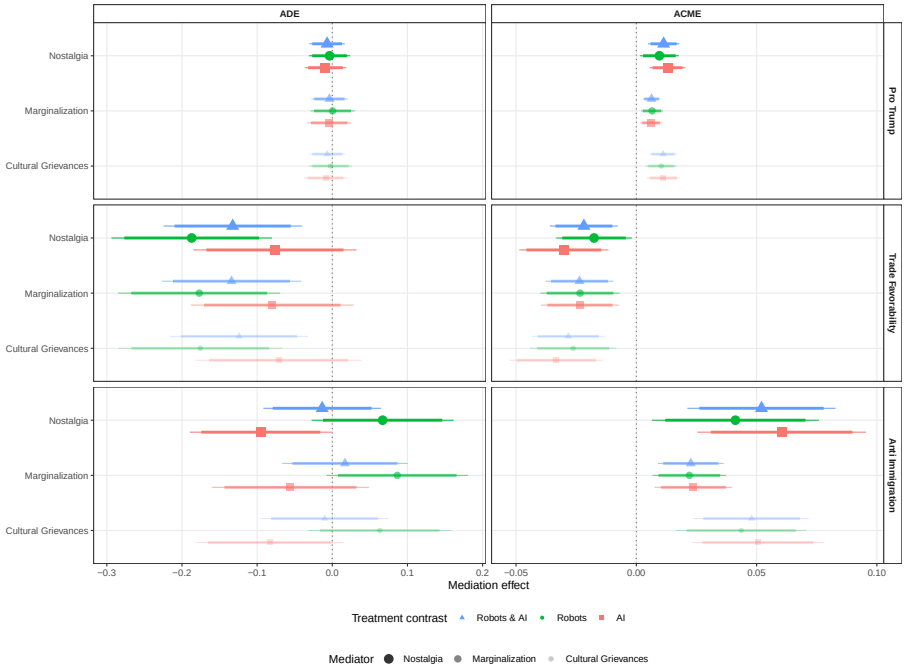


Figure E.14: Model-Based Estimates of Causal Mediation Effects Using Survey Dat (relates to Figure 4)

Note: The figure reports model-based average direct effects (ADE) and average mediated causal effects (ACME). The treatment contrasts compare exposure to information about job-threatening displacement due to robots and/or artificial intelligence with the common control condition. The treatment estimates are presented in the following order: pooled Robots & AI (blue triangles), Robots (green circles), and AI (red squares). The outcomes are support for Trump, trade favorability, and anti-immigration attitudes. "Pro Trump" measures willingness to vote for Donald Trump if he runs again for president. "Trade Favorability" measures whether respondents believe increased trade with other countries has benefited American workers. "Anti Immigration" measures preferences regarding whether the federal government should increase, decrease, or maintain the number of legal immigrants admitted to the United States; higher values indicate a stronger preference for decreasing immigration. The mediators are displayed in the following order: Nostalgia, Marginalization, and Cultural Grievances. Nostalgia measures sentimentality toward the past and concern about the loss of traditions. Marginalization measures perceptions that people like the respondent no longer represent American identity, that their values are disrespected, and that they are treated poorly in society. Cultural Grievances is the composite formed by summing the Nostalgia and Marginalization indices. Thick horizontal lines report 90% confidence intervals and thin horizontal lines report 95% confidence intervals. The samples are $N = 3{,}133$ for the pooled Robots & AI contrast, $N = 2{,}177$ for the Robots contrast, and $N = 2{,}138$ for the AI contrast.

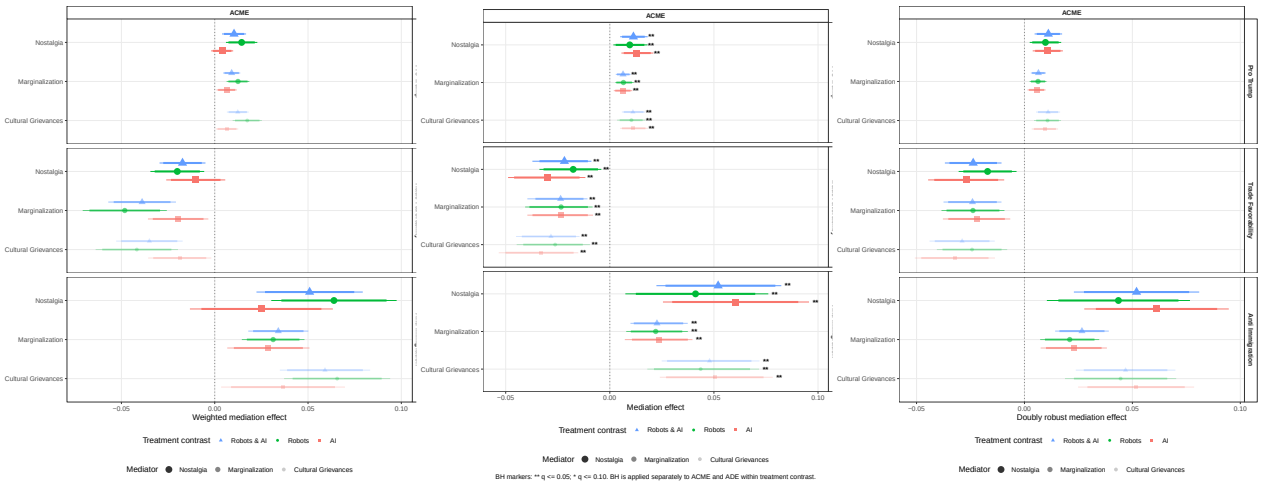
E.1 Sensitivity Analysis

	Robots			AI			Robots & AI		
Outcome	Nostalgia	Marginalization	Cultural Grievances	Nostalgia	Marginalization	Cultural Grievances	Nostalgia	Marginalization	Cultural Grievances
Pro Trump	0.306	0.177	0.263	0.284	0.175	0.249	0.302	0.178	0.261
Trade Favorability	-0.128	-0.128	-0.144	-0.149	-0.134	-0.158	-0.132	-0.129	-0.147
Anti Immigration	0.338	0.131	0.264	0.328	0.143	0.262	0.337	0.137	0.265

Table E.17: Sensitivity analysis: model-based mediation in the survey experiment.

Note: Entries report the residual correlation, ρ, at which the estimated average causal mediation effect (ACME) equals zero. Larger absolute values indicate that stronger unobserved mediator–outcome confounding would be required to explain away the mediated effect. The Pro Trump models use a probit outcome specification; Trade Favorability and Anti Immigration use linear specifications. "Robots & AI" pools the two automation-risk treatments. Refers to results in Figure E.14.

E.2 Robustness of Model-Based Mediation on Survey



(a) Survey weighted (b) BH adjusted (c) Doubly robust

Figure E.15: Robustness of the estimated average causal mediation effects (ACMEs) across alternative specifications. *Note*: The figure reports model-based average direct effects (ADE) and average causal mediated effects (ACME). The treatment contrasts compare exposure to information about job-threatening displacement due to robots and/or artificial intelligence with the common control condition. The treatment estimates are presented in the following order: pooled Robots & AI (blue triangles), Robots (green circles), and AI (red squares). The outcomes are support for Trump, trade favorability, and anti-immigration attitudes. "Pro Trump" measures willingness to vote for Donald Trump if he runs again for president. "Trade Favorability" measures whether respondents believe increased trade with other countries has benefited American workers. "Anti Immigration" measures preferences regarding whether the federal government should increase, decrease, or maintain the number of legal immigrants admitted to the United States; higher values indicate a stronger preference for decreasing immigration. The mediators are displayed in the following order: Nostalgia, Marginalization, and Cultural Grievances. Nostalgia measures sentimentality toward the past and concern about the loss of traditions. Marginalization measures perceptions that people like the respondent no longer represent American identity, that their values are disrespected, and that they are treated poorly in society. Cultural Grievances is the composite formed by summing the Nostalgia and Marginalization indices. Thick horizontal lines report 90% confidence intervals and thin horizontal lines report 95% confidence intervals.

E.3 Model-Based Inference Survey: Sequence Marginalization-Nostalgia

Automation Threat → Marginalization → Nostalgia → Populism

Although the main analysis examines marginalization and nostalgia as separate dimensions of cultural grievance, they may also be sequentially related. Perceived marginalization may heighten a sense of lost status and belonging, making an idealized past more salient. To explore this possibility, I first estimate the three constituent links in the proposed sequence using separate regressions with the same pre-treatment controls as in the main specification. The resulting chain plot does not identify indirect effects, but it provides suggestive evidence consistent with the proposed sequence: exposure to automation risk increases marginalization, higher marginalization is associated with greater nostalgia, and higher nostalgia is associated with stronger populist support (Figure E.16). However, because marginalization and nostalgia were measured in the same post-treatment survey block, these estimates do not establish temporal ordering and should be interpreted as suggestive evidence.

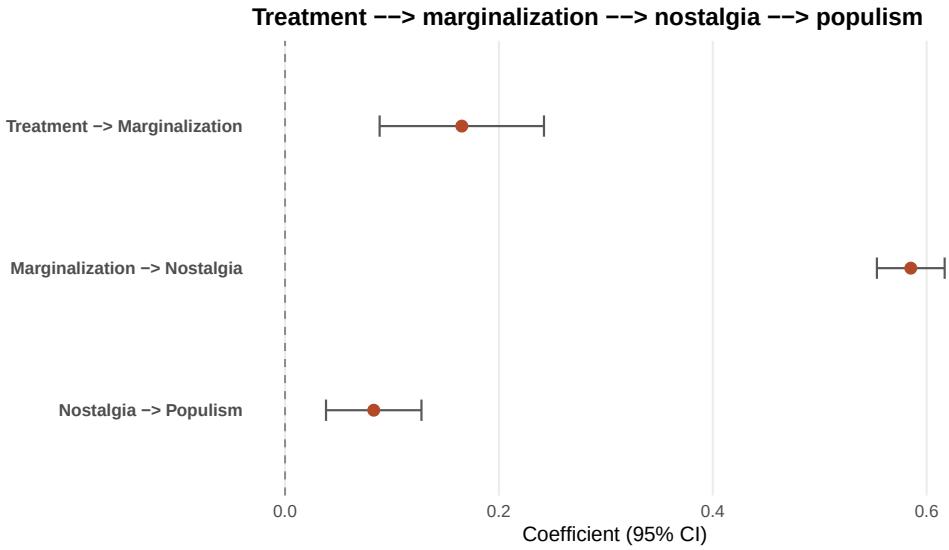


Figure E.16: Treatment → marginalization → nostalgia → populism

As an additional exploratory exercise, I then estimate the observable sequential pathway from treatment to populist attitudes using a serial mediation model[28]. The model treats exposure to automation risk as the exogenous treatment, marginalization as the first mediator, nostalgia as the second mediator, and populism as the outcome, while adjusting for the same pre-treatment controls used in the main specification. The results are consistent with the idea that the treatment first raises marginalization, which is then associated with higher nostalgia, and that nostalgia is in turn associated with greater populist support. The estimated treatment effect on marginalization is positive and statistically distinguishable from zero, the coefficient linking marginalization to nostalgia is large and positive, and the coefficient linking nostalgia to populism remains positive once marginalization is included in the outcome equation (Figure E.17).

The decomposition of effects suggests that most of the mediated relationship operates through marginalization, with a smaller but positive serial component running from marginalization to nostalgia and then to populism. At the same time, these estimates should be interpreted cautiously. Because marginalization and nostalgia are both measured in the same post-treatment survey block, this appendix does not fully identify temporal ordering between the two mediators. The serial mediation results are therefore best understood as suggestive evidence that the data are compatible with the sequence treatment → marginalization → nostalgia → populism, rather than as definitive proof of that ordering.

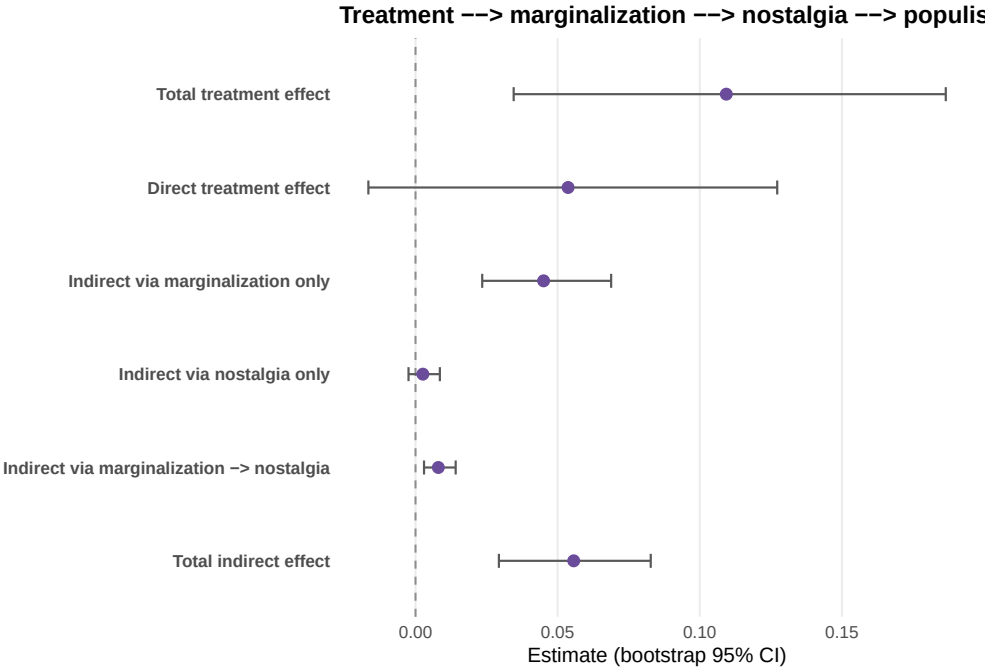


Figure E.17: Treatment → marginalization → nostalgia → populism

F Discussing Causal Mediation Analysis

F.1 Challenges

To test the mediated aspects of my theory, I must estimate the impact of encouraging the mediator on political attitudes. This task is not without challenges. The first assumption I must invoke is that the experimental encouragement of mediators exclusively impacts that specific mediator and not others (Bullock, Green, and Ha, 2010). In my case, I relied on previous work in social psychology to design the encouragement of the mediators, which I do via reflective tasks on the topics (e.g., Xia, Wang, and Santana, 2021; van Tilburg, Sedikides, and Wildschut, 2015; Newman et al., 2020; Bhattacharya, 2020). My initial approach involves estimating the relationship between M and Y.

However, a second challenge emerges as the encouragement may not impact the entire sample (Bullock, Green, and Ha, 2010; Pirlott and MacKinnon, 2016). Addressing this challenge requires accounting for non-compliers due to potential biases that may arise when recipients of the encouragement deviate from the initial assignment (Angrist, Imbens, and Rubin, 1996; Balke and Pearl, 1997). Compliers are individuals for whom the value of M changes in the expected direction upon receiving encouragement.

To address this challenge, my first attempt follows Balke and Pearl (1997), who proposed non-parametric solutions known as sharp bounds. However, these bounds proved uninformative, and to narrow them down, a set of additional assumptions for this case would be needed (Imai et al., 2011; Blackwell et al., 2023; Knox, Lowe, and Mummolo, 2020). Hence, I implement an alternative solution by adopting the 2SLS approach

[28]Using the R package `lavaan`.

(Angrist, Imbens, and Rubin, 1996) and enhancing it by incorporating a detailed compliance measurement that I formulated, given that my encouragement took the form of a task. I assessed this by observing task completion and evaluating the compliance with the prompt and the intensity. By adopting this strategy, we can better estimate the encouragement's effects, reducing biases associated with imperfect compliance and establishing a metric that considers high mediator levels and task completion.

Treatment (and encouragement) effect heterogeneity can undermine the validity of the product-of-coefficients approach to estimating the mediated causal effect (Bullock, Green, and Ha, 2010; Imai et al., 2011; Pirlott and MacKinnon, 2016). Essentially, this heterogeneity means that the effects of X on M and M on Y do not remain consistent across all participants. Imai et al. (2011) proposes a solution that encourages low and high mediator levels and employs non-parametric bounds. However, this approach does not provide point estimates but rather offers bounds, which are often uninformative (Bullock and Green, 2021), and require various levels for the mediator.[29]

To place trust in the results of the design-based experimental approach, one might consider adopting an assumption used in prior mediation studies: "monotonicity" (Bullock and Green, 2021; Knox, Lowe, and Mummolo, 2020). This assumption implies that treatments (and encouragement) consistently have either a nonnegative or nonpositive effect (i.e., there is no sign heterogeneity). In simpler terms, for all subjects i, either $M_i(1) \geq M_i(0)$ or $M_i(0) \leq M_i(1)$. However, it's crucial to acknowledge that researchers cannot empirically test this assumption. As Bullock and Green (2021) notes, "One must construct arguments grounded in theory to support them" (p.9).

This assumption of monotonicity implies that subjects facing the risk of automation would feel heightened nostalgia and a stronger sense of exclusion. The opposite response should not happen: exposure to the treatment should not result in lessened cultural grievances compared to the control group. Similarly, subjects encouraged with respect to the mediators (nostalgia and marginalization) should not exhibit less support for the populist right.

Based on these assumptions, once I estimate the relationships between X and M, as well as M and Y, I calculate the indirect effect using the product-of-coefficients. Additionally, I compute confidence intervals for the indirect effect estimates following Sobel (1982), which rely on the Delta method (see Appendix F.5 for further explanations).

Finally, I replicate the analysis as a measurement-of-mediation analysis using the observed values of M. This complements the manipulation-of-mediation analysis, and the convergence of results offers valuable insights into a mediation relationship, addressing concerns that encouraged mediators inherently become moderators (Pirlott and MacKinnon, 2016). The following sections present the results for each of these estimations.

F.2 Summary Design-Based Mediation: Contributions and Assumptions

Although the recognition of the importance of causal mechanisms has been growing, the political science domain lags behind fields like social psychology in this regard. While studies focusing on mechanisms do exist, they predominantly rely on model-based inference (measurement-of-mediation-design), either using purely observational analyses (e.g, Karpowitz, Mendelberg, and Shaker, 2012; Hays, Lim, and Spoon, 2019) or by randomizing only the treatment (Tomz and Weeks, 2020; Powers and Renshon, 2023; Young, 2019). In contrast, my research introduces a design-based analysis with randomization of the treatment and encouragement of the mediator. I have integrated the parallel encouragement design and adopted various recommendations to navigate the inherent challenges of these analyses:

1. I encouraged the specific mediators towards precise writing tasks, a method well-established in prior psychological studies.
2. To account for non-compliance post-mediator encouragement, I utilized the 2SLS approach, defining compliance indicators based on the extent of task completion and levels of the mediator. Notably, to my awareness, no prior research has integrated task completion into the compliance framework.
3. To prevent misconstruing the mediator as a moderator, I included a manipulation check for the mediators. I then scrutinized the outcomes using both the design-based and model-based approaches (i.e., measurement of mediators).
4. For potential heterogeneity effects, I meticulously delineated the conditions under which the findings are reliable, expressly identifying monotonicity as a foundational assumption.

While this research is not free of assumptions, every observational and experimental analysis carries implicit assumptions that often go unexamined. My design, in contrast, candidly discusses its assumptions, ensuring transparency throughout.

[29] In empirical simulations replicating Imai et al. (2011), these bounds were found to lack informativeness in some scenarios, and optimization was infeasible under certain random seeds.

F.3 Effects of M on Y, refers to Figure 5

2SLS models use randomized encouragement assignment, Z_i, as an instrument for the constructed uptake indicator, D_i. Their interpretation as local average treatment effects requires five conditions (Angrist, Imbens, and Rubin, 1996): (1) *relevance*, meaning that assignment changes the probability of uptake; (2) *independence*, supplied by successful random assignment; (3) *exclusion*, meaning that assignment affects each political outcome only through the defined uptake measure; (4) *monotonicity*, meaning that assignment does not reduce uptake for any respondent; and (5) consistency and no interference across respondents. Under these conditions, the coefficient identifies the average effect of encouraged uptake for respondents whose uptake is induced by assignment.

Some implications of these assumptions can be examined, but the central identifying restrictions cannot all be tested. Relevance is strongly supported by the covariate-adjusted first stage: encouragement increases uptake by 0.264 (SE = 0.012), with an excluded-instrument partial F-statistic of 477.39. Pretreatment balance tests in Appendix A.5 assess whether the randomization was implemented successfully, while the reduced-form ITT estimates in Figure 5 remain causally identified by assignment. Exclusion cannot be established statistically because the writing task could affect outcomes through reflection, salience, demand, or emotions not captured by the binary uptake measure; monotonicity is likewise a counterfactual assumption and cannot be tested directly. Because the specification has one instrument for one endogenous variable, it is exactly identified. The 2SLS results should therefore be read as complier effects.

	Dependent variable:			
	Pro Trump	Trade Favorability	Anti Immigration	Robots Concern
	(1)	(2)	(3)	(4)
Mediator Encouraged	0.102*	−0.131*	0.107*	−0.013
	(0.062)	(0.067)	(0.061)	(0.050)
Female	−0.020	−0.125***	0.028*	0.014
	(0.016)	(0.018)	(0.016)	(0.013)
White	0.087	0.015	−0.083	−0.033
	(0.139)	(0.151)	(0.136)	(0.111)
Education: highschool	0.030	0.069	−0.131	−0.031
	(0.138)	(0.150)	(0.135)	(0.110)
Education: incomplete college	0.035	0.073	−0.105	−0.032
	(0.138)	(0.151)	(0.136)	(0.111)
Education: technical	−0.025	0.125	−0.202	−0.044
	(0.138)	(0.150)	(0.135)	(0.110)
Education: college	−0.030	0.172	−0.243*	−0.078
	(0.139)	(0.151)	(0.136)	(0.111)
Education: gradschool	0.222	0.047	−0.448**	−0.290*
	(0.210)	(0.228)	(0.206)	(0.168)
Education: other	0.059***	−0.058***	0.060***	0.001
	(0.018)	(0.019)	(0.018)	(0.014)
RTI	0.001	−0.0001	0.002*	−0.0002
	(0.001)	(0.001)	(0.001)	(0.001)
Income dummy	Yes	Yes	Yes	Yes
Treat dummy	Yes	Yes	Yes	Yes
First-stage partial F	477.39	477.39	477.39	477.39
Observations	3,133	3,133	3,133	3,133
R^2	0.036	0.047	0.052	0.009
Note:				*p<0.1; **p<0.05; ***p<0.01

Table F.18: 2SLS Estimates of Encouraged Cultural-Grievance Uptake, contains models of Figure 5

Note: This table reports the 2SLS models corresponding to the red-square estimates in Figure 5. The endogenous variable is mediator uptake, defined as successful completion of the assigned writing task together with a cultural-grievance level above the main compliance threshold. Random assignment to the cultural-grievance encouragement is used as the instrument for uptake. The sample comprises 3,133 respondents, of whom 1,816 were assigned to encouragement and 1,317 were not. In the covariate-adjusted first stage, encouragement assignment increases uptake by 0.264 (SE = 0.012); the excluded-instrument partial F-statistic is 477.39. Under random assignment, exclusion, relevance, and monotonicity, the coefficients identify local average treatment effects of encouraged uptake among respondents whose uptake is induced by assignment; they do not identify an assumption-free population-average effect of cultural grievances. All outcomes are binary, so a coefficient of 0.10 represents a 10-percentage-point change. “Pro Trump” measures willingness to vote for Donald Trump; “Trade Favorability” indicates whether respondents view increased trade as beneficial to American workers; “Anti Immigration” captures preferences for reducing the number of legal immigrants admitted to the United States; and “Robots Concern” summarizes perceived risks from robots and artificial intelligence and serves as a manipulation check. All models adjust for gender, race, income, education, occupational RTI, and automation-treatment assignment. Standard errors are reported in parentheses.

Additionally, Figure F.18 shows that the encouragements worked as intended, increasing nostalgia and marginalization in the expected directions.

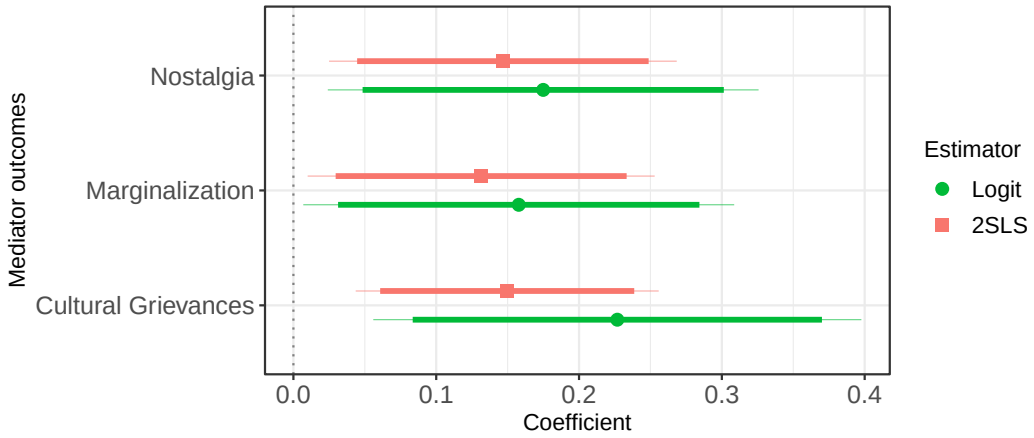


Figure F.18: Manipulation check: effects of the cultural-grievance encouragement on the mediator outcomes. Same models as in Figure 5.

Note: The positive estimates indicate that respondents assigned to the encouragement condition exhibited higher levels of nostalgia, marginalization, and the combined cultural-grievances measure. Green circles report raw logit coefficients for randomized encouragement (`encPD`). Red squares report 2SLS estimates for complier status, instrumented by randomized encouragement. All models adjust for sex, race, income, education, occupational routine-task intensity, and automation-treatment assignment. Thick and thin intervals represent 90% and 95% confidence intervals, respectively.

F.4 Robustness of the 2SLS Estimates to Alternative Definitions of Compliance

Because compliance with the encouragement is not directly observed, I examine whether the 2SLS estimates in Figure 5 depend on how compliance is defined. I consider four definitions:

1. *Subgroup-standardized mediator response and hand coding*: a respondent has a high mediator response when either nostalgia or marginalization exceeds a subgroup-specific cutoff, and hand coders determine that the response followed the prompt. Discussed in Appendix F.4.1.
2. *Subgroup-standardized mediator response and word count*: the same mediator criterion is combined with writing more than a specified number of words, replacing hand coding as the measure of task effort. Discussed in Appendix F.4.2.
3. *Combined mediator and effort criteria*: a high combined cultural-grievance score is required together with both hand-coded prompt adherence and a minimum word count. Discussed in Appendix F.4.3.
4. *Hand coding only*: compliance is defined solely by whether the written response followed the prompt. Discussed in Appendix F.4.4.

F.4.1 Alternative cutoffs for a high mediator response. To account for baseline differences in cultural grievances, I divide respondents into eight pretreatment strata defined by race (White or non-White), education (college or no college), and gender (female or male). For mediator k—nostalgia or marginalization—respondent i's score is standardized relative to the corresponding unencouraged reference group g:

$$z_{ikg} = \frac{M_{ik} - \overline{M}_{kg,0}}{s_{kg,0}},$$

where $\overline{M}_{kg,0}$ and $s_{kg,0}$ are the reference-group mean and standard deviation. Thus, $z = 1$ means that the mediator is one subgroup-specific standard deviation above the reference mean. A mediator response is classified as high when either standardized mediator exceeds cutoff c. Compliance additionally requires hand-coded prompt adherence. Figures F.19–F.20 vary c from 0.18 to 1.962; larger values define smaller, more selective complier groups. The direction and inferential interpretation of the estimates remain stable, although their absolute magnitudes increase under the stricter definitions.

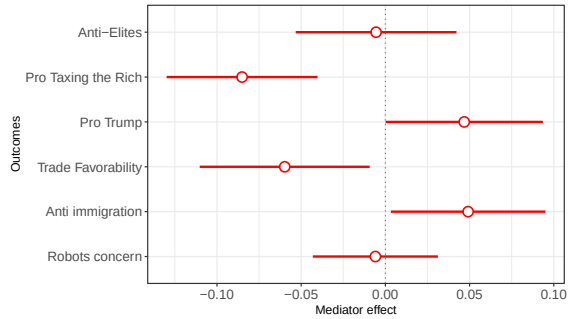


Figure F.19: Least restrictive cutoff: $c = 0.18$.

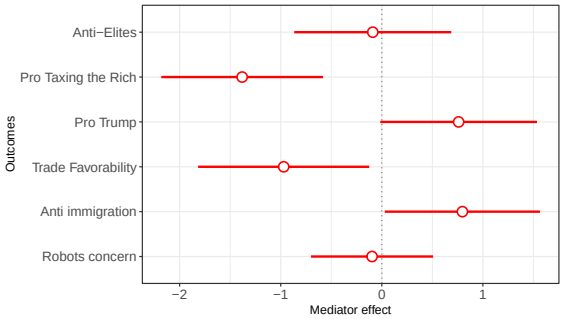


Figure F.20: Most restrictive cutoff: $c = 1.962$.

Notes: Circles are covariate-adjusted 2SLS point estimates, horizontal lines are 90% confidence intervals, and the dashed line denotes zero. Positive (negative) estimates indicate a higher (lower) probability of the outcome coded one. For binary outcomes, a coefficient of 0.102 corresponds to 10.2 percentage points. Horizontal scales differ across panels. The sample contains 3,133 respondents: 1,816 assigned to encouragement and 1,317 to no encouragement.

F.4.2 Task effort measured by word count. The second definition replaces the hand-coded measure with word count as an objective proxy for task effort. I consider thresholds of more than 38, 58, and approximately 88 words, corresponding to the 25th, 50th, and 75th percentiles. Figure F.21 shows the median specification: a subgroup-standardized mediator score above $c = 1.036$ and more than 58 words. The alternative word-count thresholds produce the same qualitative pattern.

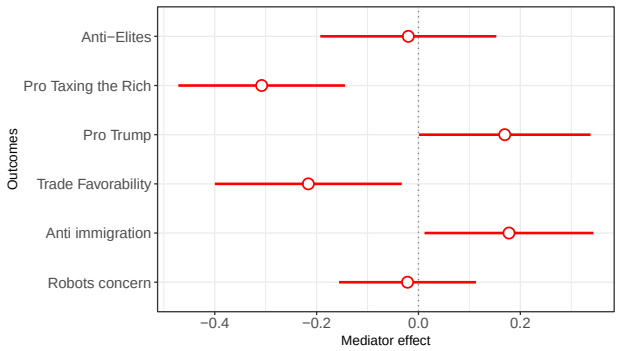


Figure F.21: Compliance defined by a high mediator response ($c = 1.036$) and writing more than the median of 58 words.

Notes: Circles are covariate-adjusted 2SLS estimates, horizontal lines are 90% confidence intervals, and the dashed line denotes zero. Random assignment to encouragement is the instrument.

F.4.3 Combined mediator and effort criteria. The third definition classifies the mediator response as high when the sum of nostalgia and marginalization exceeds 23. It also requires both hand-coded prompt adherence and more than 58 words. Figure F.22 shows that this more restrictive definition yields the same qualitative pattern, but for a narrower complier population.

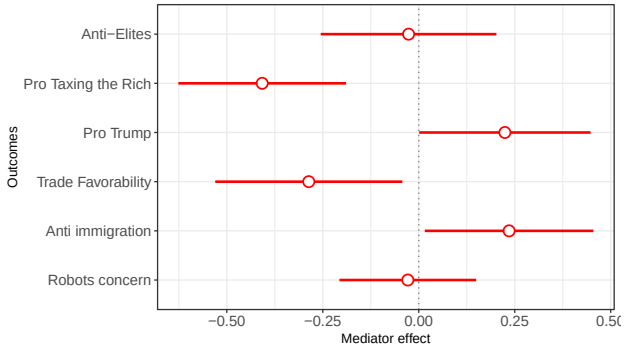


Figure F.22: Compliance defined by a combined cultural-grievance score above 23, hand-coded prompt adherence, and more than 58 words.

Notes: Circles are covariate-adjusted 2SLS estimates, horizontal lines are 90% confidence intervals, and the dashed line denotes zero. Random assignment to encouragement is the instrument.

F.4.4 Compliance based only on hand-coded prompt adherence. The final definition uses only the hand-coded assessment that the respondent followed the encouragement prompt, without imposing a

mediator or word-count threshold. As shown in Figure F.23, the estimates retain the same directions but are smaller in absolute magnitude.

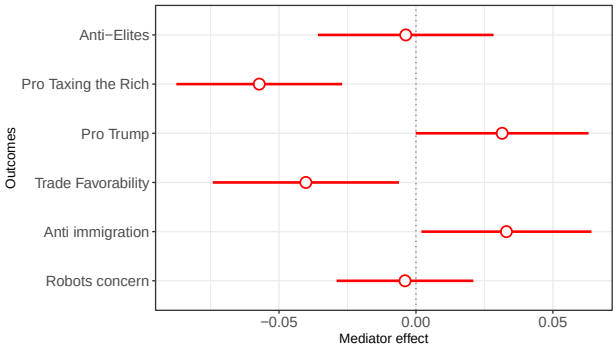


Figure F.23: Compliance defined solely by hand-coded adherence to the encouragement prompt.

Notes: Circles are covariate-adjusted 2SLS estimates, horizontal lines are 90% confidence intervals, and the dashed line denotes zero. Random assignment to encouragement is the instrument.

F.5 Causal Chain and Delta Method - Average Causal Mediation Analysis

After estimating the effects of the treatment on the mediators, and the effects of the mediators' encouragement on the political outcomes (2SLS), I use the Delta Method in order to approximate the probability distribution for the multiplication of two parameters. Let p_1 be the effects of the treatment on mediators, and p_2 be the effects of the encouragement of the mediator, then the causal mediation effect $(Y_i(t, M_i(1)) - Y_i(t, M_i(0)))$ can be estimated under the assumption of homogeneous treatment effects as $p^1 \times p^2$. Then, we can use the **Delta method** to calculate the standard error of $p_1 p_2$. This method states that an approximation of the variance of a function $g(t)$ is given by:

$$\text{Var}(g(t)) \approx \sum_{i=1}^{k} [g_i'(\theta)^2 \text{Var}(t_i)] + 2 \sum_{i>j} g_i'(\theta) g_j'(\theta) \text{Cov}(t_i, t_j).$$

The estimation of the anticipated value of $g(t)$ is expressed as:

$$\mathbb{E}(g(t)) \approx g(\theta).$$

Hence, the expectation corresponds simply to the function, $g(t)$ is defined as $g(p_1, p_2) = p_1 p_2$. Consequently, the expected value of $g(p_1, p_2) = p_1 p_2$ would directly yield $p_1 p_2$. For the computation of variance, it becomes necessary to evaluate the partial derivatives of $g(p_1, p_2)$:

$$\frac{\partial}{\partial p_1} g(p_1, p_2) = p_1 \qquad \frac{\partial}{\partial p_2} g(p_1, p_2) = p_2$$

Thus we get:

$$\text{Var}(p_1 p_2) = p_2^2 \text{Var}(p_1) + p_1^2 \text{Var}(p_2) + 2 \cdot p_1 p_2 \text{Cov}(p_1, p_2)$$

$$\text{SE}(p_1 p_2) = \sqrt{p_2^2 \text{Var}(p_1) + p_1^2 \text{Var}(p_2^2) + 2 \cdot p_1 p_2 \text{Cov}(p_1, p_2)}$$

Finally, we can estimate the uncertainty around $p_1 p_2$ using CI formula, and the estimated $\text{SE}(p_1 p_2)$

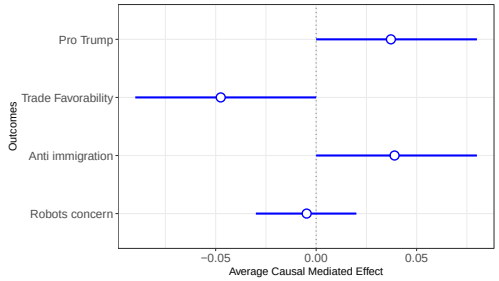


Figure F.24: Causal Mediation Effects

Note: Compliers defined as 1) high level of the mediator relative to those non-encouraged (t=1.036), and 2) task completion as hand-coders.

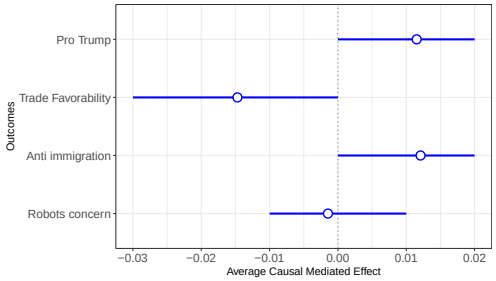


Figure F.26: Causal Mediation Effects

Note: Compliers defined as 1) high level of the mediator relative to median level in the sample, and 2) task completion as more than median words.

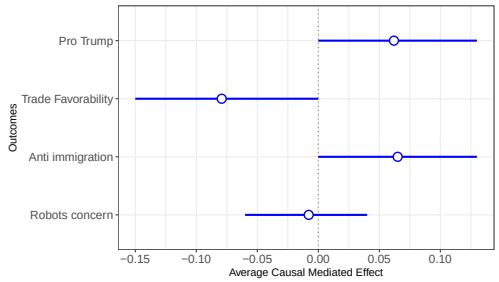


Figure F.25: Causal Mediation Effects

Note: Compliers defined as 1) high level of the mediator relative to those non-encouraged (t=1.036), and 2) task completion as hand-coders & more than median words.

Note: These figures (F.24-F.26) show the average mediated causal effect. N = 3,133, with 1,816 receiving encouragement and 1,317 not. Moreover, 1,951 received either robot or AI treatment and 1,182 the control. The estimates result from the product of coefficients (from the effects of the treatment on cultural grievances) and the effects of the encouragement on the outcomes. SE comes from Delta method.

F.6 Mediation vs moderation

Mediation and moderation target different causal quantities. Following the notation in the estimation strategy, X_i denotes respondent i's randomized assignment to automation-risk exposure, M_i denotes the post-treatment measure of cultural grievances, Z_i denotes the independently randomized grievance encouragement, and Y_i denotes a political outcome. Mediation concerns the sequential pathway $X_i \rightarrow M_i \rightarrow Y_i$: a represents the effect of automation-risk exposure on cultural grievances, and the downstream component b is inferred using the mediator encouragement under the identifying assumptions discussed in Appendix F. The product ab is therefore a pathway estimate under those assumptions, not a quantity identified by randomization alone. Moderation, by contrast, concerns treatment-effect heterogeneity—whether the effect of one variable changes across levels of another.

Regarding moderation, within each two-arm automation contrast, a factorial regression of Y_i on X_i, Z_i, and $X_i \times Z_i$ validly estimates two intention-to-treat main effects and a randomized interaction. The interaction is the difference between the encouragement effect under automation exposure and the encouragement effect under control—equivalently, a difference-in-differences across the four experimental cells. Thus, $X_i \times Z_i$ is a valid estimand of heterogeneity in the effect of encouragement assignment. Because Z_i is an encouragement rather than the mediator itself, however, this interaction is not the mediated effect ab.

The factorial interaction can consequently be informative without adjudicating the mediation claim. It can equal zero when $ab \neq 0$ if automation changes grievances and the downstream effect induced by encouragement is constant across automation conditions. Conversely, it can be nonzero because automation changes task uptake, because encouragement has salience or emotional effects not captured by the measured mediator, or because the downstream response to encouragement differs across automation conditions. None of these possibilities alone establishes the theorized $X_i \rightarrow M_i \rightarrow Y_i$ sequence. A null $X_i \times Z_i$ interaction therefore does not refute mediation, and a nonzero interaction does not establish it.

Accordingly, I do not treat mediation and moderation as a statistical "horse race." Comparing coefficient magnitudes or statistical significance across regressions containing the post-treatment mediator cannot determine which process is the better explanation and may introduce post-treatment bias (Agnolin, Colantone, and Stanig, 2024). The causal interpretation of each reported quantity remains distinct: the encouragement ITT is the effect of Z_i assignment; the 2SLS coefficient is a complier effect; and ab is the design-based pathway estimate under the additional assumptions needed to connect the encouragement-induced change in grievances to the change naturally induced by automation. A separate moderation claim concerning pre-existing grievances would require an appropriate pretreatment measure and would address a different question. Future studies could, for example, measure cultural grievances before treatment and examine whether they moderate the effects of the automation treatment.

G Study 2, Observational Evidence for Exposure to Automation on Politics and Mediators

G.1 Refers to Table 1

Following Table 1 with control variables.

	Nostalgia (Hyp. I)		Marginalization (Hyp. I)			Political Behavior (Hyp. II)
	(1)	(2)	(3)	(4)	(5)	(6)
	Life Better	Hopeful	Culture	Economy	Live	Radical Right
Frey & Osborne	-0.717***	-0.742***	-2.355***	-2.301***	-1.964***	3.560***
	(0.052)	(0.059)	(0.099)	(0.093)	(0.093)	(0.234)
Education (years)	0.009***	0.008***	0.079***	0.076***	0.059***	-0.057***
	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)	(0.007)
Age	-0.005***	-0.007***	-0.011***	-0.002***	-0.009***	-0.011***
	(0.000)	(0.000)	(0.001)	(0.001)	(0.001)	(0.001)
Female	-0.106***	-0.080***	0.059***	-0.290***	-0.031*	-0.428***
	(0.010)	(0.011)	(0.017)	(0.015)	(0.016)	(0.038)
Ethnic minority	0.007	0.079***	-0.740***	-0.716***	-0.886***	0.672***
	(0.024)	(0.030)	(0.045)	(0.046)	(0.038)	(0.161)
Constant	3.186***	2.955***	7.427***	5.791***	6.007***	-5.709***
	(0.072)	(0.272)	(0.531)	(0.189)	(0.143)	(0.489)
Country-Year FE	✓	✓	✓	✓	✓	✓
NUTS FE	✓	✓	✓	✓	✓	✓
Observations	44326	44571	150245	149680	150516	63136
R^2 (p)	0.294	0.134	0.166	0.120	0.143	0.178
AIC	1.1e+05	1.2e+05	6.7e+05	6.6e+05	6.4e+05	2.6e+04

Standard errors clustered by region-year in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table G.19: Automation, cultural attitudes, and vote choices.

Note: This figure represents the effects of exposure to automation on the outcome of interest (support for right-wing populism) and mediators. The independent variable is exposure to automation approached following Frey, Berger, and Chen (2017) Dependent variable: (1) Support for populist radical right. (2-4) Level of agreement with immigration as better for culture, economy and life. Answers range from "Not like me at all" ($= 0$) to "Very much like me" ($= 10$). (5-6) Level of agreement with "life is getting worse" and "hard to have hope about the future." Answers range from "Agree strongly" ($= 1$) to "Disagree strongly" ($= 5$). Source: ESS (1-7) data.

G.2 Study 2, Additional Tables: Robustness checks associated with Table 1.

The findings remain consistent across multiple alternative specifications. Initially, I include an extra regional-level predictor—changes in robot exposure measured by the variation in robot quantity per thousand workers over the past three years, using data from the International Federation of Robotics and methodology from Anelli, Colantone, and Stanig (2021). Table G.20 confirms that both individual risk exposure and regional robot presence are statistically significant.

Then, I add several control variables to the previous models that the literature on voting behavior suggests may affect vote choice and individual attitudes (e.g. Frey, Berger, and Chen, 2017; Gingrich, 2019; Thewissen and Rueda, 2019; Hays, Lim, and Spoon, 2019). At the individual level, I add dummy variables for being foreign-born, living in a city (urban), and being an ethnic minority. Then, I incorporate three dummy variables representing respondents' experience in the labor market: i) unemployed, ii) union membership, and iii) limited employment contract, which reflects some degree of precariousness in the respondent's linkages with the job. (The results remain the same if I exclude employment variables, which arguably may also be post-treatment.) Finally, I incorporate into the models two additional regional-level variables: i) immigrant exposure, proxied as the proportion of foreign-born respondents in the region, and ii) regional unemployment, calculated as the share of unemployed respondents in the region. I expect respondents in regions with high unemployment and immigrants to be more likely to hold anti-immigration attitudes, nostalgic sentiments, and support populist right candidates. Regarding the regional-level variables, while unemployment may increase anti-immigrant propensity and nostalgic views, the expectations regarding immigration exposure are less clear. Previous scholars have argued that it can either decrease outgroup threat predispositions or exacerbate them (Inglehart, 2018; Norris, 2004). The results remain similar across these model specifications (see Table G.21)

To assess i) whether current occupations may mask past automation dynamics (e.g., a worker that has already been displaced) and ii) the interaction of individual and regional exposure, I re-estimate previous models relying on the measure proposed by Anelli, Colantone, and Stanig (2021) as the independent variable. It is based on the predicted probabilities for an individual to be occupied in high-automatability occupations and the incorporation of robots in an individual's region. Tables G.22, and G.23 (with and without control variables) show the estimations. The results remain unchanged. A one-SD increase in individual exposure to automation leads to a decrease of about 0.31–0.26 units in pro-immigration predisposition (11 points-scale) and a decrease of about 0.14–0.12 in nostalgic sentiments (5 points-scale).

As a further robustness check, Table G.24 replaces the earlier automation measures with occupational suitability for machine learning (SML), a more AI-relevant indicator of technological exposure (Brynjolfsson, Mitchell, and Rock, 2019). All associations are statistically significant at the 1% level. These results indicate that the main findings extend beyond measures centered on robots or computerization to occupational exposure more directly related to machine learning.

To examine the link between cultural grievances and voting choices, I start with logistic regression models explaining voting choice by automation risk, mediators, and other demographic controls (refer to Table G.25).(Comparable models were replicated in Tables G.26, utilizing robot adoption in different countries as an instrumental variable for robot exposure (Anelli, Colantone, and Stanig, 2021). The results remained consistent.) All estimated coefficients displayed statistical significance and aligned with the expected direction. In the subsequent section, I further analyze this relationship by conducting a causal mediation analysis. This analysis treats exposure to automation risk as the treatment, cultural grievances (marginalization and nostalgia) as mediators, and support for the radical right as the outcome variable.

	Nostalgia (Hyp. I)		Marginalization (Hyp. I)			Political Behavior (Hyp. II)
	(1)	(2)	(3)	(4)	(5)	(6)
	Life Better	Hopeful	Culture	Economy	Live	Radical Right
Frey & Osborne	-0.717***	-0.734***	-2.375***	-2.315***	-1.987***	3.505***
	(0.052)	(0.058)	(0.100)	(0.094)	(0.094)	(0.231)
Regional Δ robots	-0.259***	-0.213**	-0.381**	-0.333*	-0.391**	1.042*
	(0.088)	(0.096)	(0.168)	(0.181)	(0.155)	(0.589)
Education (years)	0.008***	0.008***	0.076***	0.074***	0.056***	-0.057***
	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)	(0.007)
Age	-0.005***	-0.006***	-0.012***	-0.003***	-0.011***	-0.011***
	(0.000)	(0.000)	(0.001)	(0.001)	(0.001)	(0.001)
Female	-0.105***	-0.080***	0.052***	-0.295***	-0.038**	-0.426***
	(0.010)	(0.011)	(0.018)	(0.015)	(0.016)	(0.037)
Country-Year FE	✓	✓	✓	✓	✓	✓
NUTS FE	✓	✓	✓	✓	✓	✓
Observations	44674	44923	151296	150778	151615	64440
R^2 (p)	0.294	0.134	0.162	0.116	0.136	0.175
AIC	1.2e+05	1.3e+05	6.7e+05	6.6e+05	6.5e+05	2.7e+04

Standard errors clustered by region-year in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table G.20: Individual and regional exposure to automation, cultural attitudes, and vote choices.
Dependent variable: A) Support for populist radical right; B) level of agreement with immigration as better for culture, economy and life. Answers range from "Not like me at all" ($= 1$) to "Very much like me" ($= 10$). C) level of agreement with "life is getting worse" and "hard to have hope about the future." Answers range from "Agree strongly" ($= 1$) to "Disagree strongly" ($= 5$). Source: ESS (1-7) data.

	Nostalgia (Hyp. I)		Marginalization (Hyp. I)			Political Behavior (Hyp. II)
	(1)	(2)	(3)	(4)	(5)	(6)
	Life Better	Hopeful	Culture	Economy	Live	Radical Right
Frey & Osborne	-0.800***	-0.757***	-2.298***	-2.339***	-1.944***	3.640***
	(0.059)	(0.063)	(0.111)	(0.104)	(0.107)	(0.282)
Regional Δ robots	-0.242***	-0.117	-0.408**	-0.440**	-0.506***	-0.100
	(0.081)	(0.125)	(0.174)	(0.179)	(0.151)	(0.739)
Education (years)	0.014***	0.011***	0.083***	0.081***	0.065***	-0.064***
	(0.003)	(0.002)	(0.004)	(0.004)	(0.003)	(0.008)
Age	-0.004***	-0.007***	-0.008***	0.001	-0.006***	-0.012***
	(0.000)	(0.000)	(0.001)	(0.001)	(0.001)	(0.002)
Female	-0.114***	-0.085***	0.082***	-0.312***	-0.044**	-0.404***
	(0.011)	(0.012)	(0.020)	(0.017)	(0.019)	(0.044)
Urban	-0.006	-0.018	0.164***	0.133***	0.132***	-0.087*
	(0.016)	(0.016)	(0.021)	(0.022)	(0.018)	(0.053)
Union Member	-0.069***	-0.055***	0.124***	0.057**	0.055***	-0.096
	(0.015)	(0.014)	(0.021)	(0.022)	(0.019)	(0.060)
Unemployed	-0.206***	-0.206***	-0.217***	-0.413***	-0.311***	0.334**
	(0.034)	(0.037)	(0.044)	(0.042)	(0.041)	(0.145)
Ethnic minority	0.044	0.076**	-0.436***	-0.360***	-0.480***	0.678***
	(0.029)	(0.033)	(0.052)	(0.047)	(0.045)	(0.201)
Foreign Born	0.083***	0.007	0.483***	0.608***	0.702***	-0.326***
	(0.024)	(0.028)	(0.043)	(0.040)	(0.036)	(0.112)
Precarious employment contract	-0.010	-0.062***	0.054**	0.055**	0.073***	-0.064
	(0.015)	(0.018)	(0.024)	(0.022)	(0.020)	(0.073)
Regional Immigrant Exposure	-0.807	-1.767***	2.366***	-0.066	0.720	0.135
	(0.851)	(0.679)	(0.589)	(0.482)	(0.540)	(1.489)
Regional Unemployment	1.762	2.027	0.381	-0.364	0.635	8.605***
	(1.254)	(1.289)	(0.971)	(0.961)	(0.901)	(3.282)
Country-Year FE	✓	✓	✓	✓	✓	✓
NUTS FE	✓	✓	✓	✓	✓	✓
Observations	31401	31567	108641	108113	108561	48186
R^2 (p)	0.294	0.141	0.178	0.136	0.158	0.180
AIC	8.0e+04	8.8e+04	4.8e+05	4.7e+05	4.6e+05	2.0e+04

Standard errors clustered by region-year in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table G.21: Individual and regional exposure to automation, cultural attitudes, and vote choices (with additional control variables).
Dependent variable: A) Support for populist radical right; B) level of agreement with immigration as better for culture, economy and life. Answers range from "Not like me at all" ($= 1$) to "Very much like me" ($= 10$). C) level of agreement with "life is getting worse" and "hard to have hope about the future." Answers range from "Agree strongly" ($= 1$) to "Disagree strongly" ($= 5$). Source: ESS (1-7) data.

	Political Behavior	Immigration (Hyp. I)			Nostalgia (Hyp. II)	
	Radical Right	Culture	Economy	Live	Life Better	Hopeful
Sd Individual Exposure	0.857***	-0.314***	-0.332***	-0.261***	-0.144***	-0.125***
	(0.112)	(0.042)	(0.042)	(0.039)	(0.022)	(0.024)
Education (years)	-0.101***	0.116***	0.111***	0.089***	0.019***	0.020***
	(0.008)	(0.004)	(0.004)	(0.003)	(0.002)	(0.002)
Age	-0.013***	-0.009***	-0.001	-0.008***	-0.004***	-0.006***
	(0.002)	(0.001)	(0.001)	(0.001)	(0.000)	(0.000)
Female	-0.427***	0.063***	-0.285***	-0.028*	-0.102***	-0.076***
	(0.038)	(0.018)	(0.015)	(0.017)	(0.010)	(0.011)
Country-Year FE	✓	✓	✓	✓	✓	✓
NUTS FE	✓	✓	✓	✓	✓	✓
Observations	64440	151296	150778	151615	44674	44923
$R^2(p)$	0.168	0.155	0.109	0.130	0.291	0.130
AIC	2.8e+04	6.7e+05	6.7e+05	6.5e+05	1.2e+05	1.3e+05

Standard errors clustered by region-year in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table G.22: Automation, cultural attitudes, and vote choices. Using alternative proxy automation.
Independent variable: Standardized individual-level robot exposure proposed by (Anelli, Colantone, and Stanig, 2021). Dependent variable: A) Support for populist radical right; B) level of agreement with immigration as better for culture, economy and life. Answers range from "Not like me at all" ($= 1$) to "Very much like me" ($= 10$). C) level of agreement with "life is getting worse" and "hard to have hope about the future." Answers range from "Agree strongly" ($= 1$) to "Disagree strongly" ($= 5$). Source: ESS (1-7) data.

	Nostalgia (Hyp. I)		Marginalization (Hyp. I)			Political Behavior (Hyp. II)
	(1)	(2)	(3)	(4)	(5)	(6)
	Life Better	Hopeful	Culture	Economy	Live	Radical Right
Sd Individual Exposure	-0.135***	-0.127***	-0.269***	-0.306***	-0.238***	0.908***
	(0.025)	(0.026)	(0.045)	(0.046)	(0.042)	(0.137)
Education (years)	0.027***	0.023***	0.124***	0.122***	0.099***	-0.111***
	(0.003)	(0.002)	(0.004)	(0.004)	(0.004)	(0.008)
Age	-0.004***	-0.006***	-0.006***	0.003***	-0.004***	-0.014***
	(0.000)	(0.000)	(0.001)	(0.001)	(0.001)	(0.002)
Female	-0.108***	-0.079***	0.100***	-0.295***	-0.029	-0.413***
	(0.011)	(0.012)	(0.020)	(0.017)	(0.019)	(0.044)
Urban	-0.008	-0.019	0.169***	0.137***	0.135***	-0.104*
	(0.016)	(0.016)	(0.021)	(0.023)	(0.018)	(0.055)
Union Member	-0.065***	-0.051***	0.138***	0.070***	0.066***	-0.113*
	(0.015)	(0.015)	(0.023)	(0.023)	(0.020)	(0.065)
Unemployed	-0.211***	-0.211***	-0.233***	-0.430***	-0.325***	0.345**
	(0.034)	(0.038)	(0.044)	(0.042)	(0.041)	(0.144)
Ethnic minority	0.040	0.073**	-0.438***	-0.364***	-0.483***	0.655***
	(0.029)	(0.033)	(0.052)	(0.048)	(0.046)	(0.201)
Foreign Born	0.088***	0.012	0.490***	0.615***	0.708***	-0.317***
	(0.024)	(0.028)	(0.043)	(0.040)	(0.036)	(0.112)
Precarious employment contract	-0.016	-0.067***	0.037	0.038*	0.059***	-0.032
	(0.015)	(0.018)	(0.024)	(0.022)	(0.020)	(0.073)
Regional Immigrant Exposure	-0.878	-1.802***	2.503***	0.088	0.861	0.055
	(0.829)	(0.672)	(0.587)	(0.490)	(0.548)	(1.544)
Regional Unemployment	1.759	2.024	0.135	-0.618	0.402	9.073***
	(1.217)	(1.254)	(1.007)	(0.992)	(0.926)	(3.366)
Country-Year FE	✓	✓	✓	✓	✓	✓
NUTS FE	✓	✓	✓	✓	✓	✓
Observations	31401	31567	108641	108113	108561	48186
R^2 (p)	0.290	0.138	0.171	0.128	0.152	0.173
AIC	8.1e+04	8.8e+04	4.8e+05	4.7e+05	4.6e+05	2.1e+04

Standard errors clustered by region-year in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table G.23: Automation, cultural attitudes, and vote choices. Using alternative proxy for automation (with additional control variables).
Independent variable: Standardized individual-level robot exposure proposed by (Anelli, Colantone, and Stanig, 2021). Dependent variable: A) Support for populist radical right; B) level of agreement with immigration as better for culture, economy and life. Answers range from "Not like me at all" ($= 1$) to "Very much like me" ($= 10$). C) level of agreement with "life is getting worse" and "hard to have hope about the future." Answers range from "Agree strongly" ($= 1$) to "Disagree strongly" ($= 5$). Source: ESS (1-7) data.

	Nostalgia (Hyp. I)		Marginalization (Hyp. I)			Political Behavior (Hyp. II)
	Life Better	Hopeful	Culture	Economy	Live	Radical Right
Model:	(1)	(2)	(3)	(4)	(5)	(6)
	OLS	OLS	OLS	OLS	OLS	Logit
Variables						
SML	-0.1928***	-0.2235***	-0.5157***	-0.4463***	-0.4528***	0.5945***
	(0.0308)	(0.0337)	(0.0439)	(0.0416)	(0.0382)	(0.1058)
Country-Year FE	✓	✓	✓	✓	✓	✓
NUTS FE	✓	✓	✓	✓	✓	✓
Fit statistics						
Observations	36,931	37,097	121,348	120,893	121,453	53,526
R2(p)	0.29901	0.13007	0.16607	0.11749	0.14241	0.14382
AIC	95,244.0	104,697.6	536,861.6	531,704.5	517,825.5	23,225.5

Standard errors clustered by region-year in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table G.24: Effects of Occupational Suitability for Machine Learning on Cultural Grievances and Radical-Right Support.

Note: This table reports the association between occupational suitability for machine learning (SML), measured using the index developed by Brynjolfsson, Mitchell, and Rock (2019), and the three sets of outcomes corresponding to the study's hypotheses. The SOC-10 SML scores are mapped to ISCO-88 occupations for ESS rounds 1–5 and to ISCO-08 occupations for rounds 6–7 using the IBS occupational crosswalks. Columns 1–2 report the nostalgia outcomes: whether respondents believe that life is getting better and whether they feel hopeful about the future. These outcomes range from 1 to 5, with higher values indicating greater optimism. Columns 3–5 report the marginalization outcomes: evaluations of immigration's effects on the country's culture, economy, and overall quality of life. These outcomes range from 0 to 10, with higher values indicating more favorable attitudes toward immigration. Column 6 reports political behavior, measured as a binary indicator of voting for a radical-right party. Columns 1–5 report OLS coefficients, whereas Column 6 reports logit coefficients. All models use ESS survey weights and include demographic controls, country-year fixed effects, and NUTS-region fixed effects. Standard errors are clustered by region-year. Data are from ESS rounds 1–7.

Voting behavior explained by culture and automation

	(1)	(2)	(3)	(4)	(5)	(6)
	Baseline	Culture	Imm Eco	Imm Worse Life	Life Better	Hopeful
Frey & Osborne	3.587***	2.829***	2.817***	2.991***	3.905***	4.100***
	(0.337)	(0.359)	(0.363)	(0.356)	(0.814)	(0.793)
Pro-Immigration Culture		-0.361***				
		(0.017)				
Pro-Immigration Economy			-0.343***			
			(0.019)			
Pro-Immigration General				-0.415***		
				(0.022)		
Non-Nostalgic: Life Getting Better					-0.476***	
					(0.064)	
Non-Nostalgic: Hopeful Future						-0.137***
						(0.044)
Country-Year FE	✓	✓	✓	✓	✓	✓
NUTS FE	✓	✓	✓	✓	✓	✓
Observations	21889	21675	21592	21633	8655	8696
R^2 (p)	0.131	0.207	0.196	0.211	0.169	0.153
AIC	1.0e+04	9271.058	9383.128	9219.355	3967.830	4060.569

Standard errors clustered by region-year in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table G.25: Regression estimates of the determinants on support for a radical-right party.

Note: This table represents the effects of exposure to automation on the outcome of interest (support for right-wing populism) and mediators. These result show the relationship including the mediator as control variable as a robustness check. The independent variable is exposure to automation approached following Frey, Berger, and Chen (2017) Dependent variable: (1) Support for populist radical right. (2-4) Level of agreement with immigration as better for culture, economy and life. Answers range from "Not like me at all" $(= 0)$ to "Very much like me" $(= 10)$. (5-6) Level of agreement with "life is getting worse" and "hard to have hope about the future." Answers range from "Agree strongly" $(= 1)$ to "Disagree strongly" $(= 5)$. Source: ESS (6-7).

	(1)	(2)	(3)	(4)	(5)	(6)
	Baseline	Culture	Imm Eco	Imm Worse Life	Life Better	Hopeful
Sd Individual Exposure	0.028***	0.016**	0.018***	0.018***	0.028**	0.034***
	(0.007)	(0.006)	(0.007)	(0.007)	(0.011)	(0.011)
Pro-Immigration Culture		-0.015***				
		(0.001)				
Pro-Immigration Economy			-0.013***			
			(0.001)			
Pro-Immigration General				-0.015***		
				(0.001)		
Non-Nostalgic: Life Getting Better					-0.016***	
					(0.002)	
Non-Nostalgic: Hopeful Future						-0.004***
						(0.001)
Country-Year FE	✓	✓	✓	✓	✓	✓
NUTS FE	✓	✓	✓	✓	✓	✓
Observations	97035	94081	93716	94199	27866	27963
R^2 (p)	0.108	0.131	0.127	0.129	0.111	0.107
AIC	-3.4e+04	-3.6e+04	-3.5e+04	-3.6e+04	-1.3e+04	-1.2e+04

Standard errors clustered by region-year in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table G.26: IV Regression estimates of the impact of a one-SD increase in regional-level robot exposure on voting for a radical-right party.

Note: This table represents the effects of exposure to automation on the outcome of interest (support for right-wing populism) and mediators. These result show the relationship including the mediator as control variable as a robustness check. The independent variable is exposure to automation approached following Anelli, Colantone, and Stanig (2021) Dependent variable: (1) Support for populist radical right. (2-4) Level of agreement with immigration as better for culture, economy and life. Answers range from "Not like me at all" $(= 0)$ to "Very much like me" $(= 10)$. (5-6) Level of agreement with "life is getting worse" and "hard to have hope about the future." Answers range from "Agree strongly" $(= 1)$ to "Disagree strongly" $(= 5)$. Source: ESS (1-7).

G.3 Observational: Causal mediation analysis

G.3.1 Refers to Figure 6 . **Automation proxied as Frey & Osborne**

	(1) Culture	(2) Imm Eco	(3) Imm Worse Life	(4) Hopeless	(5) Worse Life
Frey & Osborne	0.095***	0.097***	0.098***	0.117***	0.106***
	(0.016)	(0.015)	(0.016)	(0.023)	(0.021)
Education (years)	-0.001*	-0.001**	-0.001***	-0.002***	-0.002***
	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)
Age	-0.001***	-0.001***	-0.001***	-0.001***	-0.001***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Female	-0.020***	-0.027***	-0.023***	-0.023***	-0.024***
	(0.004)	(0.004)	(0.004)	(0.005)	(0.005)
Ethnic minority	0.026***	0.028***	0.025***	0.030***	0.029***
	(0.007)	(0.008)	(0.008)	(0.009)	(0.008)
Country-Year FE	✓	✓	✓	✓	✓
NUTS FE	✓	✓	✓	✓	✓
Observations	28690	28576	28638	14531	14496
R^2 (p)	0.110	0.106	0.110	0.097	0.102
AIC	-4800.000	-4500.000	-4700.000	-5300.000	-5400.000

Standard errors clustered by region-year in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table G.27: Mediated effects of Risk of automation on electoral support for the radical right (2nd stage).
Note: This table represents the Average Mediated Causal Effect (ACME) and Average Direct Effect (ADE) of exposure to automation on the outcome of interest, which is support for right-wing populism. The treatment variable, exposure to automation, is approached following Frey, Berger, and Chen (2017). The dependent variable is support for the populist radical right. The mediators include proxies for nostalgia and marginalization. For marginalization, the relevant question measures the level of agreement with the statement that immigration is beneficial for culture, economy, and life. Answers range from "Not like me at all" $(= 0)$ to "Very much like me" $(= 10)$. For nostalgia, it measures the level of agreement with statements such as "life is getting worse" and "it's hard to have hope for the future," with answers ranging from "Agree strongly" $(= 1)$ to "Disagree strongly" $(= 5)$. Source: ESS 6-7. Refers to Figure 6.

Type	**Estimate**	**CI Lower**	**CI Upper**	**Mediator**
ACME	0.036	0.030	0.042	Immigration: Culture
ADE	0.085	0.053	0.117	Immigration: Culture
Total Effect	0.121	0.090	0.153	Immigration: Culture
Proportion Mediated	29.7%	0.222	0.409	Immigration: Culture
ACME	0.034	0.029	0.041	Immigration: Economics
ADE	0.087	0.055	0.118	Immigration: Economics
Total Effect	0.122	0.091	0.153	Immigration: Economics
Proportion Mediated	28.3%	0.212	0.400	Immigration: Economics
ACME	0.032	0.027	0.038	Immigration: Live
ADE	0.087	0.055	0.119	Immigration: Live
Total Effect	0.119	0.087	0.150	Immigration: Live
Proportion Mediated	27.3%	0.202	0.378	Immigration: Live
ACME	0.016	0.012	0.020	Nostalgia: Life Worse
ADE	0.106	0.074	0.139	Nostalgia: Life Worse
Total Effect	0.122	0.090	0.155	Nostalgia: Life Worse
Proportion Mediated	13.0%	0.091	0.189	Nostalgia: Life Worse
ACME	0.005	0.002	0.008	Nostalgia: Hopeless
ADE	0.117	0.085	0.148	Nostalgia: Hopeless
Total Effect	0.122	0.091	0.153	Nostalgia: Hopeless
Proportion Mediated	3.8%	0.018	0.067	Nostalgia: Hopeless

Table G.28: Mediation Analysis
Note: This table represents the Average Mediated Causal Effect (ACME) and Average Direct Effect (ADE) of exposure to automation on the outcome of interest, which is support for right-wing populism. The treatment variable, exposure to automation, is approached following Frey, Berger, and Chen (2017). The dependent variable is support for the populist radical right. The mediators include proxies for nostalgia (indicated in red tones) and marginalization (indicated in yellow tones). For marginalization, the relevant question measures the level of agreement with the statement that immigration is beneficial for culture, economy, and life. Answers range from "Not like me at all" $(= 0)$ to "Very much like me" $(= 10)$. For nostalgia, it measures the level of agreement with statements such as "life is getting worse" and "it's hard to have hope for the future," with answers ranging from "Agree strongly" $(= 1)$ to "Disagree strongly" $(= 5)$. Source: ESS 6-7. Refers to Figure 6.

G.3.2 Sensitivity Analysis associated to Figure 6 . Table G.29 present sensitivity analysis.

Mechanism	Mediator	Support for Radical Right ρ_0
Immigration	Country's cultural life	-0.4
	Country's economy	-0.4
	Living in the country	-0.4
Nostalgia	Life is getting worse	-0.1
	Lack of hope for the future	-0.1

Table G.29: Sensitivity analyses. Estimated using the "Medsens" statistical package in Stata (Hicks and Tingley, 2011).
Note: This table represents the sensitivity for the ACME presented in Figure 6.

G.3.3 Robustness Check Mediation. Following alternative specifications of the mediation analysis.

	(1) Culture	(2) Imm Eco	(3) Imm Worse Life	(4) Hopeless	(5) Worse Life
Frey & Osborne	0.097***	0.097***	0.099***	0.126***	0.112***
	(0.016)	(0.016)	(0.017)	(0.024)	(0.023)
Regional Δ robots	-0.088**	-0.094***	-0.079**	0.047**	0.030
	(0.037)	(0.035)	(0.038)	(0.023)	(0.023)
Education (years)	-0.001***	-0.002***	-0.002***	-0.003***	-0.003***
	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)
Age	-0.001***	-0.001***	-0.001***	-0.001***	-0.001***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Female	-0.020***	-0.028***	-0.023***	-0.025***	-0.026***
	(0.005)	(0.005)	(0.005)	(0.006)	(0.006)
Urban	-0.005	-0.006	-0.008*	-0.010	-0.010*
	(0.004)	(0.005)	(0.005)	(0.006)	(0.006)
Union Member	0.001	-0.002	-0.000	0.000	-0.000
	(0.004)	(0.004)	(0.004)	(0.005)	(0.005)
Unemployed	0.009	0.004	0.005	-0.000	-0.001
	(0.010)	(0.010)	(0.010)	(0.011)	(0.011)
Ethnic minority	0.026***	0.028***	0.026***	0.028**	0.028***
	(0.009)	(0.010)	(0.009)	(0.011)	(0.011)
Foreign Born	-0.004	-0.003	-0.001	-0.003	-0.000
	(0.006)	(0.006)	(0.006)	(0.007)	(0.007)
Precarious employment contract	-0.005	-0.005	-0.003	-0.003	-0.005
	(0.006)	(0.006)	(0.006)	(0.008)	(0.009)
Regional Immigrant Exposure	-0.018	-0.045	-0.019	-0.039	-0.167
	(0.184)	(0.187)	(0.180)	(0.123)	(0.141)
Regional Unemployment	-0.158	-0.244	-0.229	0.237	0.285
	(0.257)	(0.270)	(0.256)	(0.162)	(0.175)
Country-Year FE	✓	✓	✓	✓	✓
NUTS FE	✓	✓	✓	✓	✓
Observations	21863	21763	21791	10540	10505
R^2 (p)	0.112	0.109	0.112	0.091	0.098
AIC	-2400.000	-2300.000	-2400.000	-2700.000	-2800.000

Standard errors clustered by region-year in parentheses
$^{*}\ p < 0.1$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$

Table G.30: Mediated effects of Risk of automation on electoral support for the radical right (2nd stage, with additional control variables). *Note*: The treatment variable, exposure to automation, is approached following Frey, Berger, and Chen (2017), regional robots exposure comes from Anelli, Colantone, and Stanig (2021). This model contains additional control variables. The dependent variable and mediators are similar to the main model. Source: ESS 6-7.

Automation proxied as Anelli, et al

	(1) Culture	(2) Imm Eco	(3) Imm Worse Life	(4) Hopeless	(5) Worse Life
Individual Exposure	0.499**	0.510**	0.576***	0.448*	0.840***
	(0.206)	(0.204)	(0.214)	(0.256)	(0.214)
Country-Year FE	✓	✓	✓	✓	✓
NUTS FE	✓	✓	✓	✓	✓
Observations	28810	28698	28763	14587	14603
R^2 (p)	0.108	0.105	0.109	0.094	0.092
AIC	-4800.000	-4600.000	-4800.000	-5300.000	-5300.000

Standard errors clustered by region-year in parentheses
$^{*}\ p < 0.1$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$

Table G.31: Mediated effects of Risk of automation on electoral support for the radical right (2nd stage). *Note*: This table represents the Average Mediated Causal Effect (ACME) and Average Direct Effect (ADE) of exposure to automation on the outcome of interest, which is support for right-wing populism. The treatment variable, exposure to automation, is approached following Anelli, Colantone, and Stanig (2021). The dependent variable is support for the populist radical right. The mediators include proxies for nostalgia and marginalization. For marginalization, the relevant question measures the level of agreement with the statement that immigration is beneficial for culture, economy, and life. Answers range from "Not like me at all" ($= 0$) to "Very much like me" ($= 10$). For nostalgia, it measures the level of agreement with statements such as "life is getting worse" and "it's hard to have hope for the future," with answers ranging from "Agree strongly" ($= 1$) to "Disagree strongly" ($= 5$). Source: ESS 6-7.

	(1) Culture	(2) Imm Eco	(3) Imm Worse Life	(4) Hopeless	(5) Worse Life
Individual Exposure	0.818***	0.506**	0.569**	1.032***	0.916***
	(0.208)	(0.221)	(0.230)	(0.274)	(0.259)
Age	-0.001***	-0.001***	-0.001***	-0.000***	-0.001***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Female	-0.022***	-0.030***	-0.025***	-0.028***	-0.029***
	(0.005)	(0.005)	(0.005)	(0.006)	(0.007)
Urban	-0.007	-0.006	-0.008*	-0.015**	-0.015**
	(0.005)	(0.005)	(0.005)	(0.007)	(0.007)
Union Member	-0.000	-0.002	-0.001	-0.004	-0.004
	(0.004)	(0.004)	(0.004)	(0.005)	(0.005)
Unemployed	0.011	0.005	0.005	-0.000	-0.001
	(0.010)	(0.010)	(0.010)	(0.012)	(0.011)
Ethnic minority	0.026***	0.029***	0.027***	0.027**	0.028***
	(0.009)	(0.010)	(0.009)	(0.011)	(0.011)
Foreign Born	-0.004	-0.003	-0.001	-0.003	-0.000
	(0.006)	(0.006)	(0.006)	(0.007)	(0.007)
Precarious employment contract	-0.004	-0.004	-0.002	-0.001	-0.004
	(0.006)	(0.006)	(0.006)	(0.008)	(0.009)
Regional Immigrant Exposure	0.035	0.021	0.039	-0.077	-0.201
	(0.179)	(0.181)	(0.173)	(0.173)	(0.153)
Regional Unemployment	-0.209	-0.286	-0.264	0.200	0.228
	(0.271)	(0.285)	(0.270)	(0.171)	(0.171)
Country-Year FE	✓	✓	✓	✓	✓
NUTS FE	✓	✓	✓	✓	✓
Observations	21863	21763	21791	10562	10528
R^2 (p)	0.109	0.108	0.111	0.081	0.089
AIC	-2400.000	-2300.000	-2400.000	-2600.000	-2700.000

Standard errors clustered by region-year in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table G.32: Mediated effects of Risk of automation on electoral support for the radical right (2nd stage, with additional control variables).
Note: This table represents the Average Mediated Causal Effect (ACME) and Average Direct Effect (ADE) of exposure to automation on the outcome of interest, which is support for right-wing populism. The treatment variable, exposure to automation, is approached following Anelli, Colantone, and Stanig (2021). The dependent variable is support for the populist radical right. The mediators include proxies for nostalgia and marginalization. For marginalization, the relevant question measures the level of agreement with the statement that immigration is beneficial for culture, economy, and life. Answers range from "Not like me at all" ($= 0$) to "Very much like me" ($= 10$). For nostalgia, it measures the level of agreement with statements such as "life is getting worse" and "it's hard to have hope for the future," with answers ranging from "Agree strongly" ($= 1$) to "Disagree strongly" ($= 5$). Source: ESS 6-7.

G.4 Evidence Post-AI Shock (ESS 11 - 2023)

	(1) FO: Radical Right	(2) FO: Culture	(3) FO: Economy	(4) FO: Live	(5) SML: Radical Right	(6) SML: Culture	(7) SML: Economy	(8) SML: Live
Frey & Osborne	0.3402***	-0.5177***	-0.4266***	-0.3933***				
	(0.1181)	(0.0668)	(0.0708)	(0.0670)				
SML					0.4451**	-0.6204***	-0.4569***	-0.4290***
					(0.1897)	(0.1173)	(0.1244)	(0.1259)
Demographic	✓	✓	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓	✓	✓
Region FE	✓	✓	✓	✓	✓	✓	✓	✓
Observations	11227	18487	18429	18421	11389	18773	18717	18702
R^2 (p)	0.11429	0.16616	0.12316	0.15863	0.11398	0.16280	0.12049	0.15665
AIC	8260.6	82785.8	81277.9	80017.6	8284.6	84101.2	82474.4	81234.9

Standard errors clustered by region in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table G.33: Exposure to Automation Risk, Radical Right Support, and Marginalization
Note: This table replicates the direct-effect analysis using ESS round 11. Columns 1–4 use the Frey and Osborne (2017) measure of occupational exposure to computerization; Columns 5–8 use the Suitability for Machine Learning (SML) measure from Brynjolfsson, Mitchell, and Rock (2019). The dependent variable in Columns 1 and 5 is a binary indicator for support for a radical right party, coded from reported party vote. Columns 2–4 and 6–8 use three ESS immigration items capturing whether immigration is perceived as good or bad for culture, the economy, and the country as a place to live; higher values indicate more favorable views toward immigration. All models include demographic controls, ESS weights, and region fixed effects; standard errors are clustered at the regional level. Because ESS 11 does not include the nostalgia items used in the earlier rounds, this replication focuses on radical-right support and immigration attitudes only.

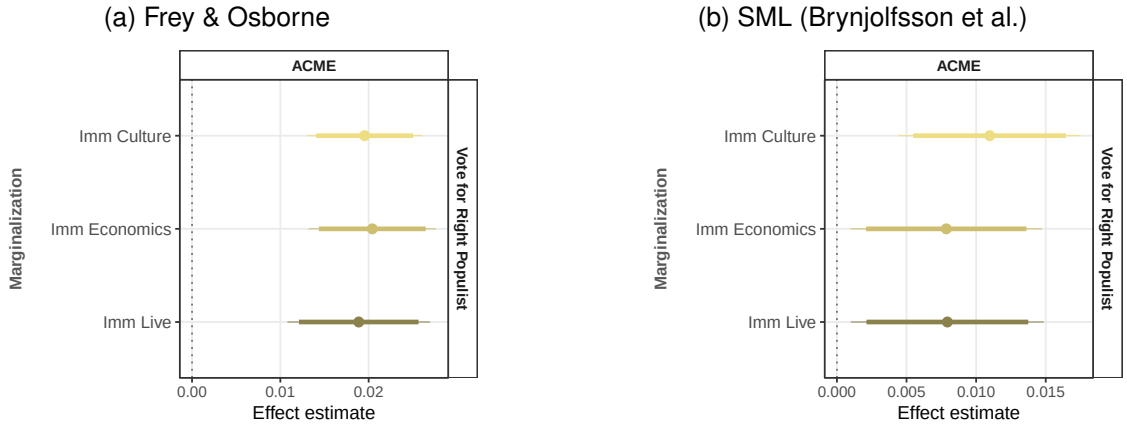


Figure G.27: Mediated effect of automation through cultural beliefs on political behavior: ESS11 alternative automation measures.

Note: Panel (a) uses the Frey and Osborne (2017) probability of computerization as the independent variable and reports the ESS wave 11 mediation analysis. Panel (b) uses the SML exposure measure by Brynjolfsson, Mitchell, and Rock (2019) and reports the ESS11 mediation analysis. The outcome is support for a radical-right party (dummy). The mediators are the three immigration items: whether immigration is good or bad for the country's cultural life, economy, and place to live; responses range from 0 to 10, with higher values indicating more pro-immigration attitudes. ESS 11 does not include the nostalgia items used in the main text, so only the immigration mediators are shown here. Data come from ESS wave 11. For panel (a), the mediation-model sample sizes are $N = 11{,}722$ for culture, $N = 11{,}677$ for economy, and $N = 11{,}675$ for place to live; the corresponding proportions mediated are 51.6%, 55.5%, and 50.1%. For panel (b), the mediation-model sample sizes are $N = 11{,}934$ for culture, $N = 11{,}889$ for economy, and $N = 11{,}884$ for place to live; the corresponding proportions mediated are 52.6%, 33.8%, and 39.0%.

H Research Ethics

This study was reviewed and granted an exemption by the Institutional Review Board of XXX on March 16, 2023 (STUDY22120089).

Voluntary informed consent was obtained by all human subjects. It was obtained electronically and was built into the survey flow. Subjects were free to decline participation. Prior to providing consent, subjects were informed about foreseeable risks, the lack of direct benefits associated with the research, whether and how identities and data will be protected, compensation, and the voluntary nature of the study, and were provided relevant contact information from the University's IRB and researcher.

The motivation of the study was not provided at the instruction step to avoid biasing subjects' responses. There was no more than minimal harm to subjects, and at the consent step individuals were told that they would be "fully debriefed about the study's purpose and procedures after your participation is complete." At the end of the study, subjects were fully debriefed about purposes and procedures.

I worked with a survey company (CloudResearch CloudResearch (2024), and the survey was prepared using Qualtrics) that does market research and compensates survey takers.

Instruction blocks: If you are a citizen of the United States, 18 years or older, and part of the workforce (currently working or looking for a job), we're inviting you to participate in a research study designed, among other things, to help a news organization decide what content it should feature in a news website about social change. You will be fully debriefed about the study's purpose and procedures after your participation is complete.

We estimate that answering the survey will take about 15-18 minutes. We will pay participants who pass a simple attention check about basic facts of the survey $1.50 for questionnaires that are at least 90% complete. You will be given a completion code for payment. There are no direct benefits from participation. There is a minimal risk of breach of confidentiality. We are not collecting any personally identifiable information, but you will be asked your zip code in order to identify the area of the country in which you live. In addition, your participation is voluntary. You can exit the survey at any time, and you may choose not to answer sensitive questions, but you cannot withdraw from the study after submitting the questionnaire.

If you are eligible and agree to participate in this survey, please click I AGREE TO PARTICIPATE below.

You may contact the Human Subjects Protection Advocate of the IRB Office, XXXXXX to discuss problems, concerns, and questions; obtain information; offer input; or discuss situations in the event that the research team is unavailable. Thank you very much for your time and for considering our request

The debriefing block: **Purpose of the Study:** This study is about attitudes toward technological change, and its consequences on political choices such as political engagement. This survey collects baseline demographic and political data and explores the effects of exposure to automation risks on support for redistribution, integration to the global economy, and voting behavior.

In order to test the project's hypotheses, the survey asked you to help a new news website about social change, and read two news. One group, read news related to job losses, and other one about new technologies. Please note that neither the news website nor individuals or companies mentioned do not in fact exist. We apologize for the use of a fictitious article.

Confidentiality: Your responses will be kept strictly confidential. No personally identifying information has been collected during the process of the survey (e.g., name, exact address). If you have any concerns please contact the researcher, XXXX at XXX.

I Survey Questionnaire

See below: